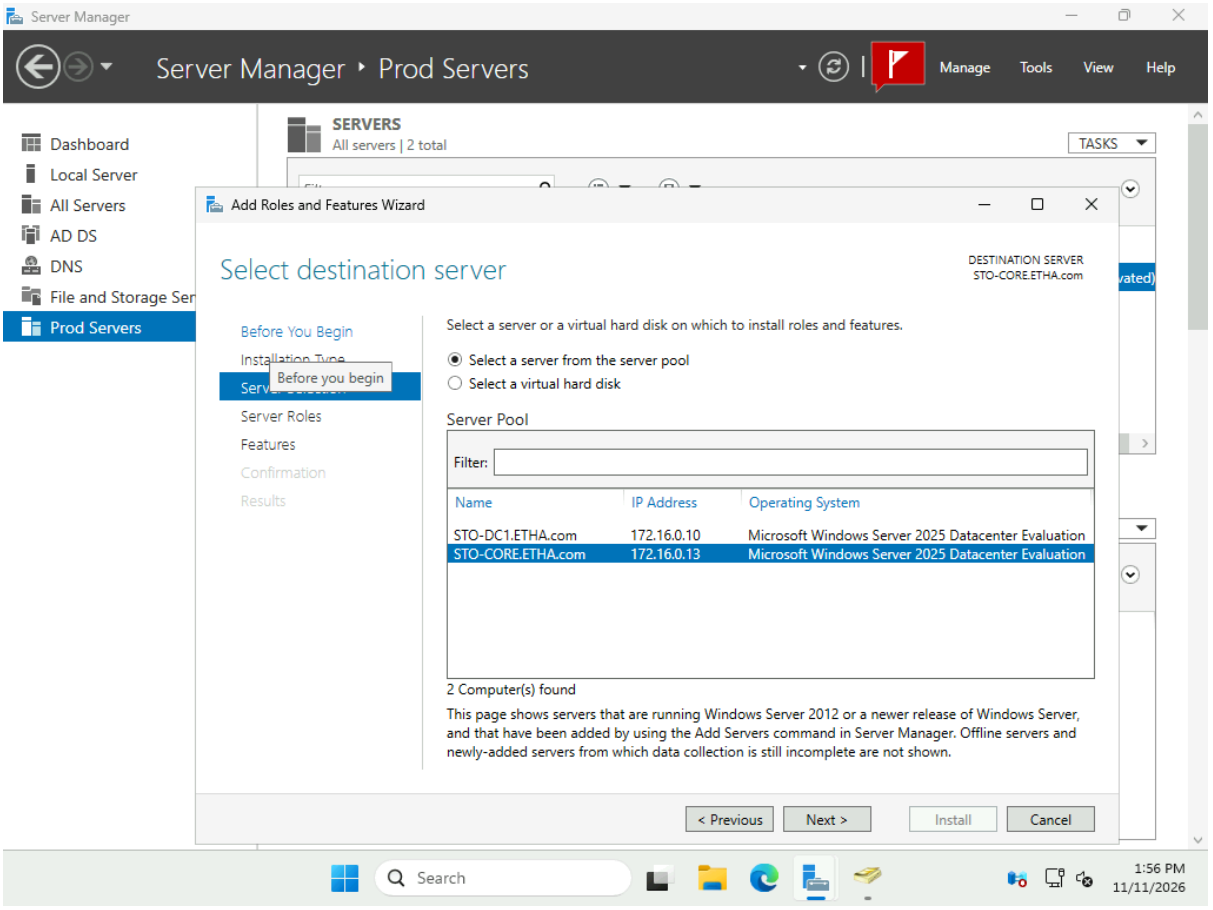
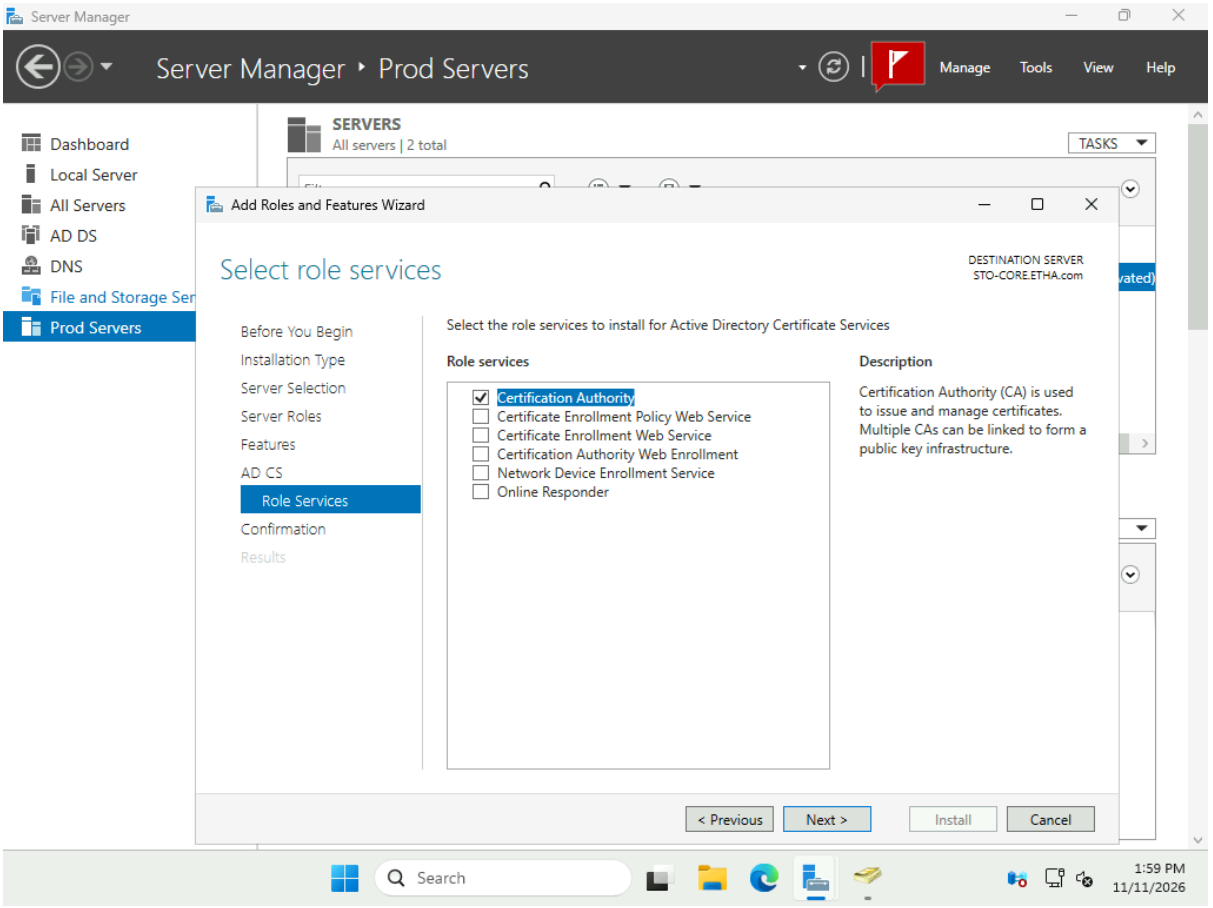
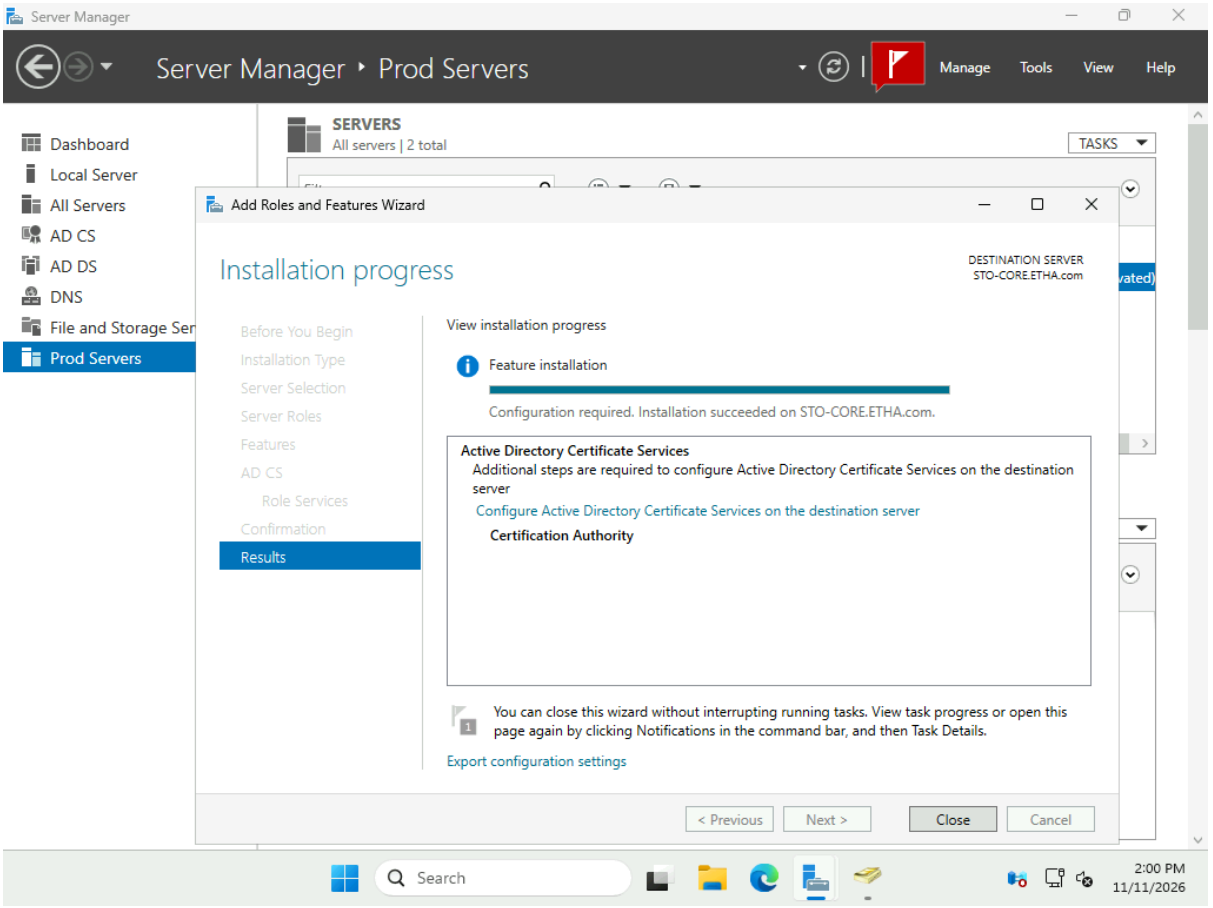
Deploying and Configuring a CA Hierarchy

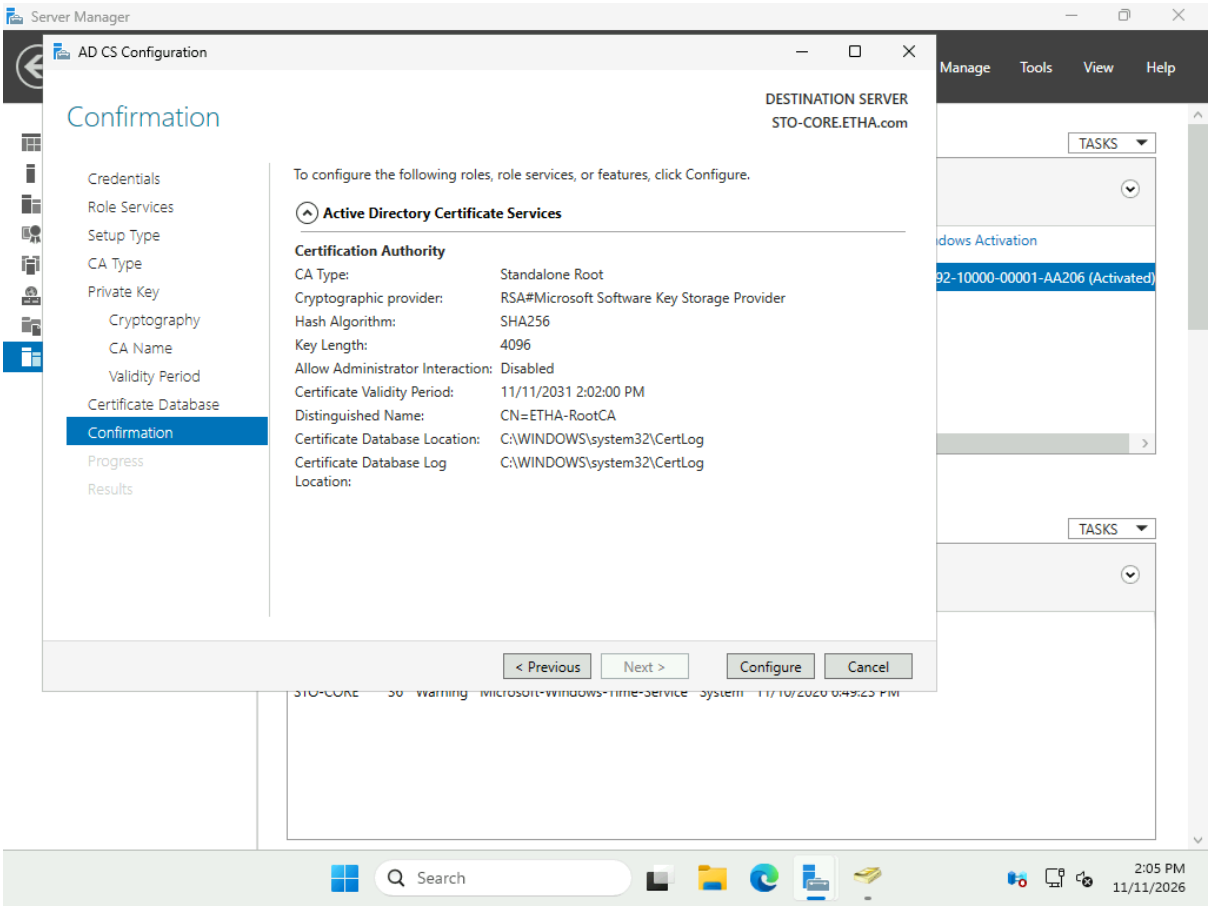
Practice 1: Deploying a Stand-Alone Root CA

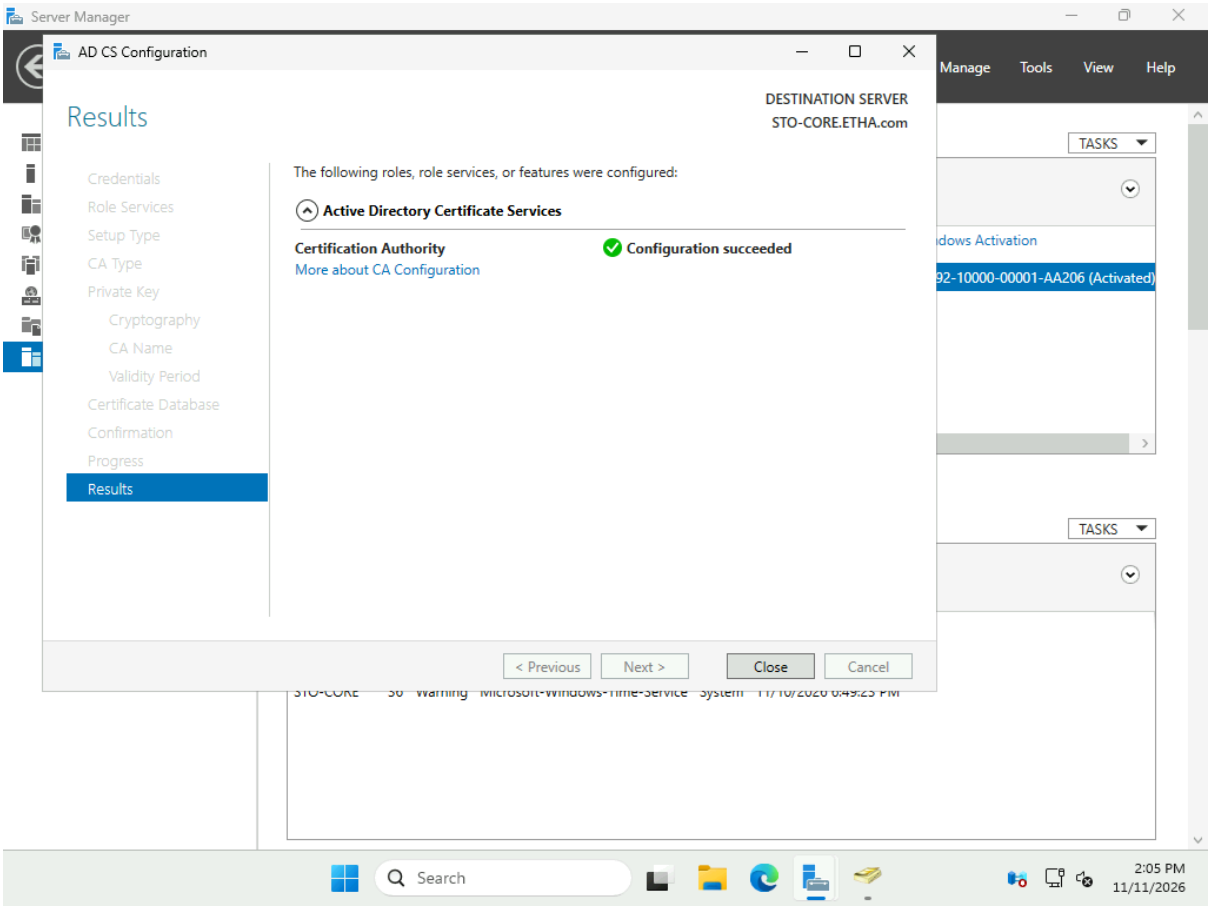
Task 1: Install and configure Active Directory Certificate Services on STO-CORE



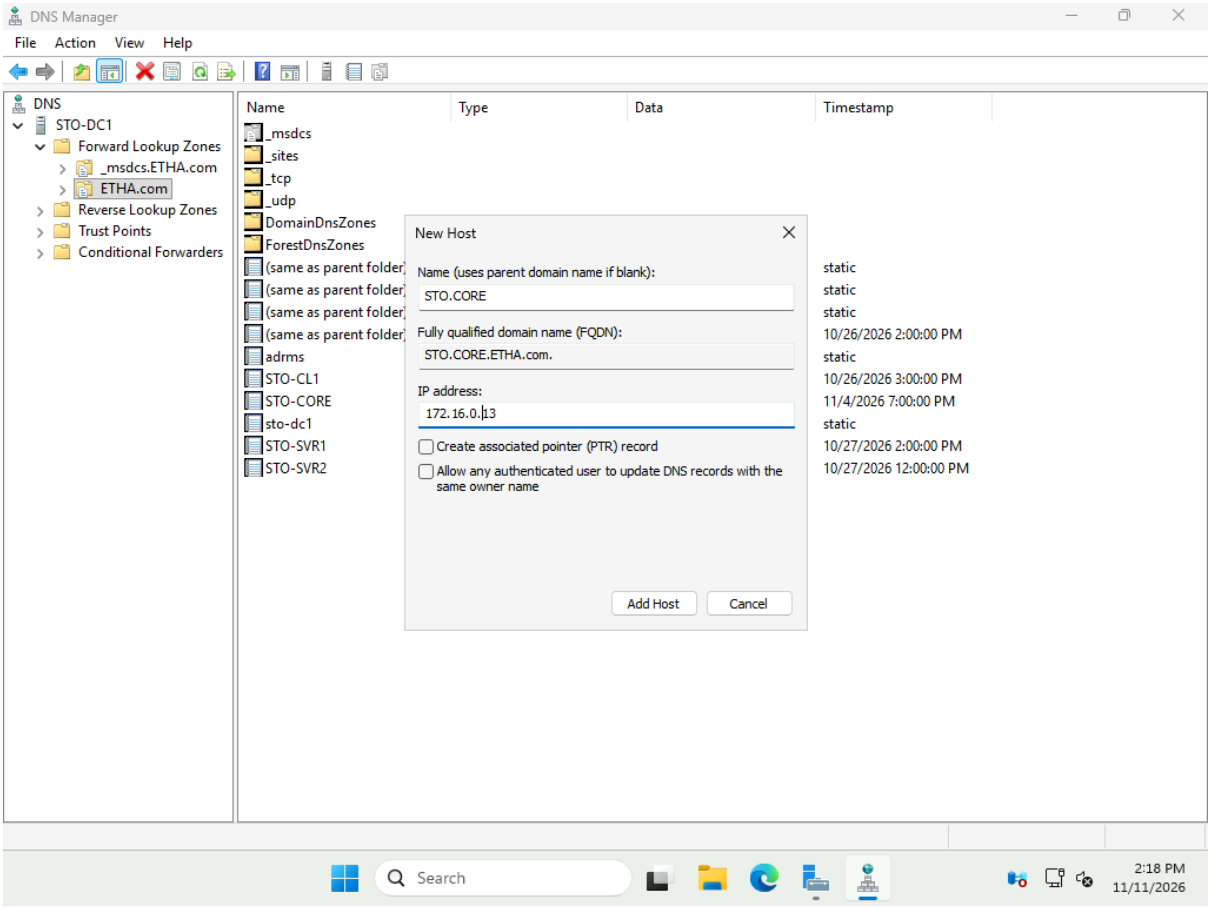

Server Manager
Server Manager • Prod Servers
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Prod Servers
SERVERS
All servers | 2 total
TASKS
Add Roles and Features Wizard
Select role services
DESTINATION SERVER
STO-CORE.ETHA.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD CS
Role Services
Confirmation
Results
Select the role services to install for Active Directory Certificate Services
Role services
Certification Authority
Certificate Enrollment Policy Web Service
Certificate Enrollment Web Service
Certification Authority Web Enrollment
Network Device Enrollment Service
Online Responder
Description
Certification Authority (CA) is used to issue and manage certificates. Multiple CAs can be linked to form a public key infrastructure.
< Previous
Next >
Install
Cancel
Search
1:59 PM
11/11/2026

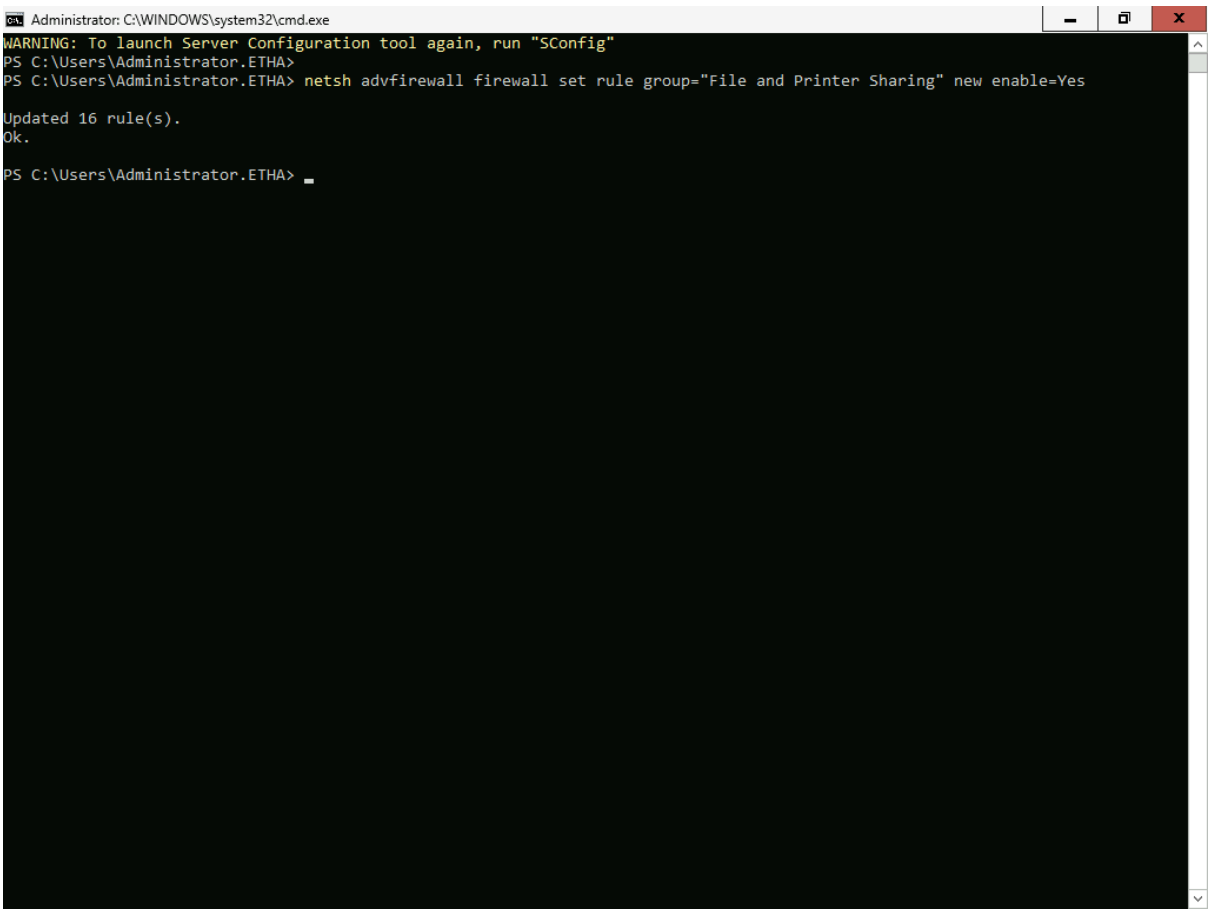

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Server Manager ‣ Prod Servers
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Local Server
All Servers
AD CS
AD DS
DNS
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Prod Servers
SERVERS
All servers | 2 total
TASKS
Add Roles and Features Wizard
Installation progress
DESTINATION SERVER
STO-CORE.ETHA.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD CS
Role Services
Confirmation
Results
View installation progress
Feature installation
Configuration required. Installation succeeded on STO-CORE.ETHA.com.
Active Directory Certificate Services
Additional steps are required to configure Active Directory Certificate Services on the destination server
Configure Active Directory Certificate Services on the destination server
Certification Authority
You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.
Export configuration settings
< Previous
Next >
Close
Cancel
Search
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Confirmation
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STO-CORE.ETHA.com
Credentials
Role Services
Setup Type
CA Type
Private Key
Cryptography
CA Name
Validity Period
Certificate Database
Confirmation
Progress
Results
To configure the following roles, role services, or features, click Configure.
Active Directory Certificate Services
Certification Authority
CA Type: Standalone Root
Cryptographic provider: RSA#Microsoft Software Key Storage Provider
Hash Algorithm: SHA256
Key Length: 4096
Allow Administrator Interaction: Disabled
Certificate Validity Period: 11/11/2031 2:02:00 PM
Distinguished Name: CN=ETHA-RootCA
Certificate Database Location: C:\WINDOWS\system32\CertLog
Certificate Database Log Location: C:\WINDOWS\system32\CertLog
< Previous
Next >
Configure
Cancel
Manage
Tools
View
Help
TASKS
Search
2:05 PM
11/11/2026



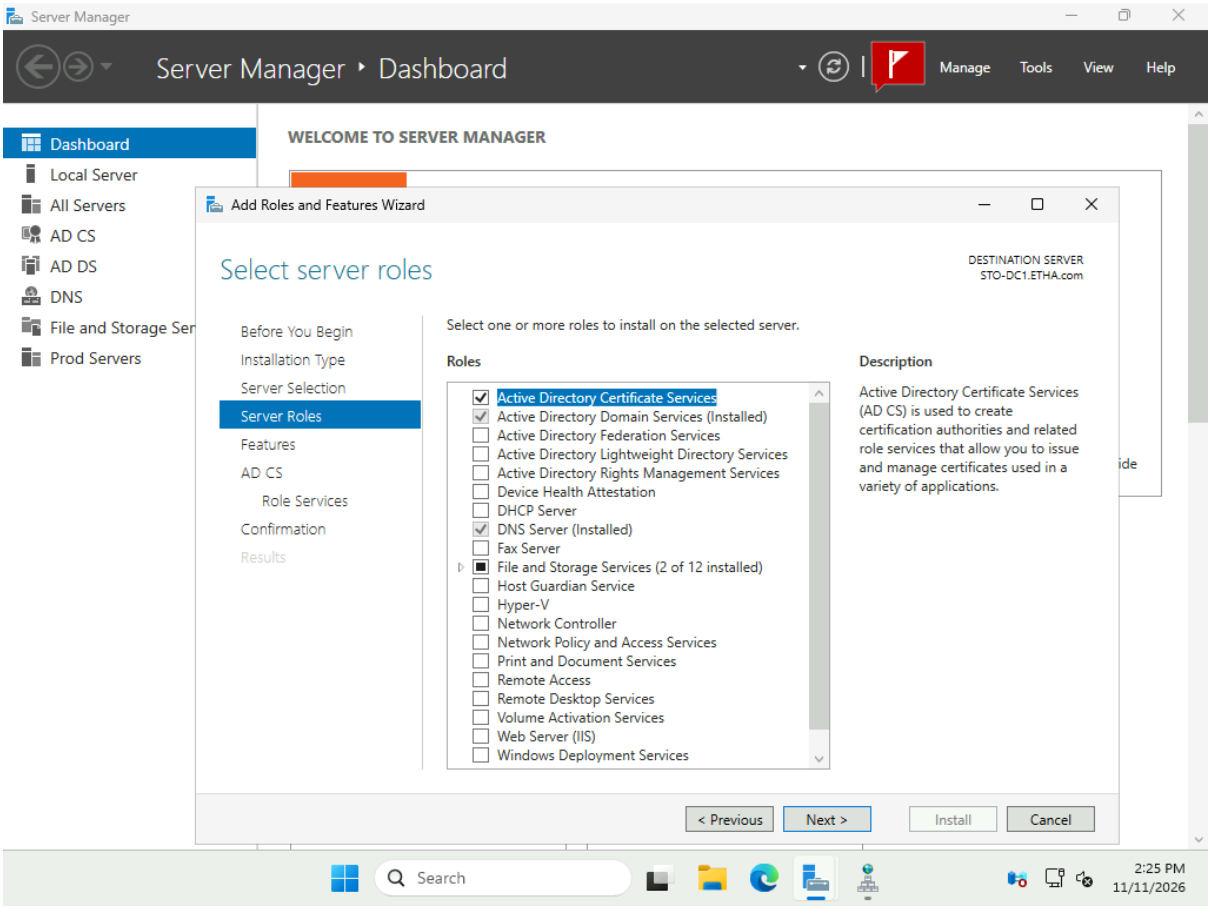
Task 2: Creating a DNS host record for STO-CORE and configure sharing on STO-CORE

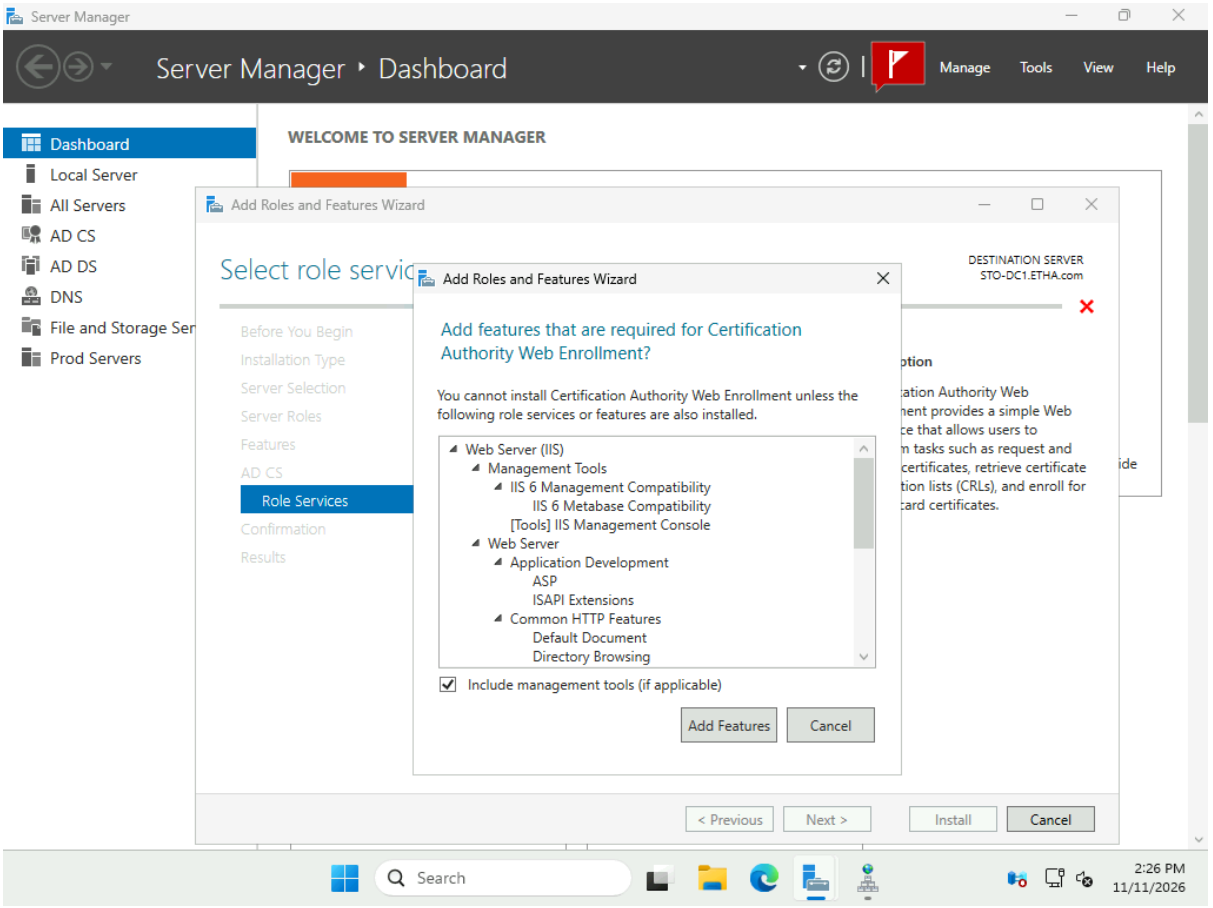

DNS Manager
File Action View Help
DNS
STO-DC1
Forward Lookup Zones
_msdcs.ETHA.com
ETHA.com
Reverse Lookup Zones
Trust Points
Conditional Forwarders
Name
Type
Data
Timestamp
_msdcs
_sites
_tcp
_udp
DomainDnsZones
ForestDnsZones
(same as parent folder)
(same as parent folder)
(same as parent folder)
(same as parent folder)
adrms
STO-CL1
STO-CORE
sto-dc1
STO-SVR1
STO-SVR2
static
static
static
10/26/2026 2:00:00 PM
static
10/26/2026 3:00:00 PM
11/4/2026 7:00:00 PM
static
10/27/2026 2:00:00 PM
10/27/2026 12:00:00 PM
New Host
Name (uses parent domain name if blank):
STO.CORE
Fully qualified domain name (FQDN):
STO.CORE.ETHA.com.
IP address:
172.16.0.13
Create associated pointer (PTR) record
Allow any authenticated user to update DNS records with the same owner name
Add Host
Cancel
Search
2:18 PM
11/11/2026

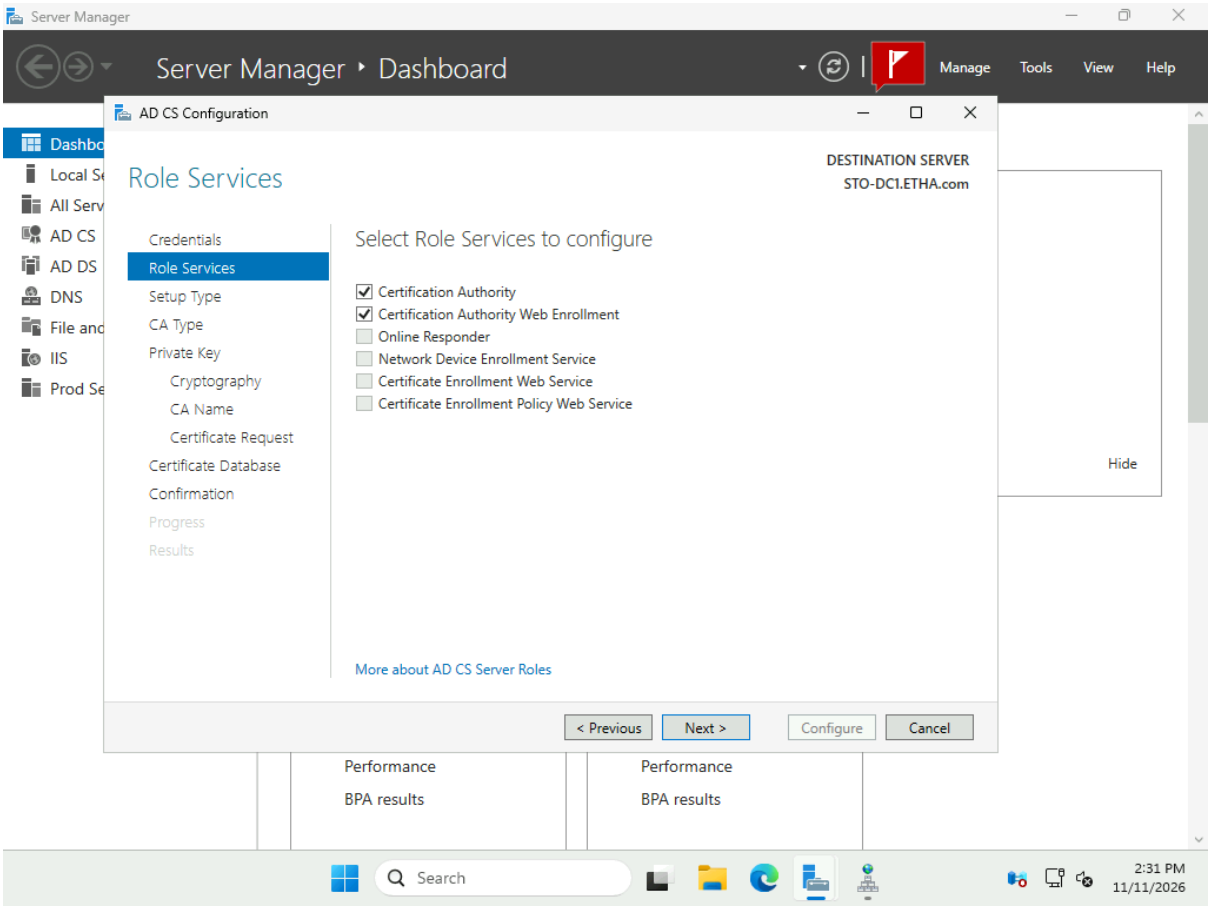


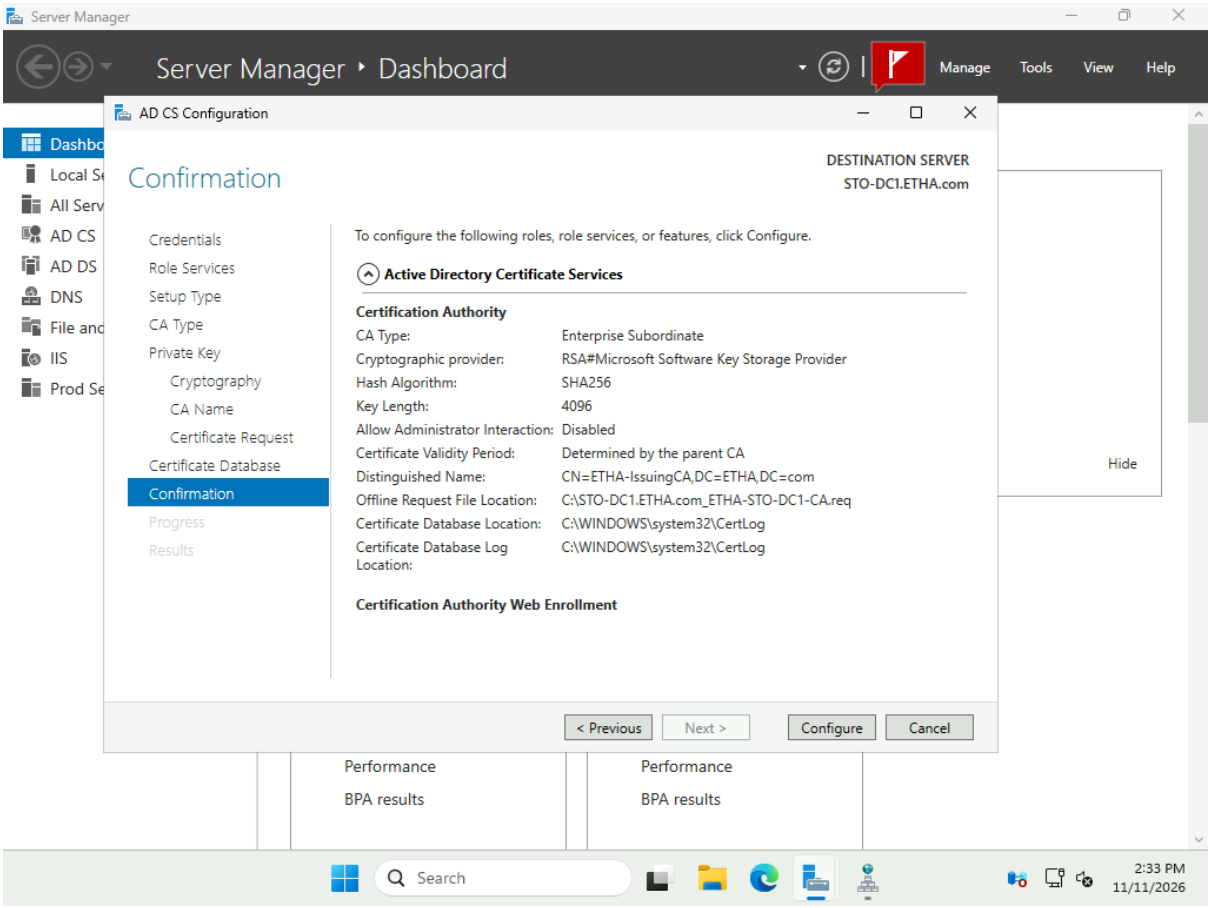
Practice 2: Deploying an Enterprise Subordinate CA

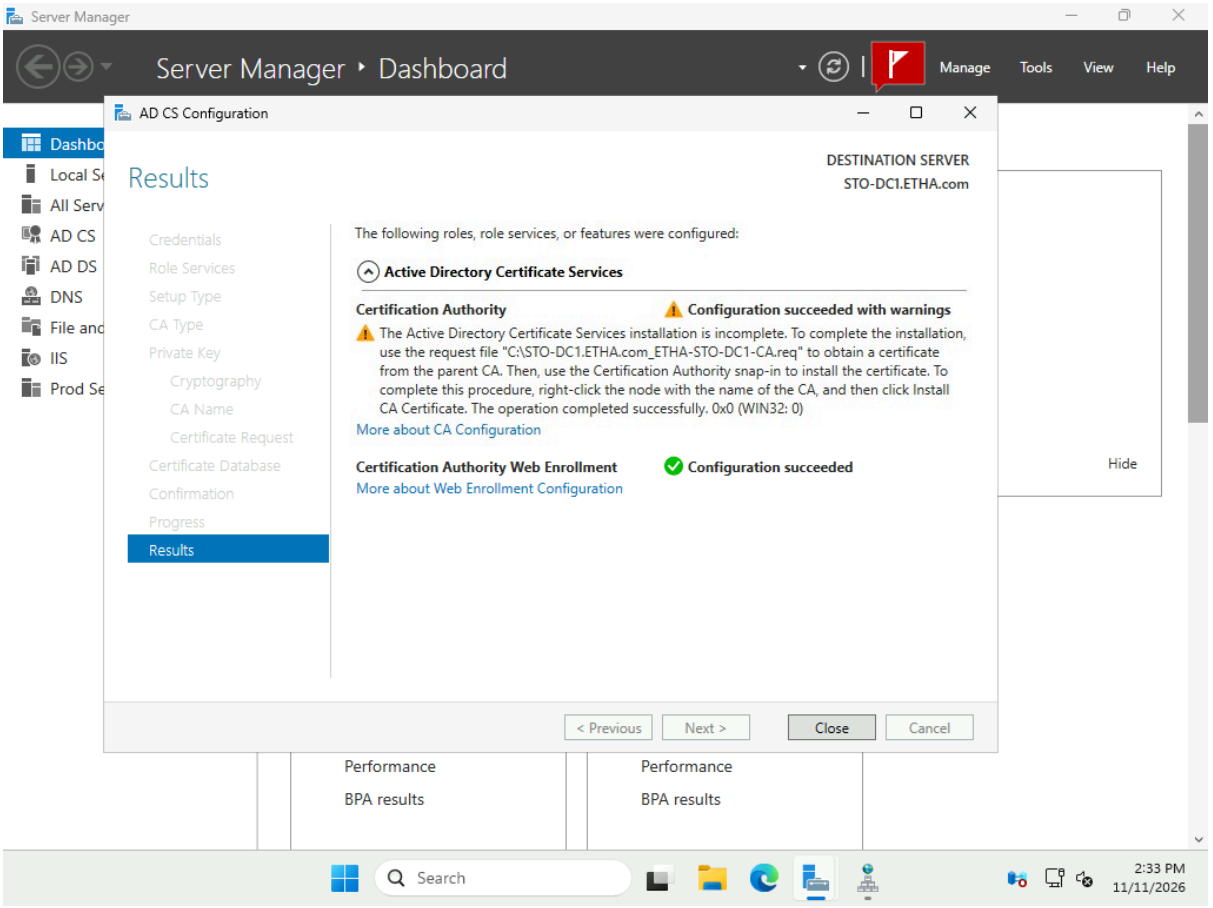
Task 3: Install and configure AD CS on STO-DC1

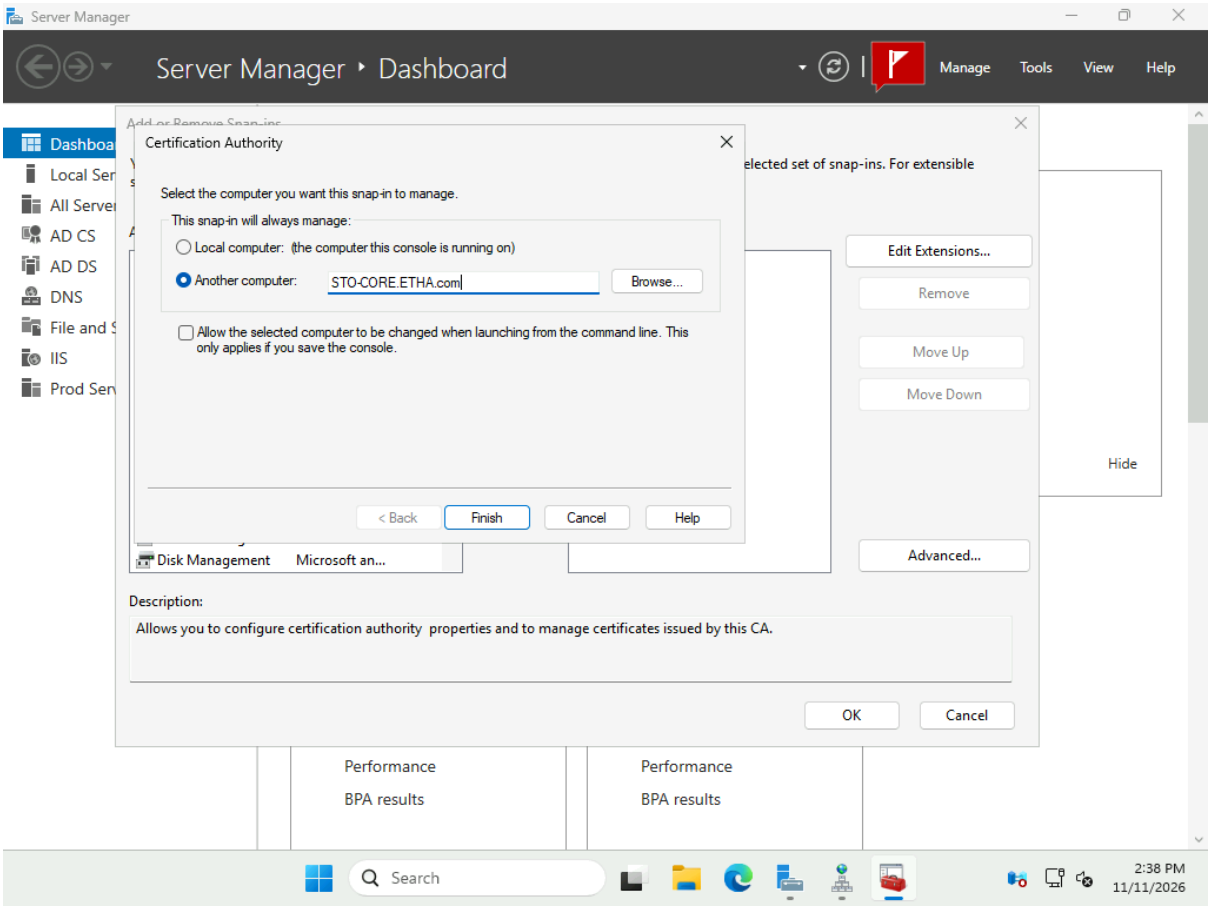

Server Manager
Server Manager ‣ Dashboard
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AD DS
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File and Storage Ser
Prod Servers
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Select server roles
DESTINATION SERVER
STO-DC1.ETHA.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD CS
Role Services
Confirmation
Results
Select one or more roles to install on the selected server.
Roles
Active Directory Certificate Services
Active Directory Domain Services (Installed)
Active Directory Federation Services
Active Directory Lightweight Directory Services
Active Directory Rights Management Services
Device Health Attestation
DHCP Server
DNS Server (Installed)
Fax Server
File and Storage Services (2 of 12 installed)
Host Guardian Service
Hyper-V
Network Controller
Network Policy and Access Services
Print and Document Services
Remote Access
Remote Desktop Services
Volume Activation Services
Web Server (IIS)
Windows Deployment Services
Description
Active Directory Certificate Services (AD CS) is used to create certification authorities and related role services that allow you to issue and manage certificates used in a variety of applications.
< Previous
Next >
Install
Cancel
Search
2:25 PM
11/11/2026

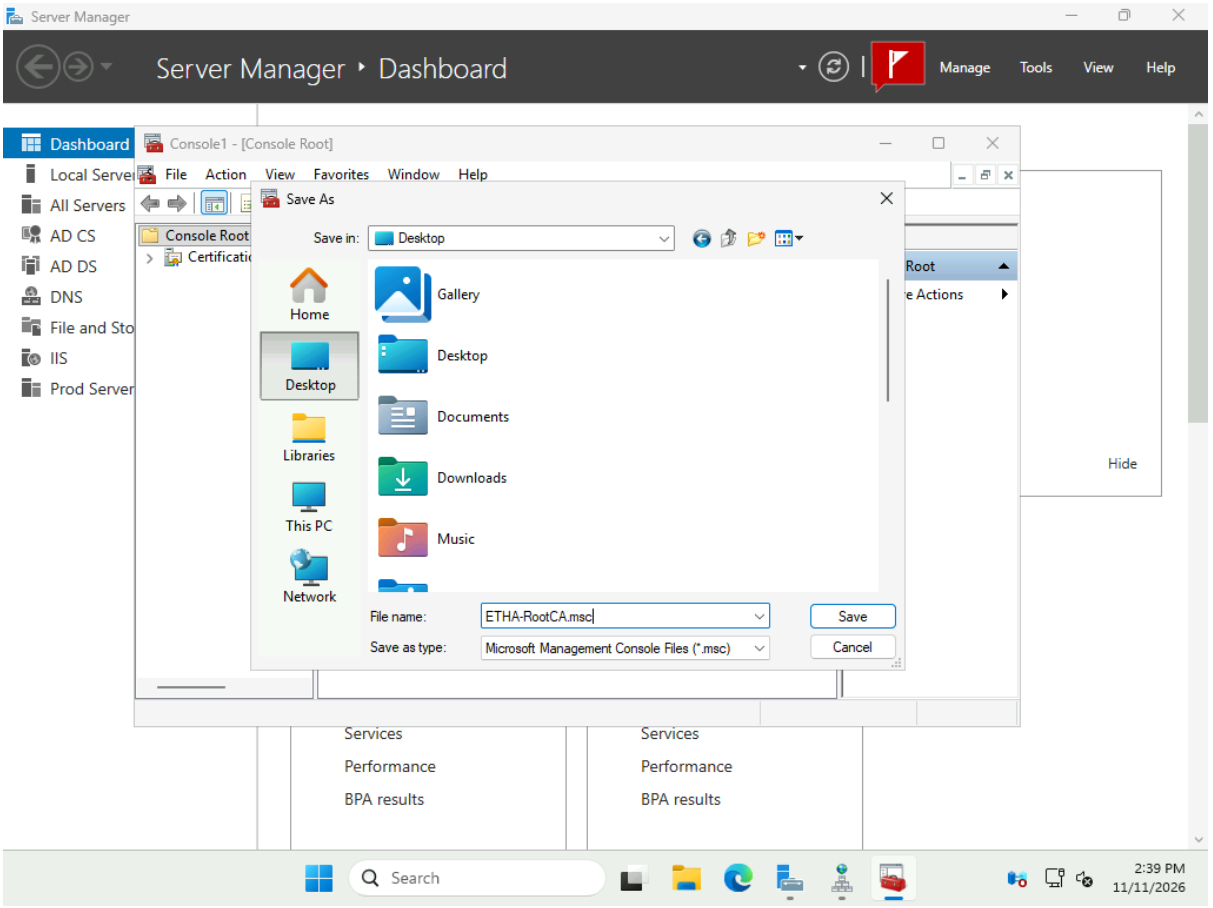

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Prod Servers
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Select role servic
DESTINATION SERVER
STO-DC1.ETHA.com
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AD CS
Role Services
Confirmation
Results
Add Roles and Features Wizard
Add features that are required for Certification Authority Web Enrollment?
You cannot install Certification Authority Web Enrollment unless the following role services or features are also installed.
Web Server (IIS)
Management Tools
IIS 6 Management Compatibility
IIS 6 Metabase Compatibility
[Tools] IIS Management Console
Web Server
Application Development
ASP
ISAPI Extensions
Common HTTP Features
Default Document
Directory Browsing
Include management tools (if applicable)
Add Features
Cancel
< Previous
Next >
Install
Cancel
Search
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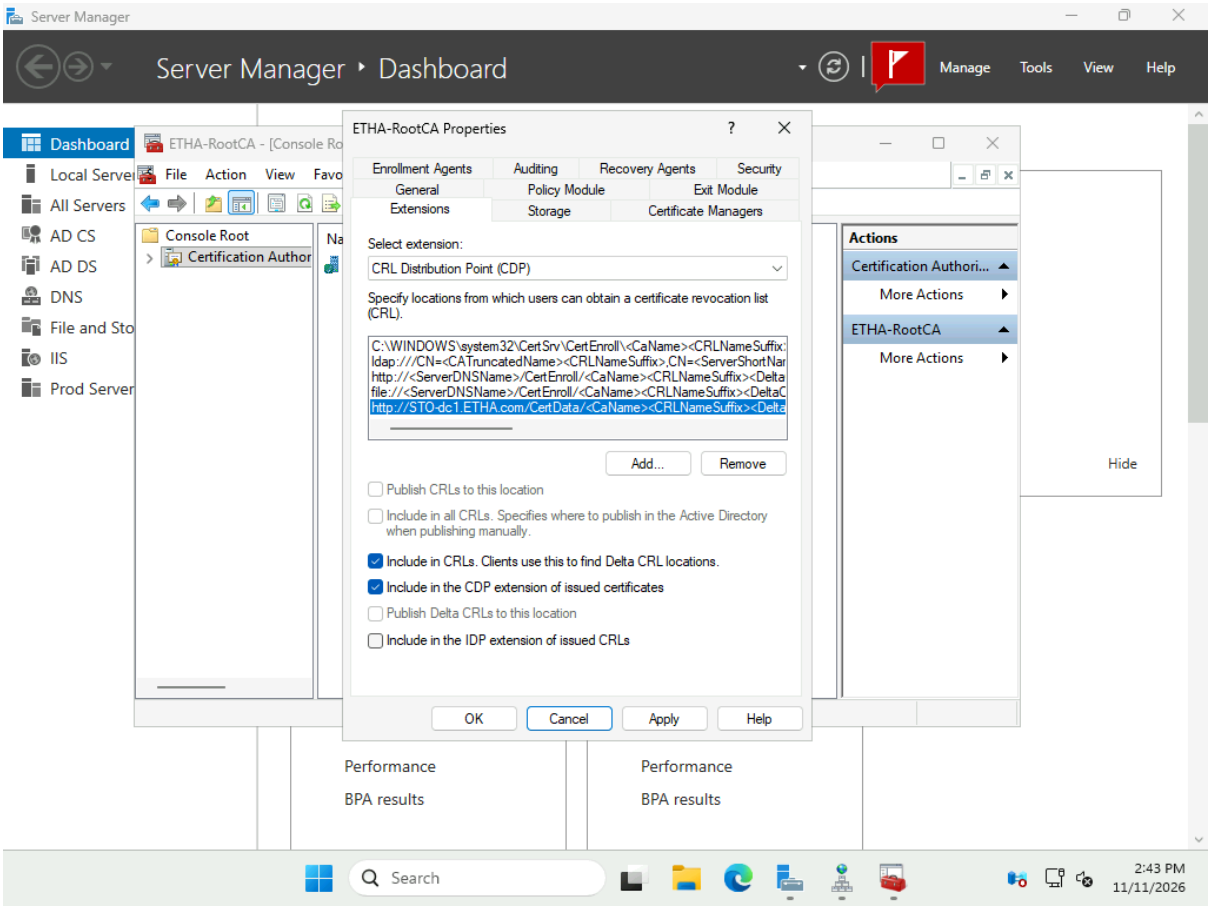

Server Manager
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AD CS Configuration
Role Services
DESTINATION SERVER
STO-DC1.ETHA.com
Credentials
Role Services
Setup Type
CA Type
Private Key
Cryptography
CA Name
Certificate Request
Certificate Database
Confirmation
Progress
Results
Select Role Services to configure
Certification Authority
Certification Authority Web Enrollment
Online Responder
Network Device Enrollment Service
Certificate Enrollment Web Service
Certificate Enrollment Policy Web Service
More about AD CS Server Roles
< Previous
Next >
Configure
Cancel
Hide
Performance
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BPA results
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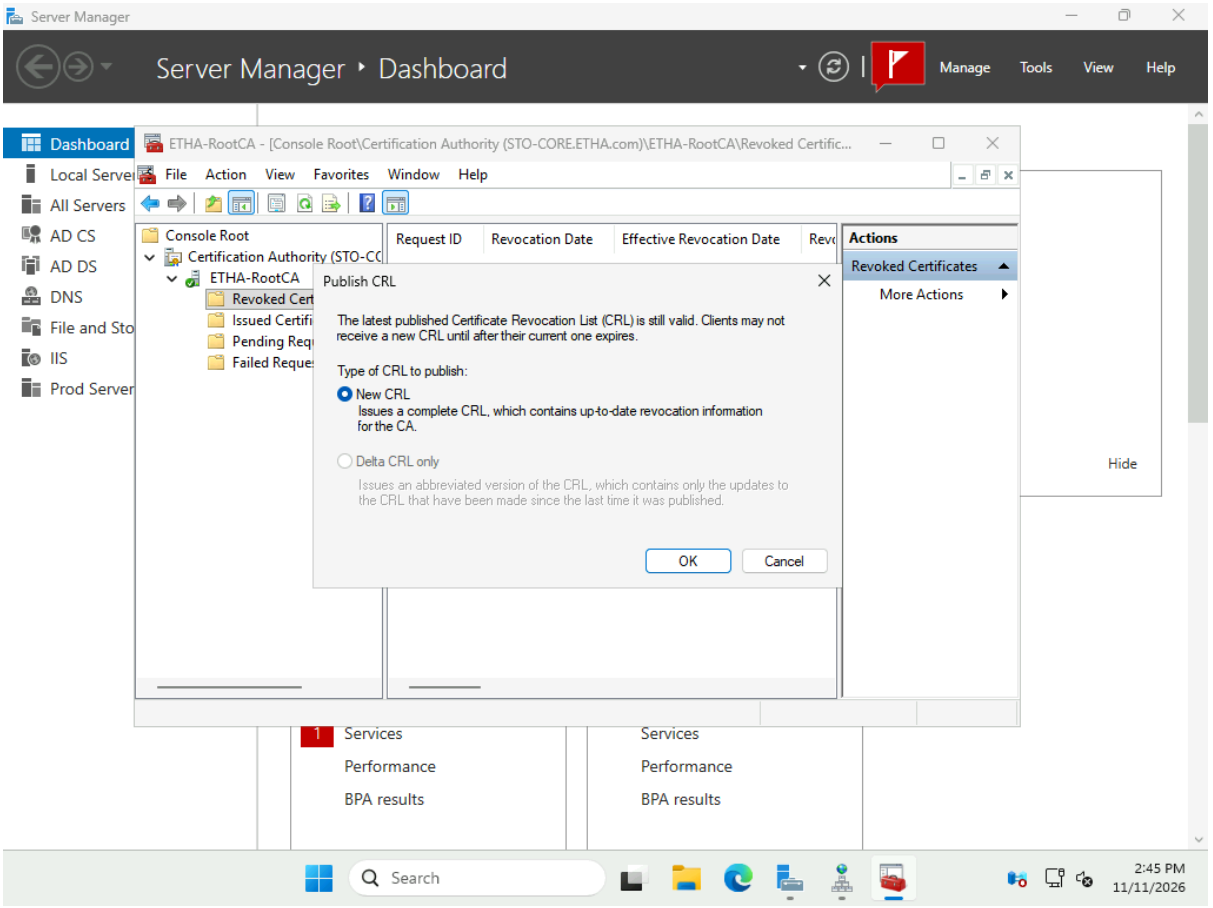

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Confirmation
DESTINATION SERVER
STO-DC1.ETHA.com
Credentials
Role Services
Setup Type
CA Type
Private Key
Cryptography
CA Name
Certificate Request
Certificate Database
Confirmation
Progress
Results
To configure the following roles, role services, or features, click Configure.
Active Directory Certificate Services
Certification Authority
CA Type: Enterprise Subordinate
Cryptographic provider: RSA#Microsoft Software Key Storage Provider
Hash Algorithm: SHA256
Key Length: 4096
Allow Administrator Interaction: Disabled
Certificate Validity Period: Determined by the parent CA
Distinguished Name: CN=ETHA-IssuingCA,DC=ETHA,DC=com
Offline Request File Location: C:\STO-DC1.ETHA.com_ETHA-STO-DC1-CA.req
Certificate Database Location: C:\WINDOWS\system32\CertLog
Certificate Database Log Location: C:\WINDOWS\system32\CertLog
Certification Authority Web Enrollment
< Previous
Next >
Configure
Cancel
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Performance
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Performance
BPA results
Search
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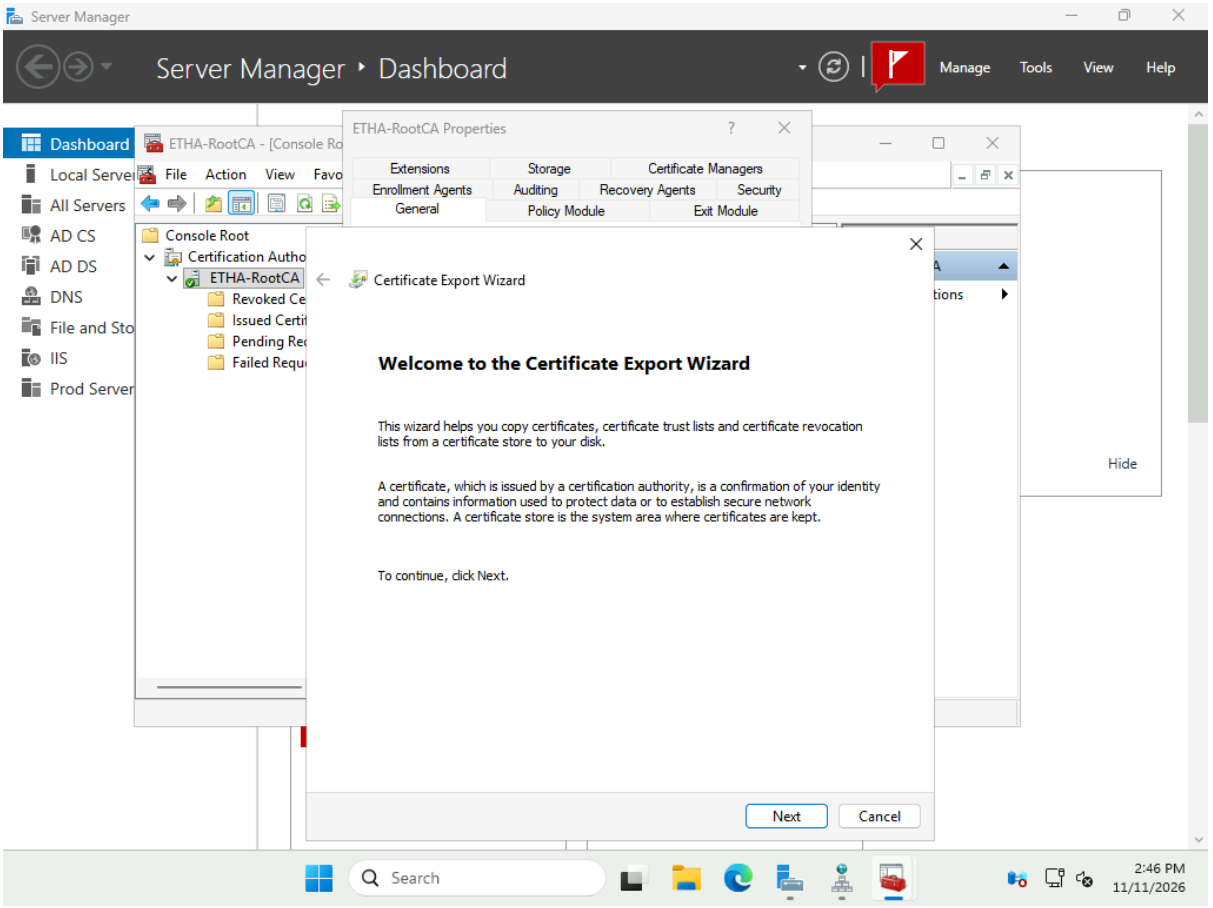

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STO-DC1.ETHA.com
Credentials
Role Services
Setup Type
CA Type
Private Key
Cryptography
CA Name
Certificate Request
Certificate Database
Confirmation
Progress
Results
The following roles, role services, or features were configured:
Active Directory Certificate Services
Certification Authority
Configuration succeeded with warnings
The Active Directory Certificate Services installation is incomplete. To complete the installation, use the request file "C:\STO-DC1.ETHA.com_ETHA-STO-DC1-CA.req" to obtain a certificate from the parent CA. Then, use the Certification Authority snap-in to install the certificate. To complete this procedure, right-click the node with the name of the CA, and then click Install CA Certificate. The operation completed successfully. 0x0 (WIN32: 0)
More about CA Configuration
Certification Authority Web Enrollment
Configuration succeeded
More about Web Enrollment Configuration
< Previous
Next >
Close
Cancel
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Performance
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BPA results
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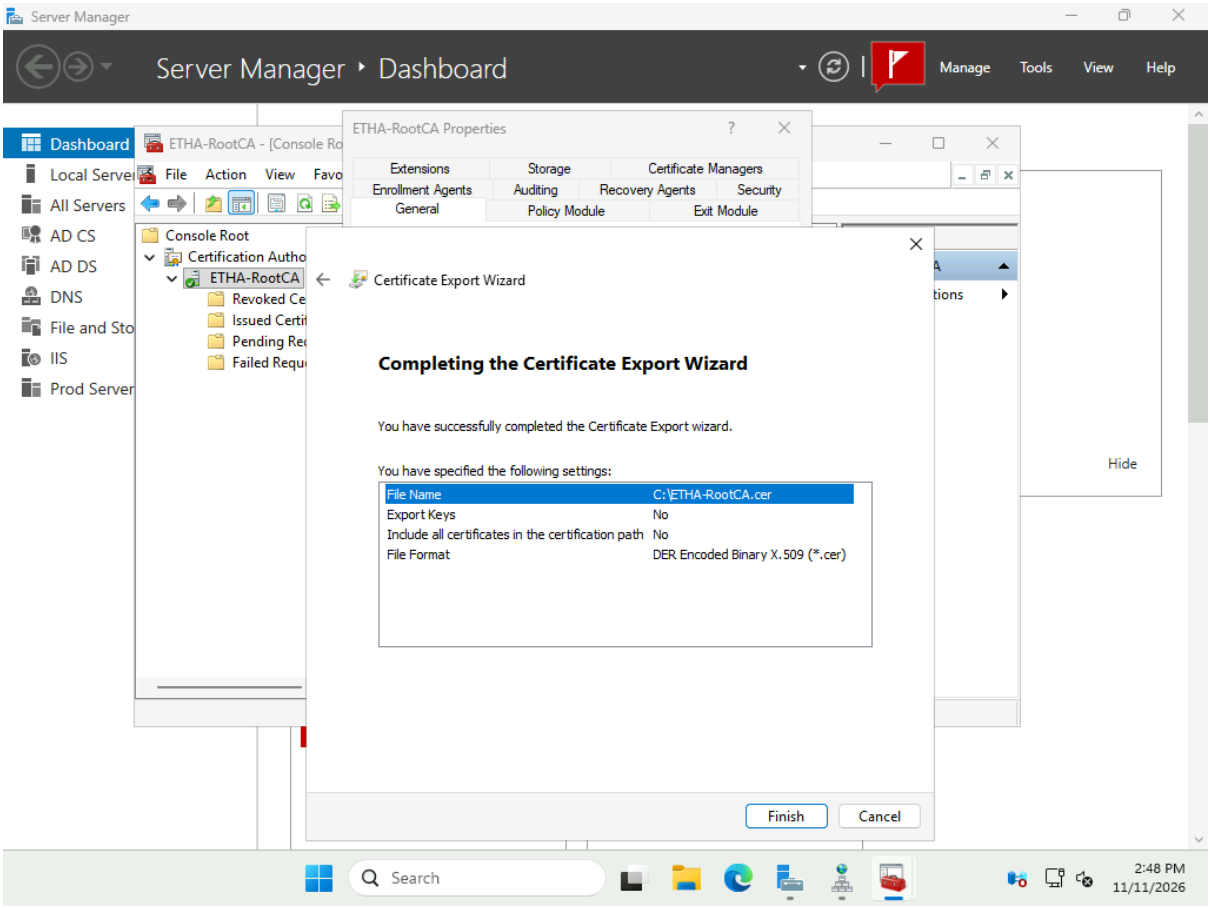

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Local Ser
All Serve
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AD DS
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IIS
Prod Ser
Certification Authority
Select the computer you want this snap-in to manage.
This snap-in will always manage:
Local computer: (the computer this console is running on)
Another computer:
STO-CORE.ETHA.com
Browse...
Allow the selected computer to be changed when launching from the command line. This only applies if you save the console.
< Back
Finish
Cancel
Help
elected set of snap-ins. For extensible
Edit Extensions...
Remove
Move Up
Move Down
Hide
Advanced...
Disk Management
Microsoft an...
Description:
Allows you to configure certification authority properties and to manage certificates issued by this CA.
OK
Cancel
Performance
BPA results
Performance
BPA results
Search
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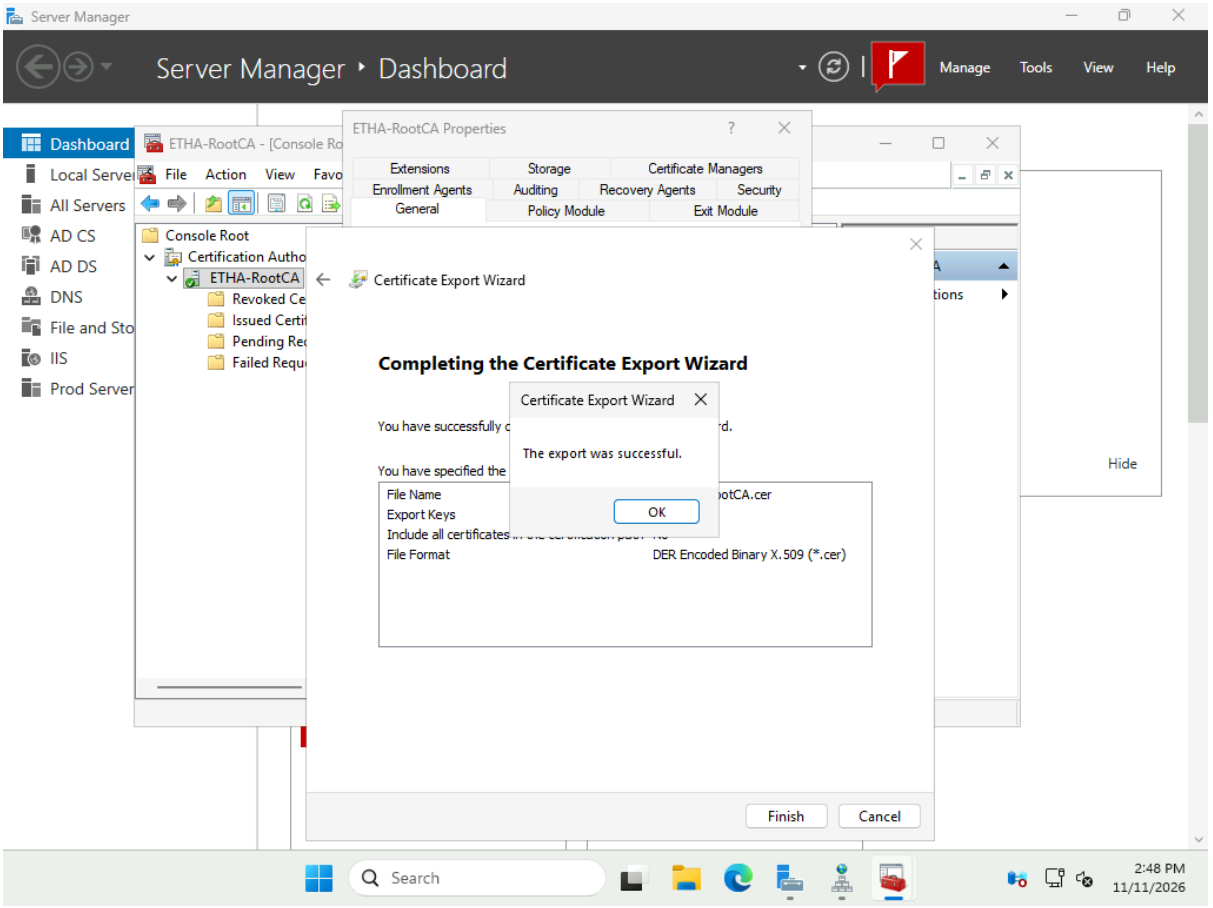

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Console1 - [Console Root]
File
Action
View
Favorites
Window
Help
Console Root
Save As
Save in:
Desktop
Home
Desktop
Libraries
This PC
Network
Gallery
Desktop
Documents
Downloads
Music
File name:
ETHA-RootCA.msc
Save as type:
Microsoft Management Console Files (*.msc)
Save
Cancel
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Console Root
Certification Author
ETHA-RootCA Properties
Enrollment Agents
Auditing
Recovery Agents
Security
General
Policy Module
Exit Module
Extensions
Storage
Certificate Managers
Select extension:
CRL Distribution Point (CDP)
Specify locations from which users can obtain a certificate revocation list (CRL).
C:\WINDOWS\system32\CertSrv\CertEnroll\<CaName><CRLNameSuffix>
ldap:///CN=<CATruncatedName><CRLNameSuffix>,CN=<ServerShortNar
http://<ServerDNSName>/CertEnroll/<CaName><CRLNameSuffix><Delta
file://<ServerDNSName>/CertEnroll/<CaName><CRLNameSuffix><DeltaC
http://STO-dc1.ETHA.com/CertData/<CaName><CRLNameSuffix><Delta
Add...
Remove
Publish CRLs to this location
Include in all CRLs. Specifies where to publish in the Active Directory when publishing manually.
Include in CRLs. Clients use this to find Delta CRL locations.
Include in the CDP extension of issued certificates
Publish Delta CRLs to this location
Include in the IDP extension of issued CRLs
OK
Cancel
Apply
Help
Actions
Certification Authori...
More Actions
ETHA-RootCA
More Actions
Hide
Performance
BPA results
Performance
BPA results
Search
2:43 PM
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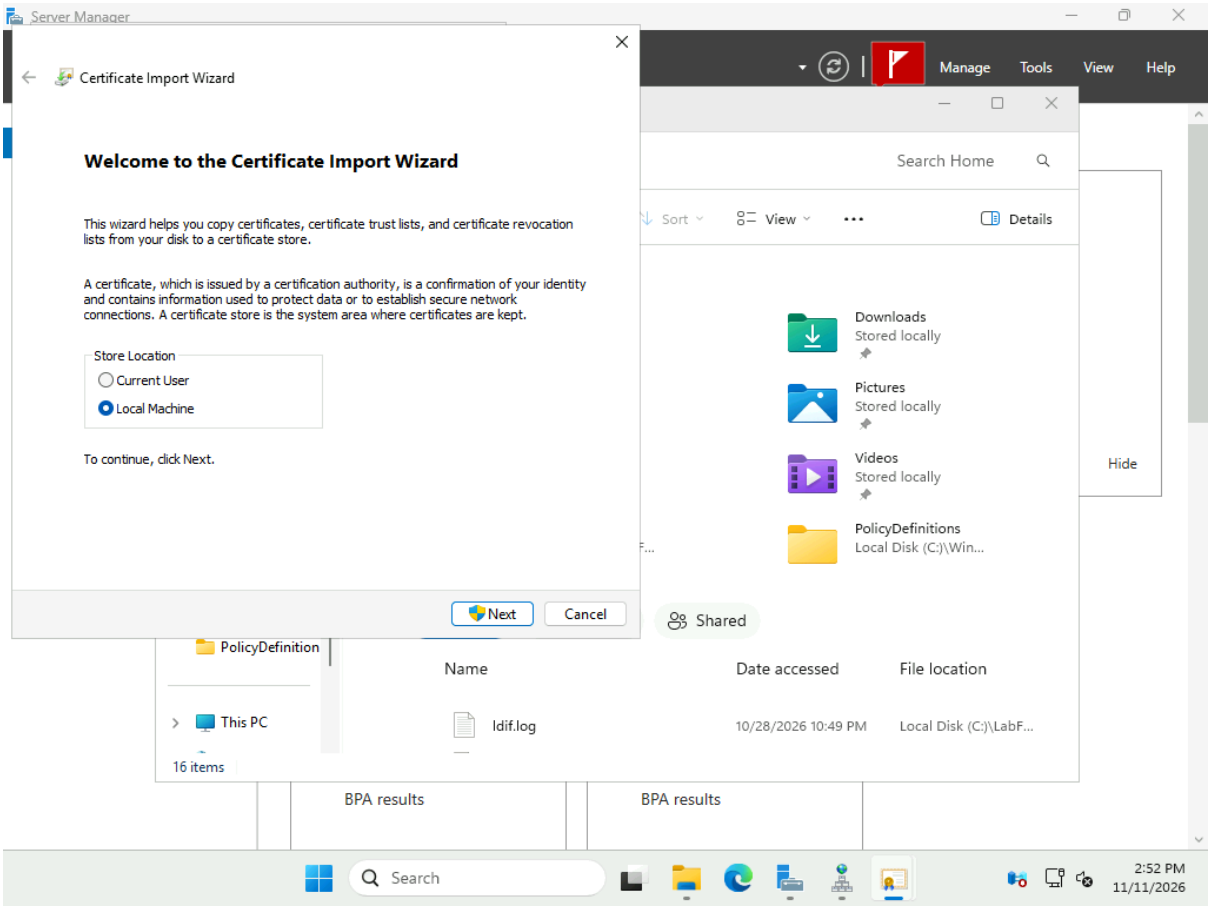

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Prod Server
ETHA-RootCA - [Console Root\Certification Authority (STO-CORE.ETHA.com)\ETHA-RootCA\Revoked Certific...
File
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Favorites
Window
Help
Console Root
Certification Authority (STO-C
ETHA-RootCA
Revoked Cert
Issued Certifi
Pending Req
Failed Reque
Request ID
Revocation Date
Effective Revocation Date
Actions
Revoked Certificates
More Actions
Publish CRL
The latest published Certificate Revocation List (CRL) is still valid. Clients may not receive a new CRL until after their current one expires.
Type of CRL to publish:
New CRL
Issues a complete CRL, which contains up-to-date revocation information for the CA.
Delta CRL only
Issues an abbreviated version of the CRL, which contains only the updates to the CRL that have been made since the last time it was published.
OK
Cancel
Hide
Services
Performance
BPA results
Services
Performance
BPA results
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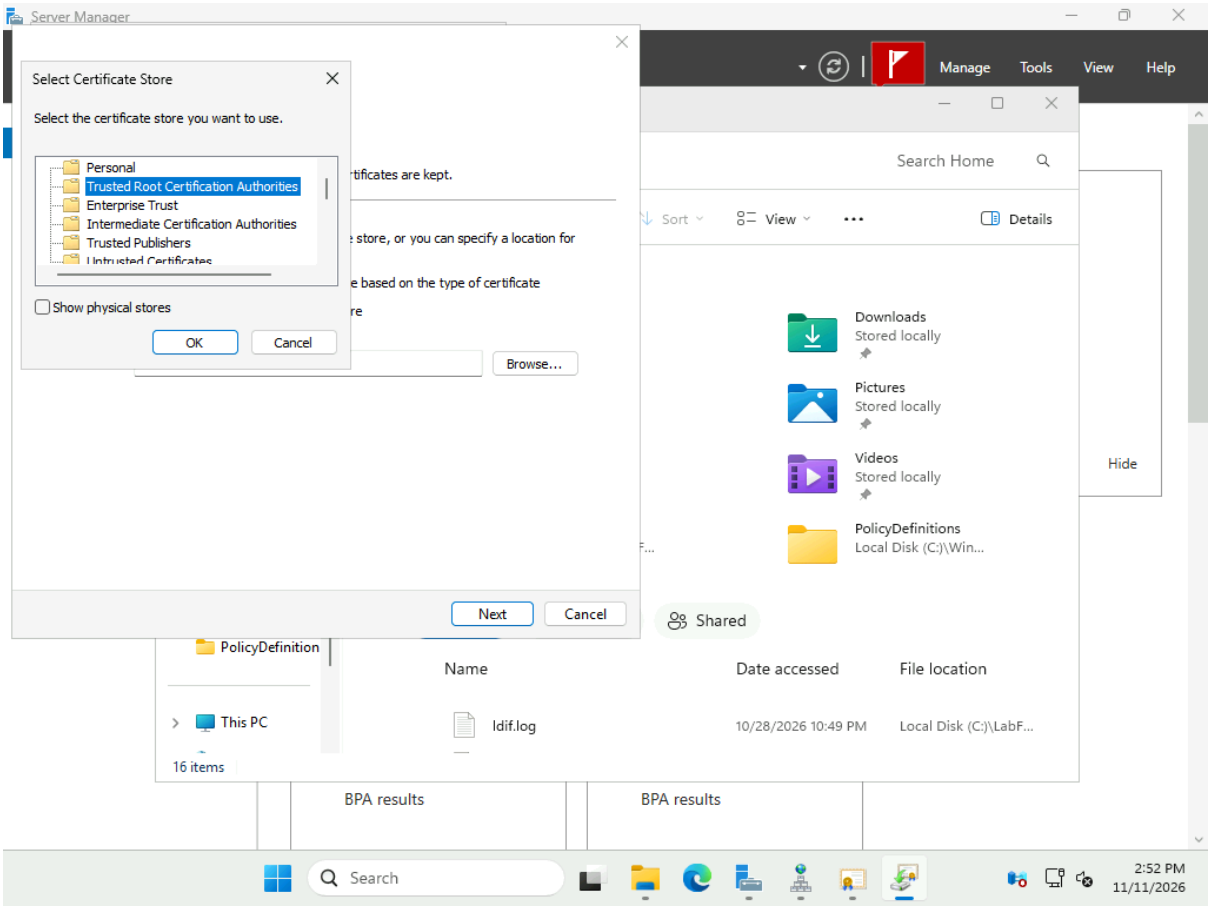

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ETHA-RootCA - [Console Ro
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Console Root
Certification Autho
ETHA-RootCA
Revoked Ce
Issued Certi
Pending Re
Failed Requ
ETHA-RootCA Properties
Extensions
Storage
Certificate Managers
Enrollment Agents
Auditing
Recovery Agents
Security
General
Policy Module
Exit Module
Certificate Export Wizard
Welcome to the Certificate Export Wizard
This wizard helps you copy certificates, certificate trust lists and certificate revocation lists from a certificate store to your disk.
A certificate, which is issued by a certification authority, is a confirmation of your identity and contains information used to protect data or to establish secure network connections. A certificate store is the system area where certificates are kept.
To continue, click Next.
Next
Cancel
Hide
Search
2:46 PM
11/11/2026

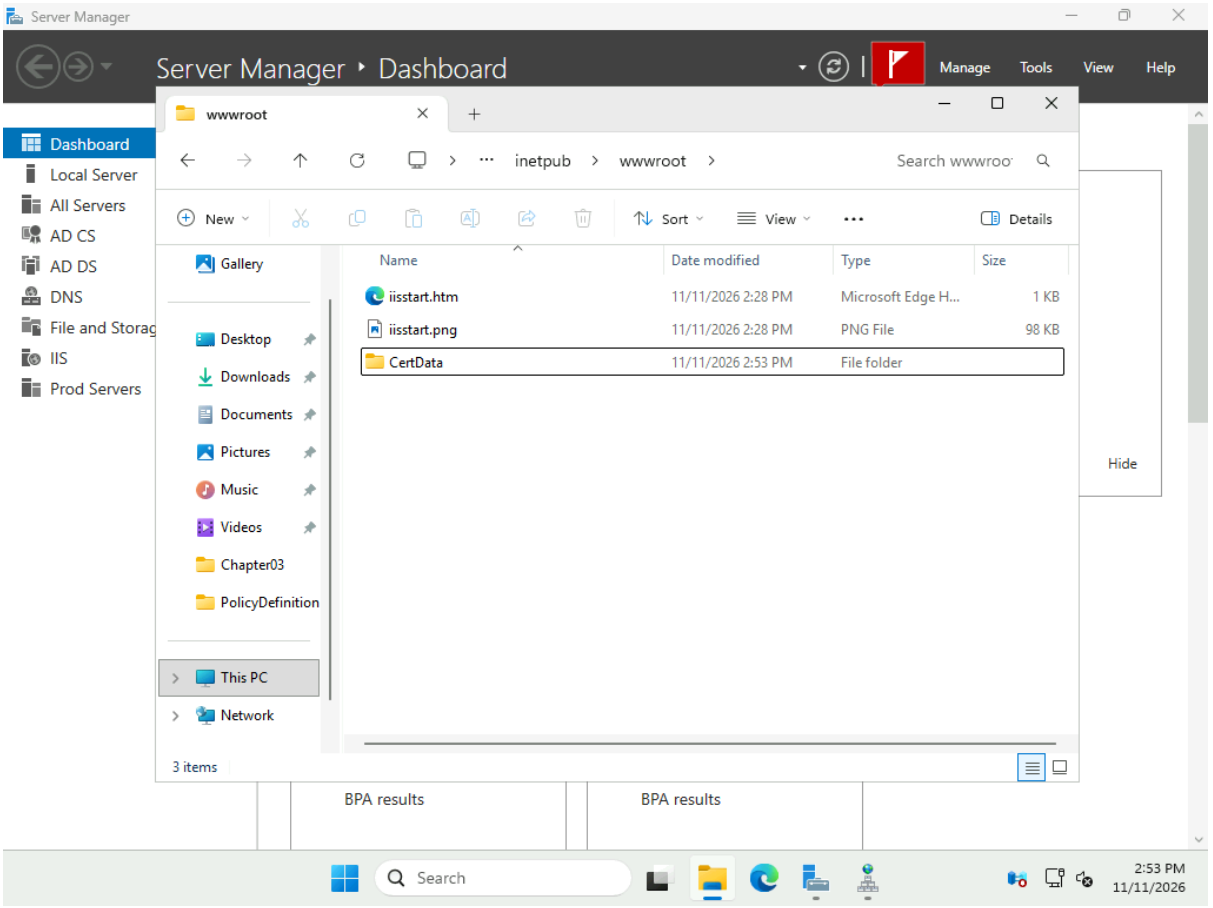

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Prod Server
ETHA-RootCA - [Console Ro
File
Action
View
Favo
Console Root
Certification Autho
ETHA-RootCA
Revoked Ce
Issued Certi
Pending Re
Failed Requ
ETHA-RootCA Properties
Extensions
Storage
Certificate Managers
Enrollment Agents
Auditing
Recovery Agents
Security
General
Policy Module
Exit Module
Certificate Export Wizard
Completing the Certificate Export Wizard
You have successfully completed the Certificate Export wizard.
You have specified the following settings:
File Name
C:\ETHA-RootCA.cer
Export Keys
No
Include all certificates in the certification path
No
File Format
DER Encoded Binary X.509 (*.cer)
Finish
Cancel
Hide
Search
2:48 PM
11/11/2026

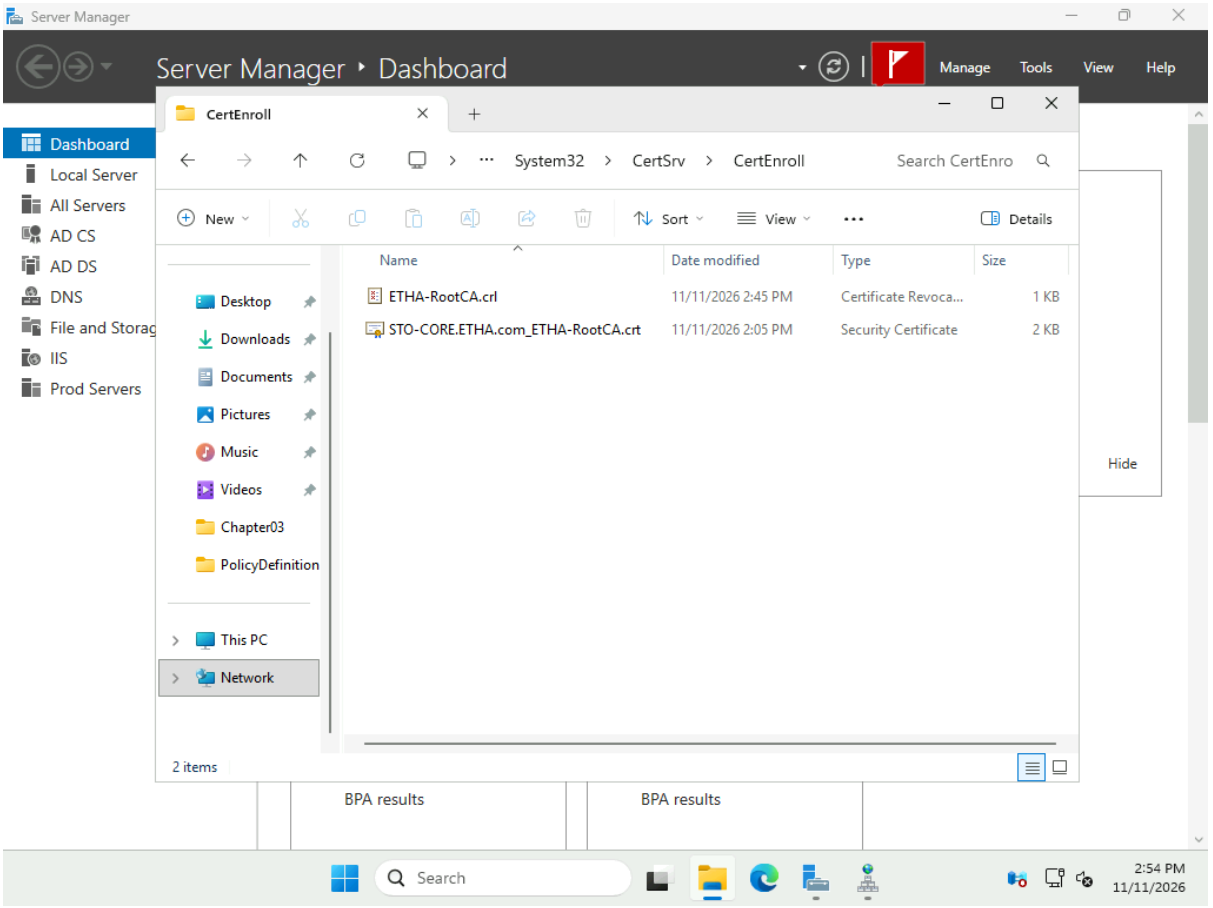


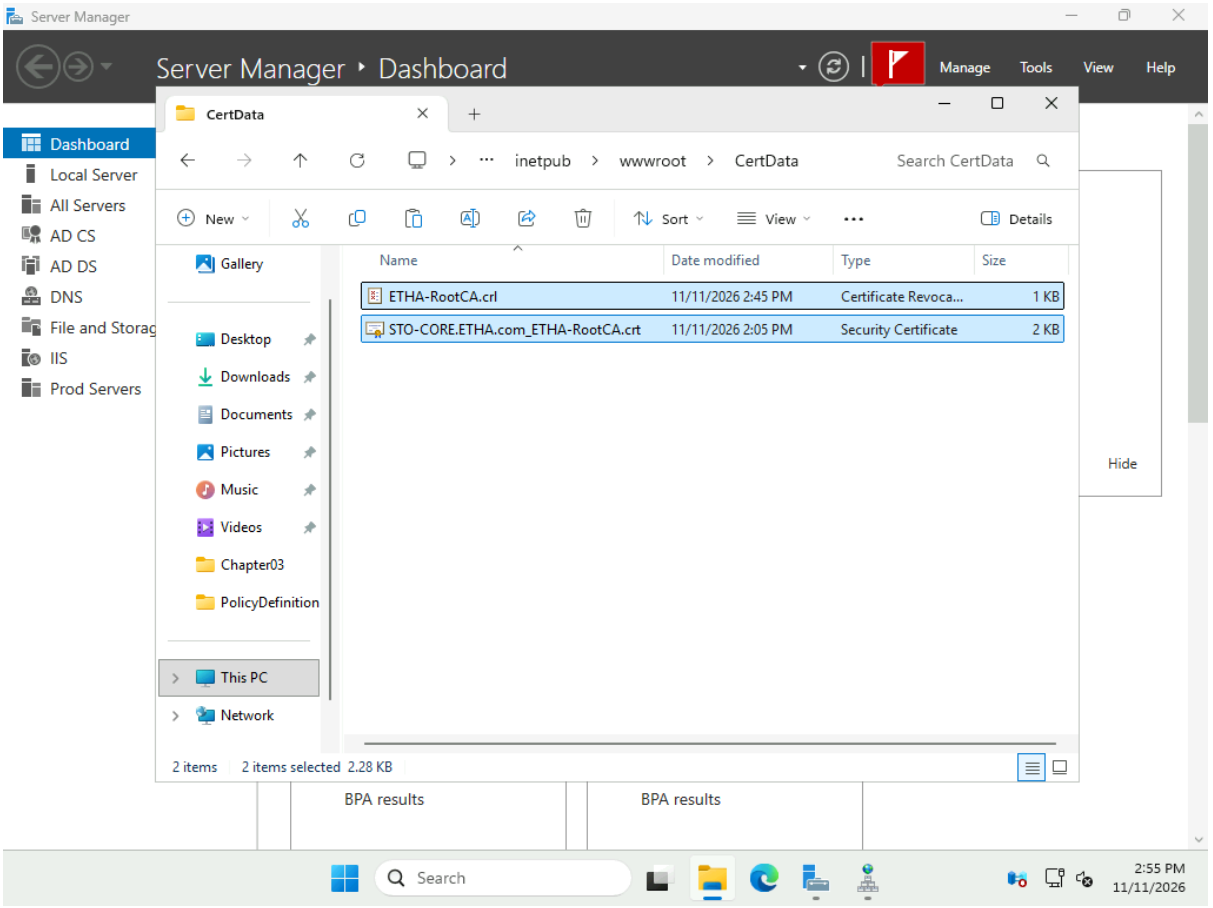
Task 4: Install a subordinate CA certificate

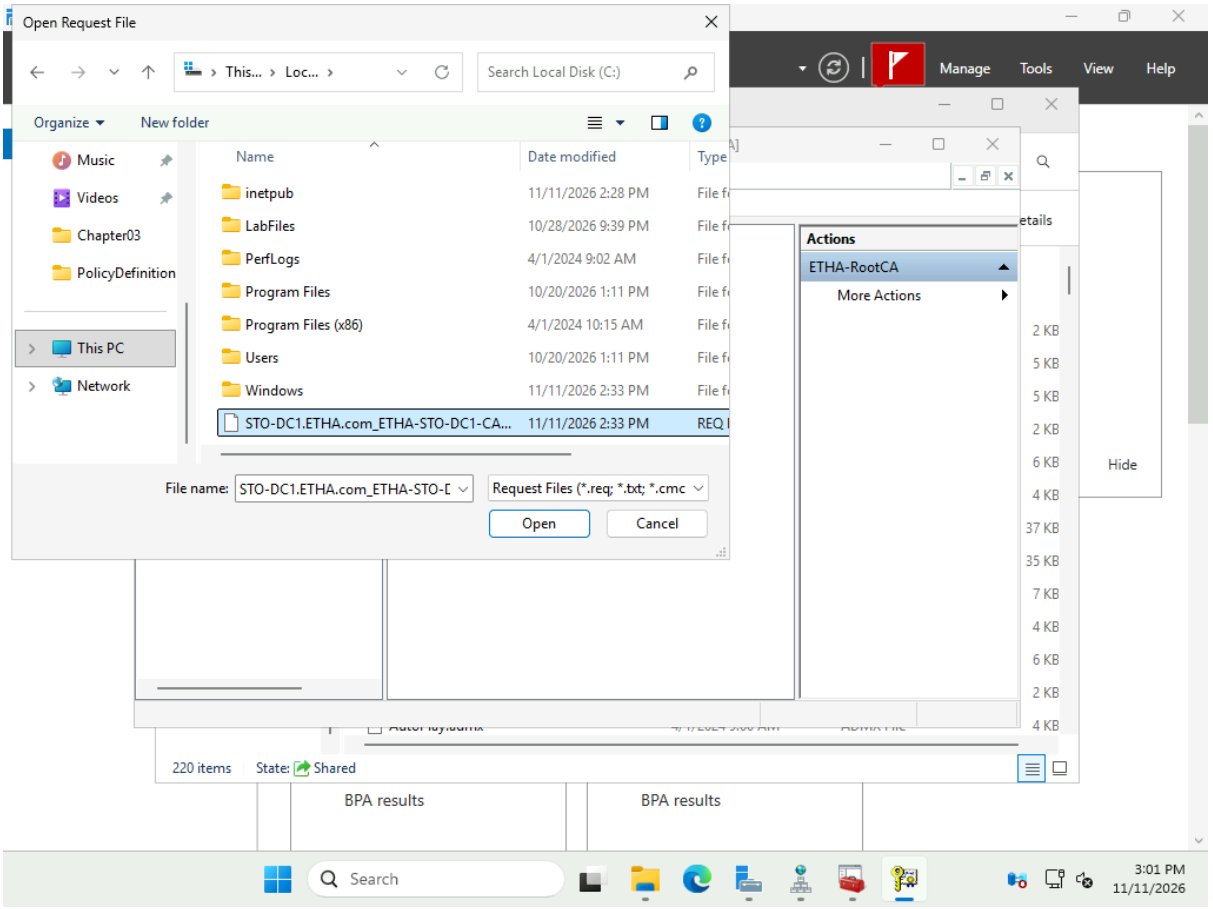

Server Manager
Manage
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Certificate Import Wizard
Welcome to the Certificate Import Wizard
This wizard helps you copy certificates, certificate trust lists, and certificate revocation lists from your disk to a certificate store.
A certificate, which is issued by a certification authority, is a confirmation of your identity and contains information used to protect data or to establish secure network connections. A certificate store is the system area where certificates are kept.
Store Location
Current User
Local Machine
To continue, click Next.
Next
Cancel
Search Home
Sort
View
Details
Downloads
Stored locally
Pictures
Stored locally
Videos
Stored locally
PolicyDefinitions
Local Disk (C:)\Win...
Hide
Shared
PolicyDefinition
This PC
16 items
Name
Date accessed
File location
ldif.log
10/28/2026 10:49 PM
Local Disk (C:)\LabF...
BPA results
BPA results
Search
2:52 PM
11/11/2026

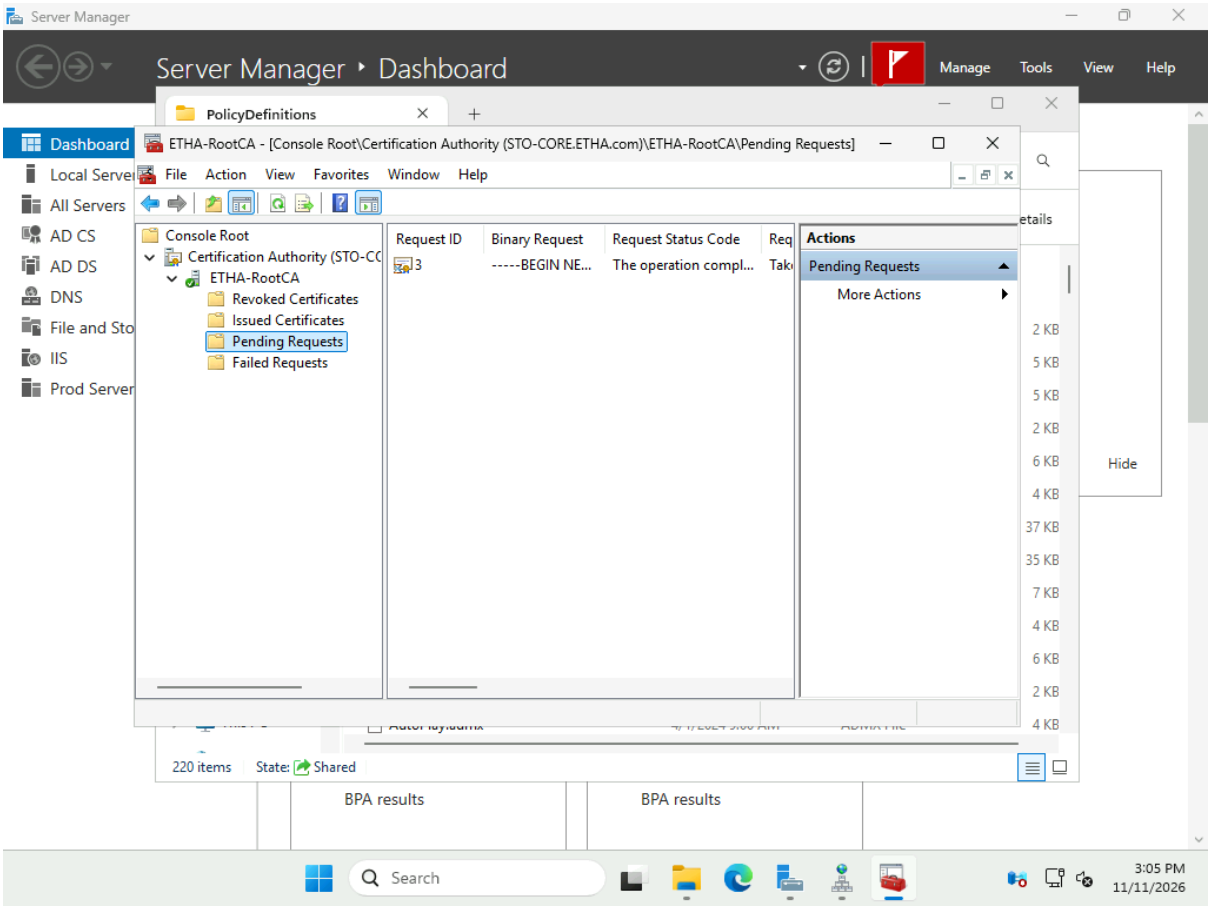

Server Manager
Manage
Tools
View
Help
Select Certificate Store
Select the certificate store you want to use.
Personal
Trusted Root Certification Authorities
Enterprise Trust
Intermediate Certification Authorities
Trusted Publishers
Show physical stores
OK
Cancel
Browse...
Next
Cancel
Search Home
Sort
View
Details
Downloads
Stored locally
Pictures
Stored locally
Videos
Stored locally
PolicyDefinitions
Local Disk (C:)\Win...
Hide
Shared
PolicyDefinition
This PC
16 items
Name
Date accessed
File location
ldif.log
10/28/2026 10:49 PM
Local Disk (C:)\LabF...
BPA results
BPA results
Search
2:52 PM
11/11/2026

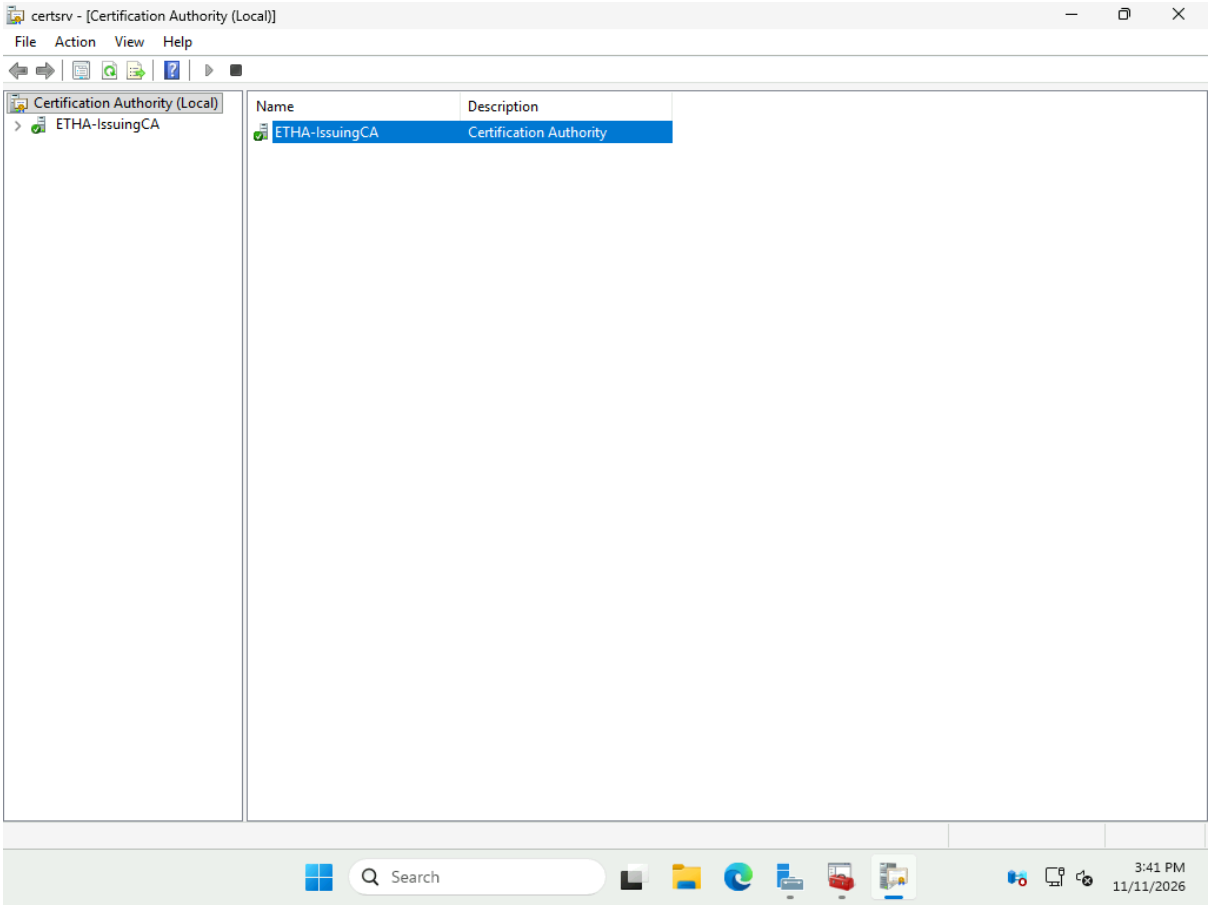

Server Manager
Server Manager • Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storag
IIS
Prod Servers
wwwroot
inetpub
wwwroot
Search wwwroo
New
Sort
View
Details
Gallery
Desktop
Downloads
Documents
Pictures
Music
Videos
Chapter03
PolicyDefinition
This PC
Network
Name
Date modified
Type
Size
iisstart.htm
11/11/2026 2:28 PM
Microsoft Edge H...
1 KB
iisstart.png
11/11/2026 2:28 PM
PNG File
98 KB
CertData
11/11/2026 2:53 PM
File folder
3 items
Hide
BPA results
BPA results
Search
2:53 PM
11/11/2026


Server Manager
Server Manager • Dashboard
Manage
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Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storag
IIS
Prod Servers
CertEnroll
System32 › CertSrv › CertEnroll
Search CertEnro
New
Sort
View
Details
Name
Date modified
Type
Size
ETHA-RootCA.crl
11/11/2026 2:45 PM
Certificate Revoca...
1 KB
STO-CORE.ETHA.com_ETHA-RootCA.crt
11/11/2026 2:05 PM
Security Certificate
2 KB
Desktop
Downloads
Documents
Pictures
Music
Videos
Chapter03
PolicyDefinition
This PC
Network
2 items
Hide
BPA results
BPA results
Search
2:54 PM
11/11/2026


Server Manager
Server Manager • Dashboard
Manage
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Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storag
IIS
Prod Servers
CertData
inetpub
wwwroot
CertData
Search CertData
New
Sort
View
Details
Gallery
Desktop
Downloads
Documents
Pictures
Music
Videos
Chapter03
PolicyDefinition
This PC
Network
Name
Date modified
Type
Size
ETHA-RootCA.crl
11/11/2026 2:45 PM
Certificate Revoca...
1 KB
STO-CORE.ETHA.com_ETHA-RootCA.crt
11/11/2026 2:05 PM
Security Certificate
2 KB
2 items
2 items selected 2.28 KB
Hide
BPA results
BPA results
Search
2:55 PM
11/11/2026

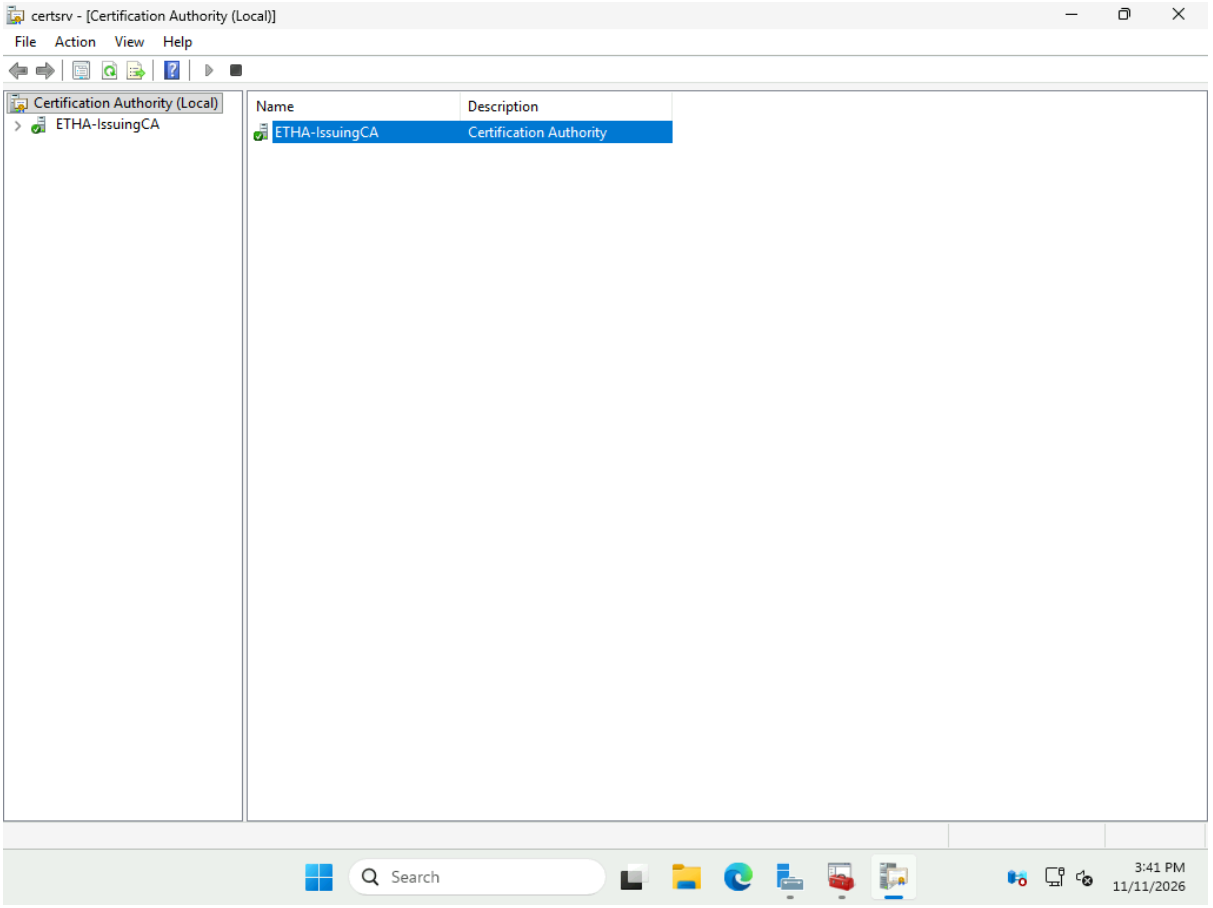

Open Request File
This... › Loc... ›
Search Local Disk (C:)
Organize
New folder
Music
Videos
Chapter03
PolicyDefinition
This PC
Network
Name
Date modified
Type
inetpub
11/11/2026 2:28 PM
LabFiles
10/28/2026 9:39 PM
PerfLogs
4/1/2024 9:02 AM
Program Files
10/20/2026 1:11 PM
Program Files (x86)
4/1/2024 10:15 AM
Users
10/20/2026 1:11 PM
Windows
11/11/2026 2:33 PM
STO-DC1.ETHA.com_ETHA-STO-DC1-CA...
11/11/2026 2:33 PM
File name:
STO-DC1.ETHA.com_ETHA-STO-D
Request Files (*.req; *.txt; *.cmc
Open
Cancel
Manage
Tools
View
Help
Actions
ETHA-RootCA
More Actions
Hide
220 items
State: Shared
BPA results
BPA results
Search
3:01 PM
11/11/2026

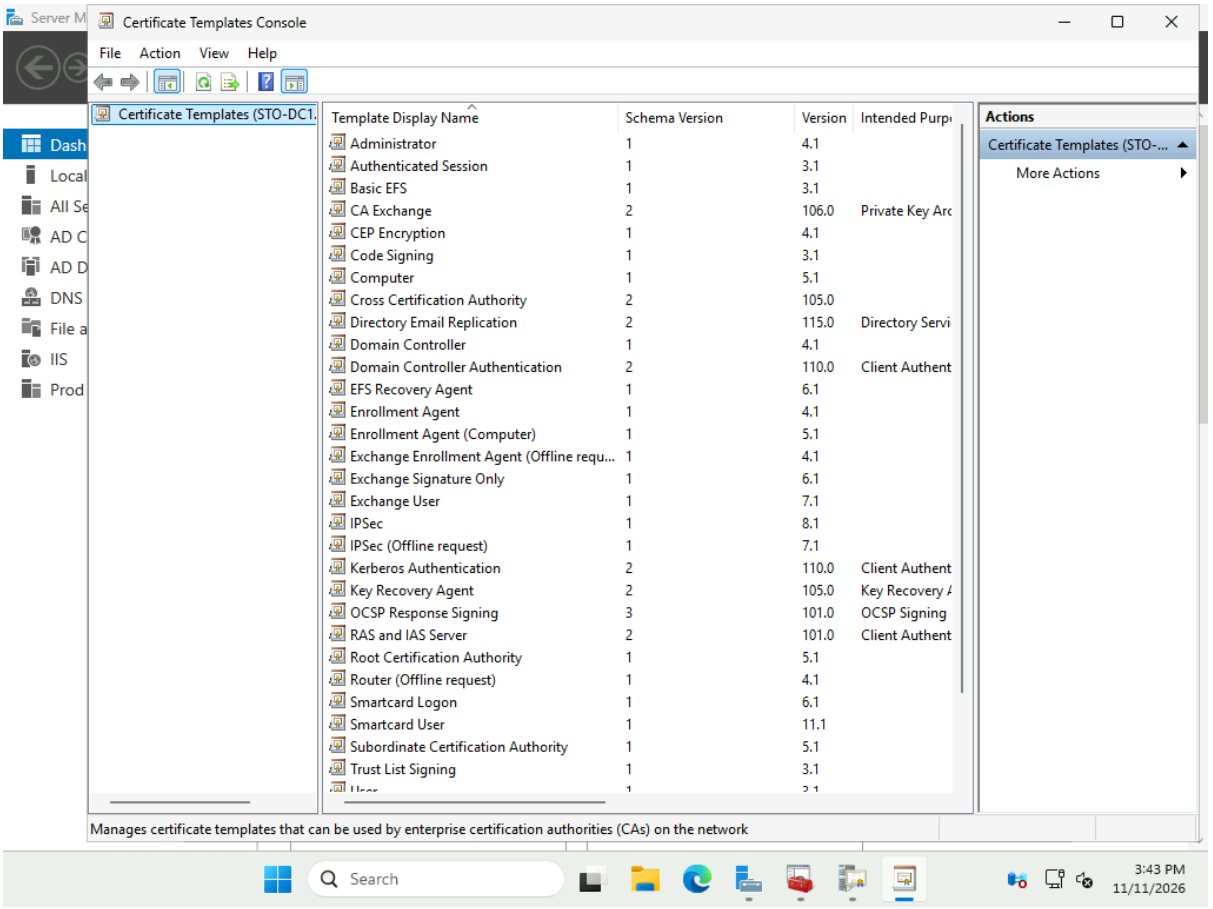

Server Manager
Server Manager • Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Sto
IIS
Prod Server
PolicyDefinitions
ETHA-RootCA - [Console Root\Certification Authority (STO-CORE.ETHA.com)\ETHA-RootCA\Pending Requests]
File
Action
View
Favorites
Window
Help
Console Root
Certification Authority (STO-CC
ETHA-RootCA
Revoked Certificates
Issued Certificates
Pending Requests
Failed Requests
Request ID
Binary Request
Request Status Code
Req
3
-----BEGIN NE...
The operation compl...
Tak
Actions
Pending Requests
More Actions
Hide
220 items
State: Shared
BPA results
BPA results
Search
3:05 PM
11/11/2026


certsrv - [Certification Authority (Local)]
File
Action
View
Help
Certification Authority (Local)
ETHA-IssuingCA
Name
Description
ETHA-IssuingCA
Certification Authority
Search
3:41 PM
11/11/2026

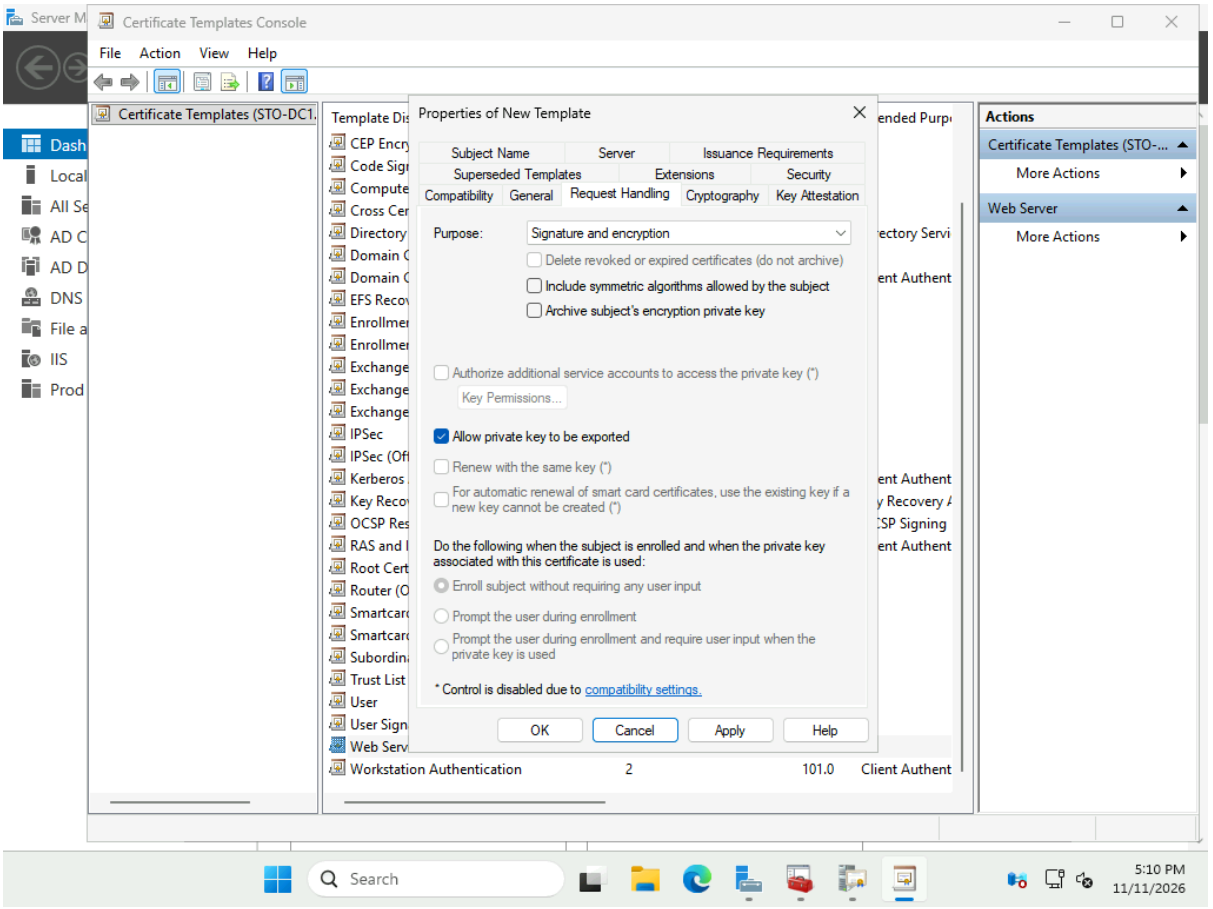
Practice 3: Configuring Certificate Templates

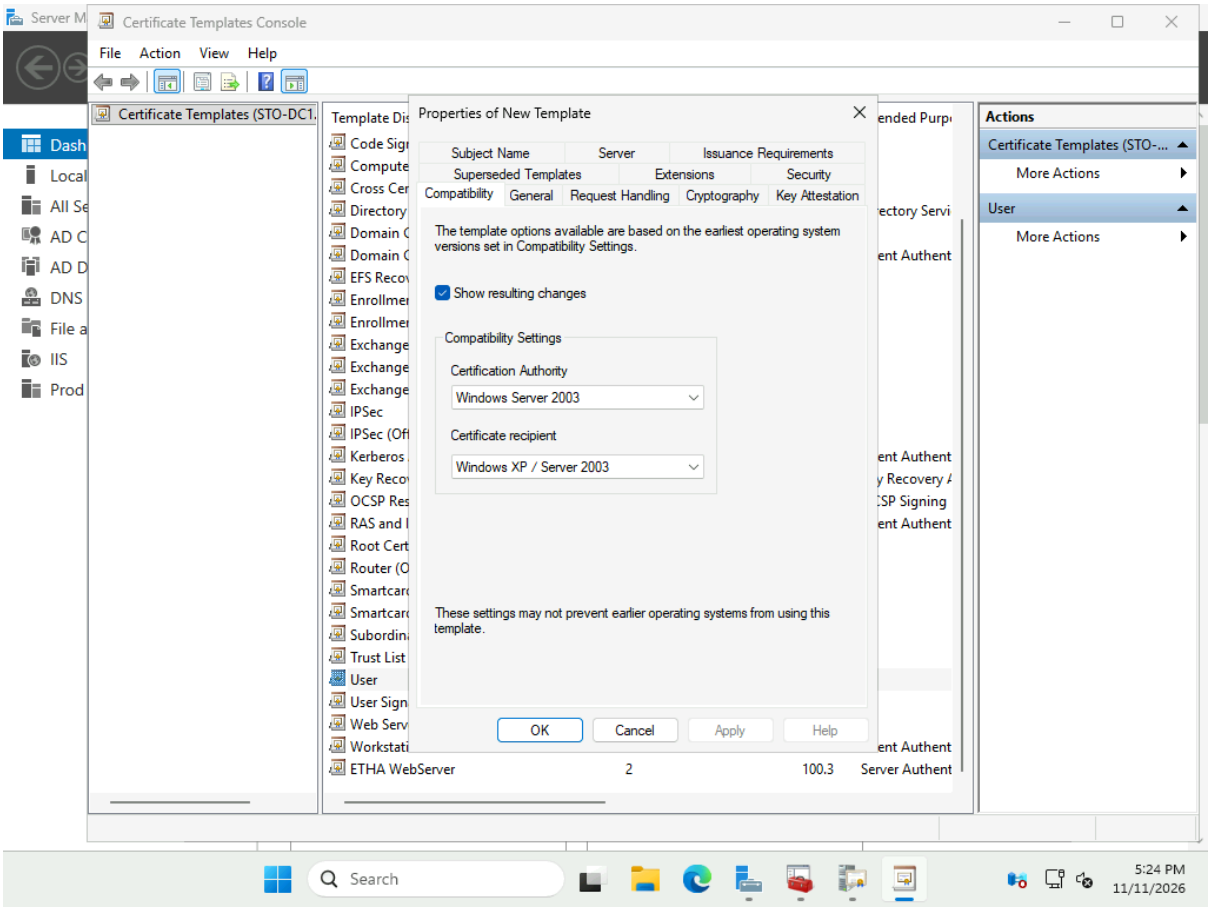
Task 5: Create a new template based on the web server template

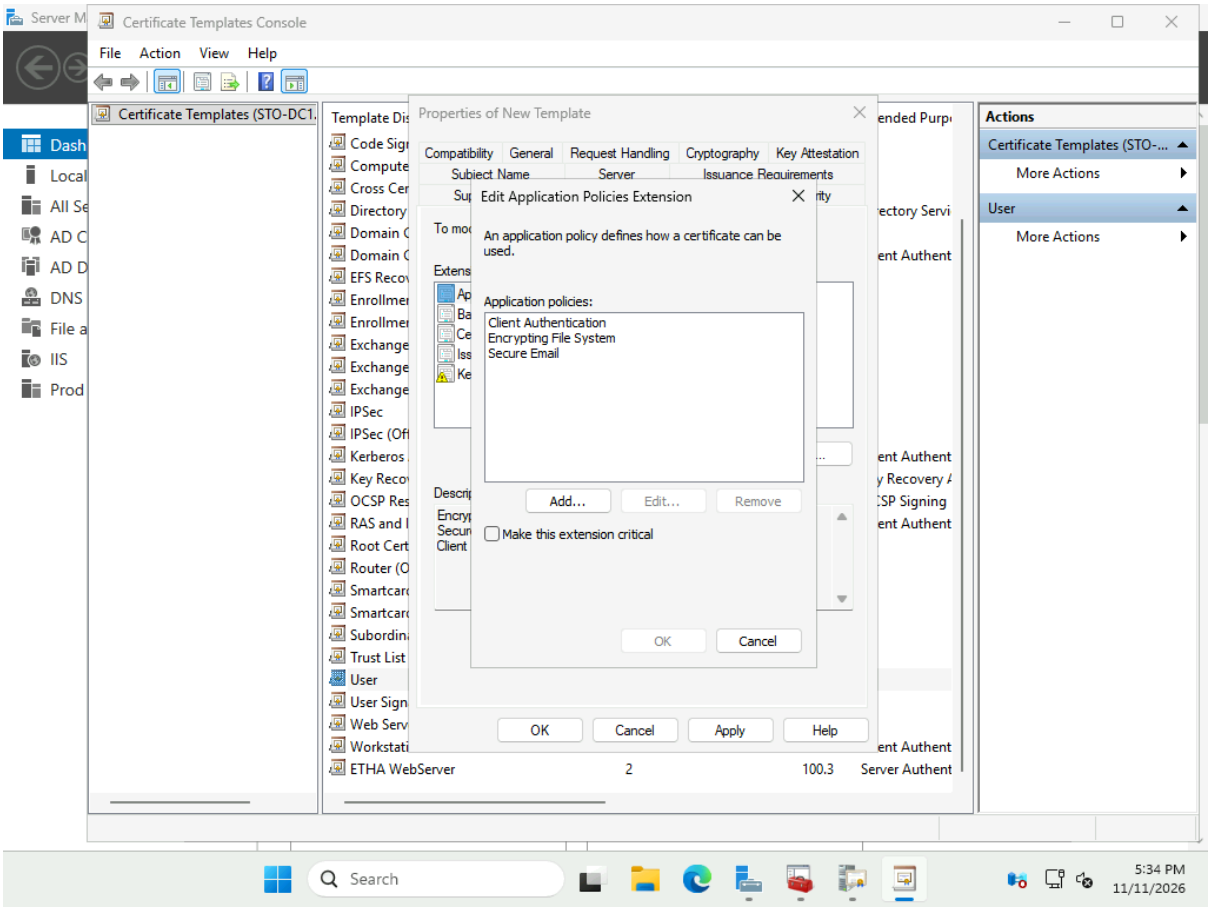


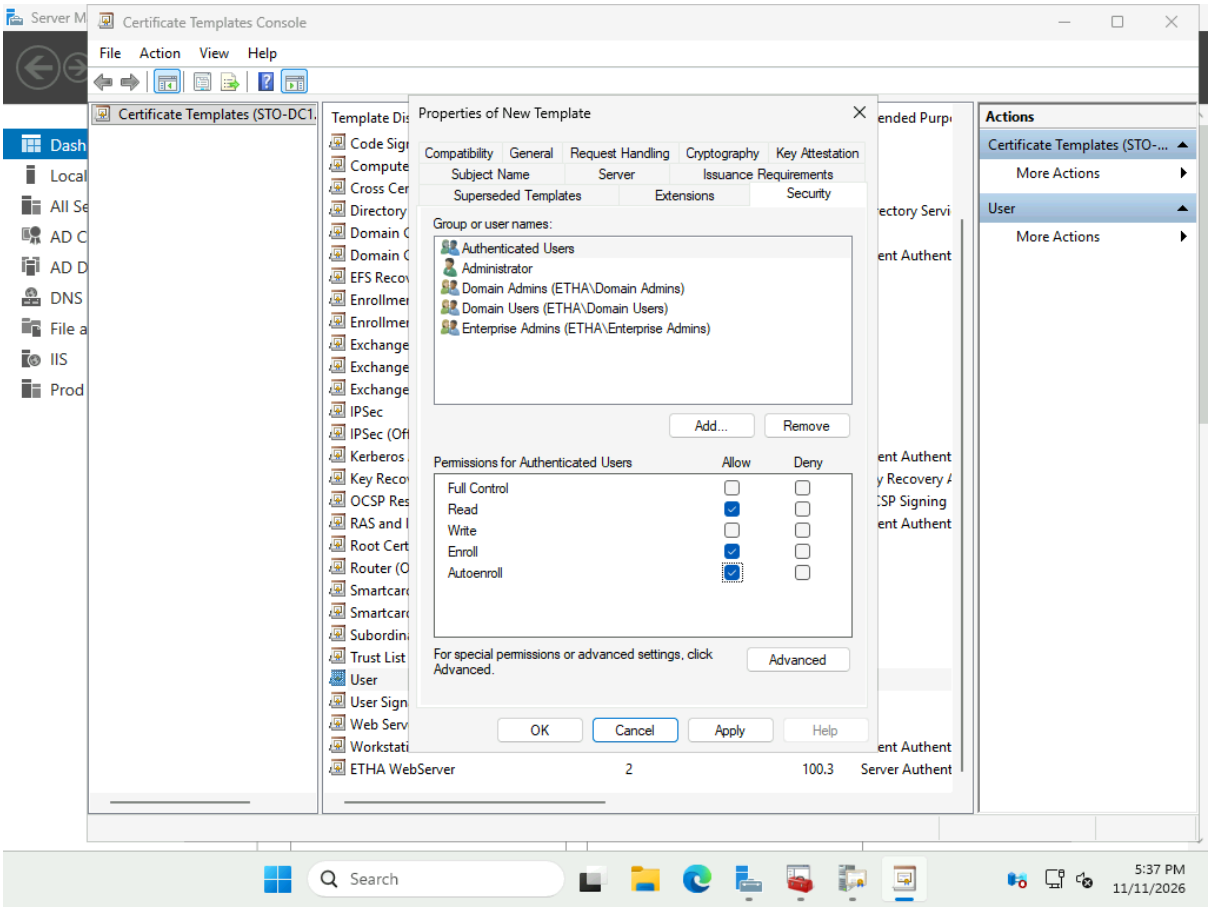


Task 6: Create a new template for users that includes smart card logon

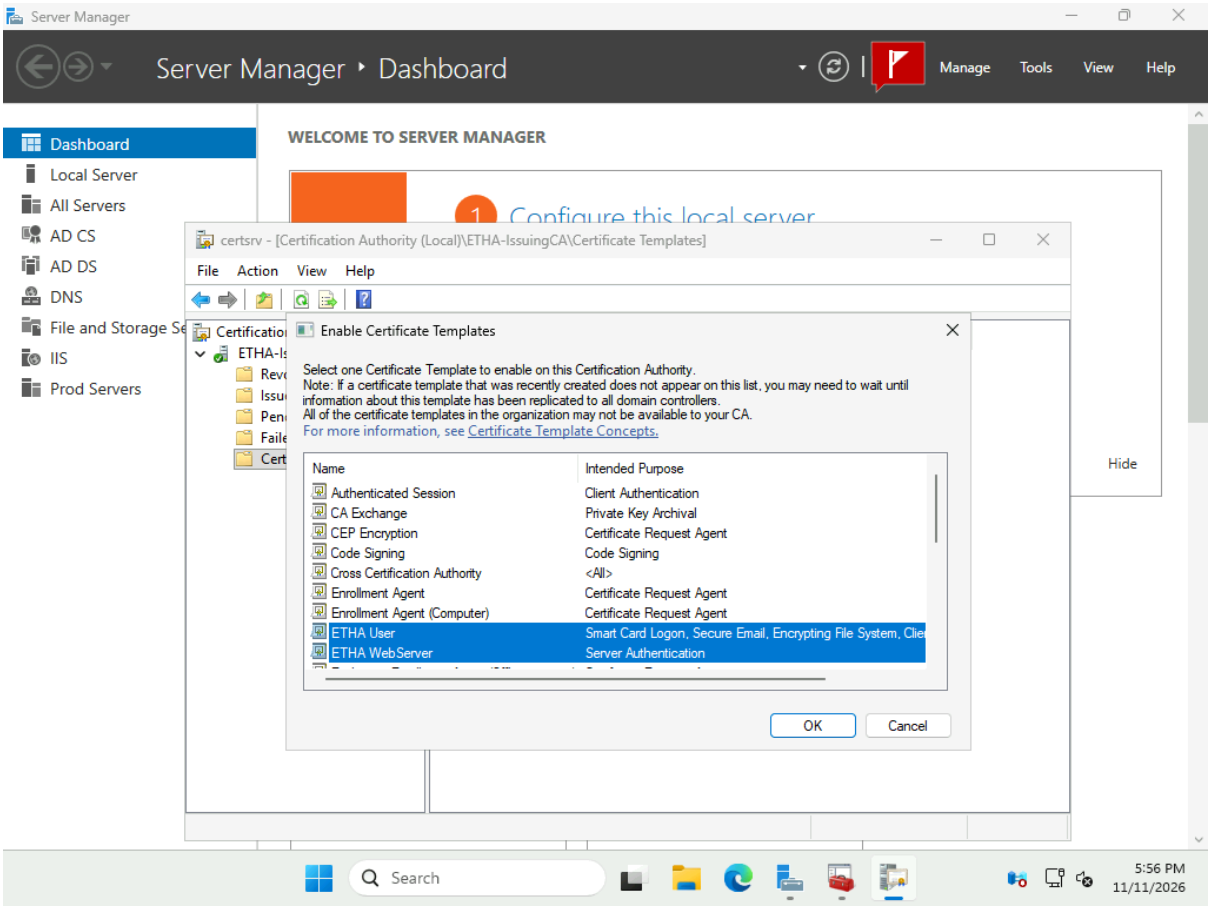

Certificate Templates Console
File Action View Help
Certificate Templates (STO-DC1...
Properties of New Template
Subject Name Server Issuance Requirements
Superseded Templates Extensions Security
Compatibility General Request Handling Cryptography Key Attestation
Purpose: Signature and encryption
Delete revoked or expired certificates (do not archive)
Include symmetric algorithms allowed by the subject
Archive subject's encryption private key
Authorize additional service accounts to access the private key (*)
Key Permissions...
Allow private key to be exported
Renew with the same key (*)
For automatic renewal of smart card certificates, use the existing key if a new key cannot be created (*)
Do the following when the subject is enrolled and when the private key associated with this certificate is used:
Enroll subject without requiring any user input
Prompt the user during enrollment
Prompt the user during enrollment and require user input when the private key is used
* Control is disabled due to compatibility settings.
OK Cancel Apply Help
Workstation Authentication 2 101.0 Client Authent
Actions
Certificate Templates (STO-...
More Actions
Web Server
More Actions
Search
5:10 PM
11/11/2026


Certificate Templates Console
File Action View Help
Certificate Templates (STO-DC1...
Properties of New Template
Subject Name Server Issuance Requirements
Superseded Templates Extensions Security
Compatibility General Request Handling Cryptography Key Attestation
The template options available are based on the earliest operating system versions set in Compatibility Settings.
Show resulting changes
Compatibility Settings
Certification Authority
Windows Server 2003
Certificate recipient
Windows XP / Server 2003
These settings may not prevent earlier operating systems from using this template.
OK Cancel Apply Help
Actions
Certificate Templates (STO-...
More Actions
User
More Actions
ETHA WebServer 2 100.3 Server Authent
Search
5:24 PM
11/11/2026

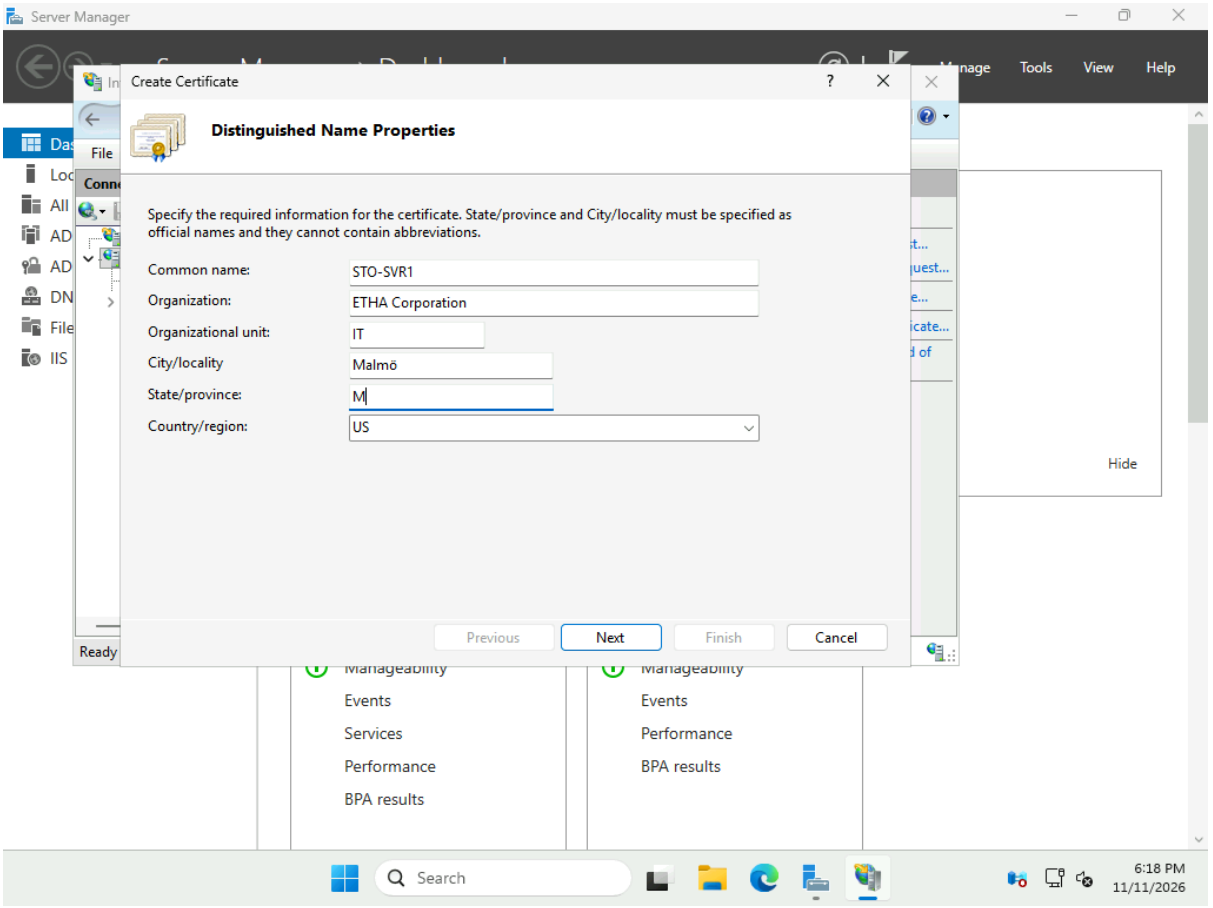

Certificate Templates Console
File Action View Help
Certificate Templates (STO-DC1...
Properties of New Template
Compatibility General Request Handling Cryptography Key Attestation
Subject Name Server Issuance Requirements
Edit Application Policies Extension
An application policy defines how a certificate can be used.
Application policies:
Client Authentication
Encrypting File System
Secure Email
Add... Edit... Remove
Make this extension critical
OK Cancel
OK Cancel Apply Help
Actions
Certificate Templates (STO-...
More Actions
User
More Actions
ETHA WebServer 2 100.3 Server Authent
Search
5:34 PM
11/11/2026

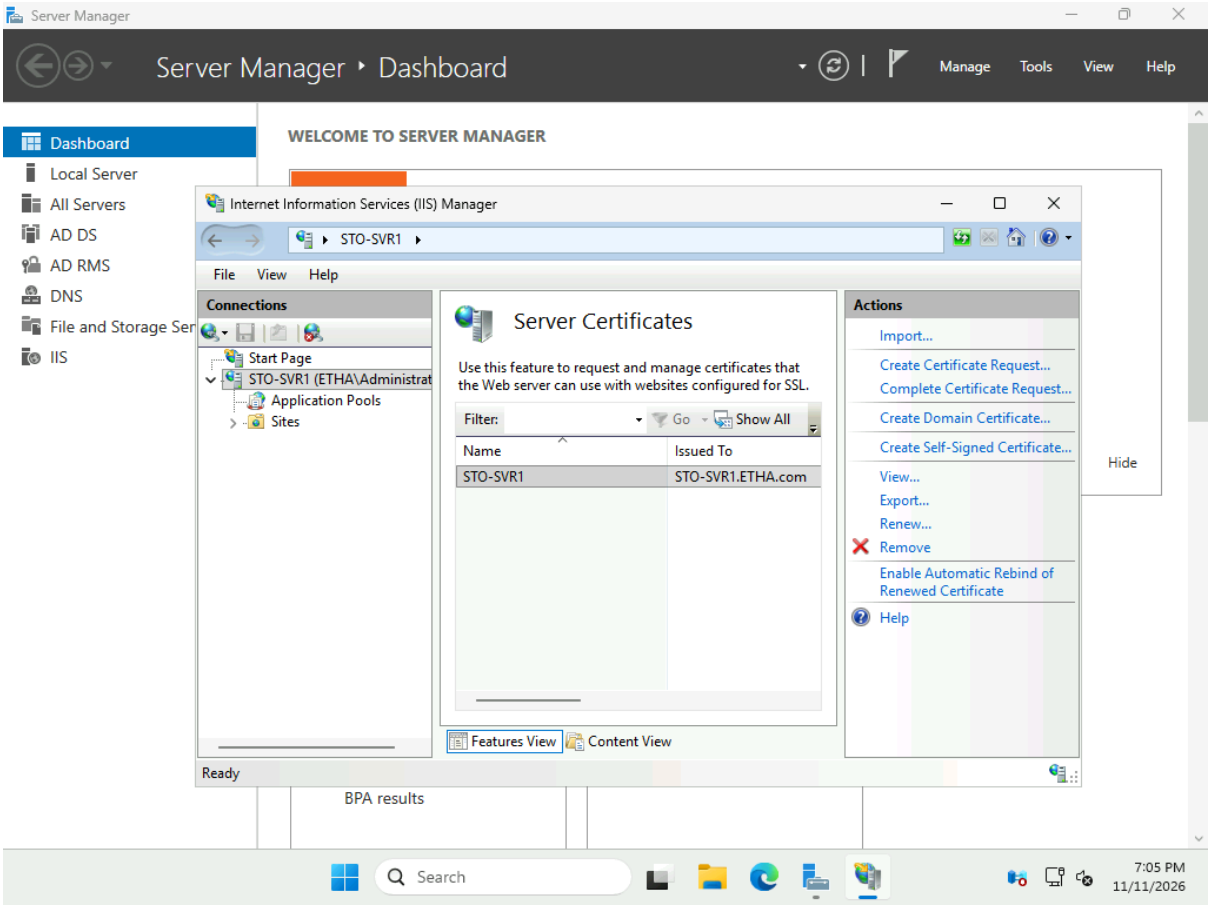


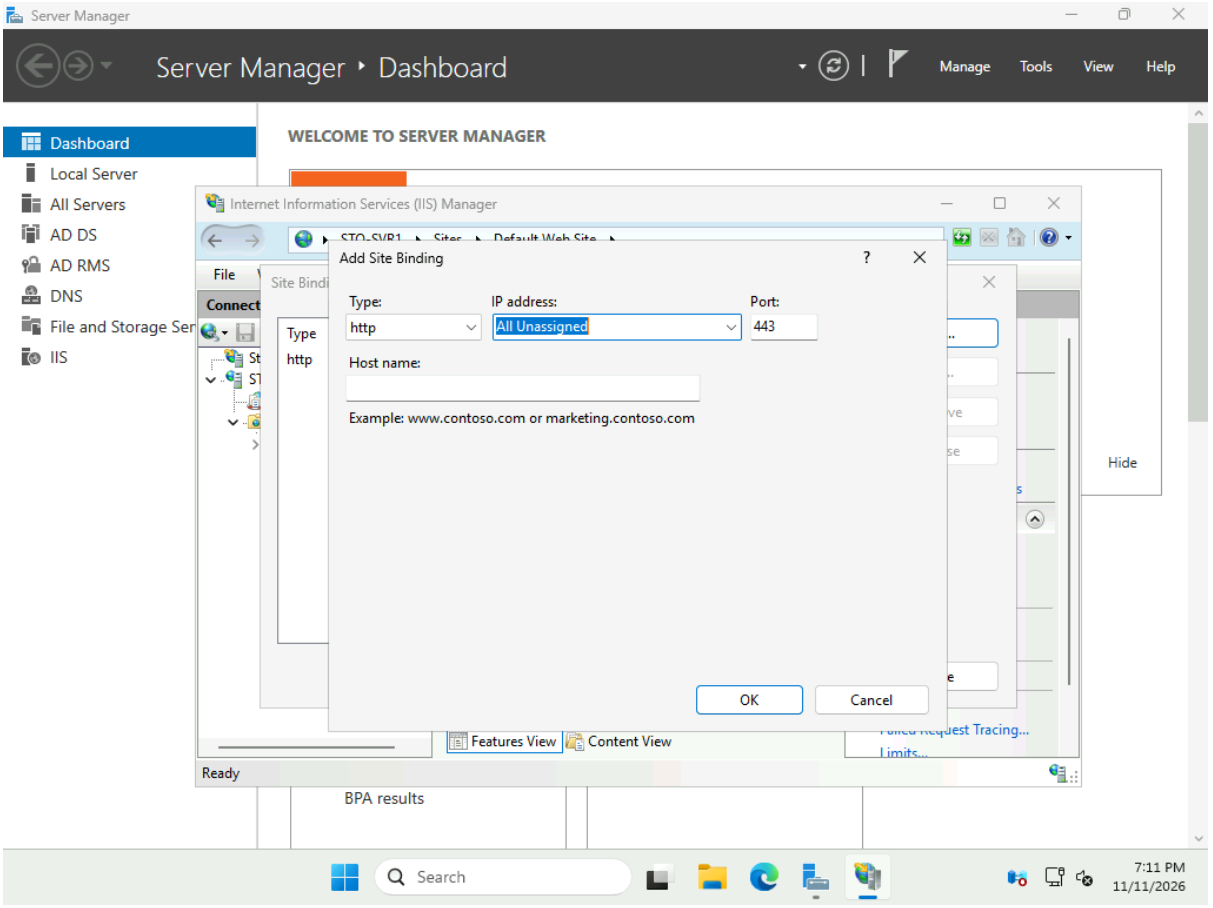
Task 7: Configure the templates so that they can be issued



Task 8: Update the web server certificate on the STO-SVR1 web server

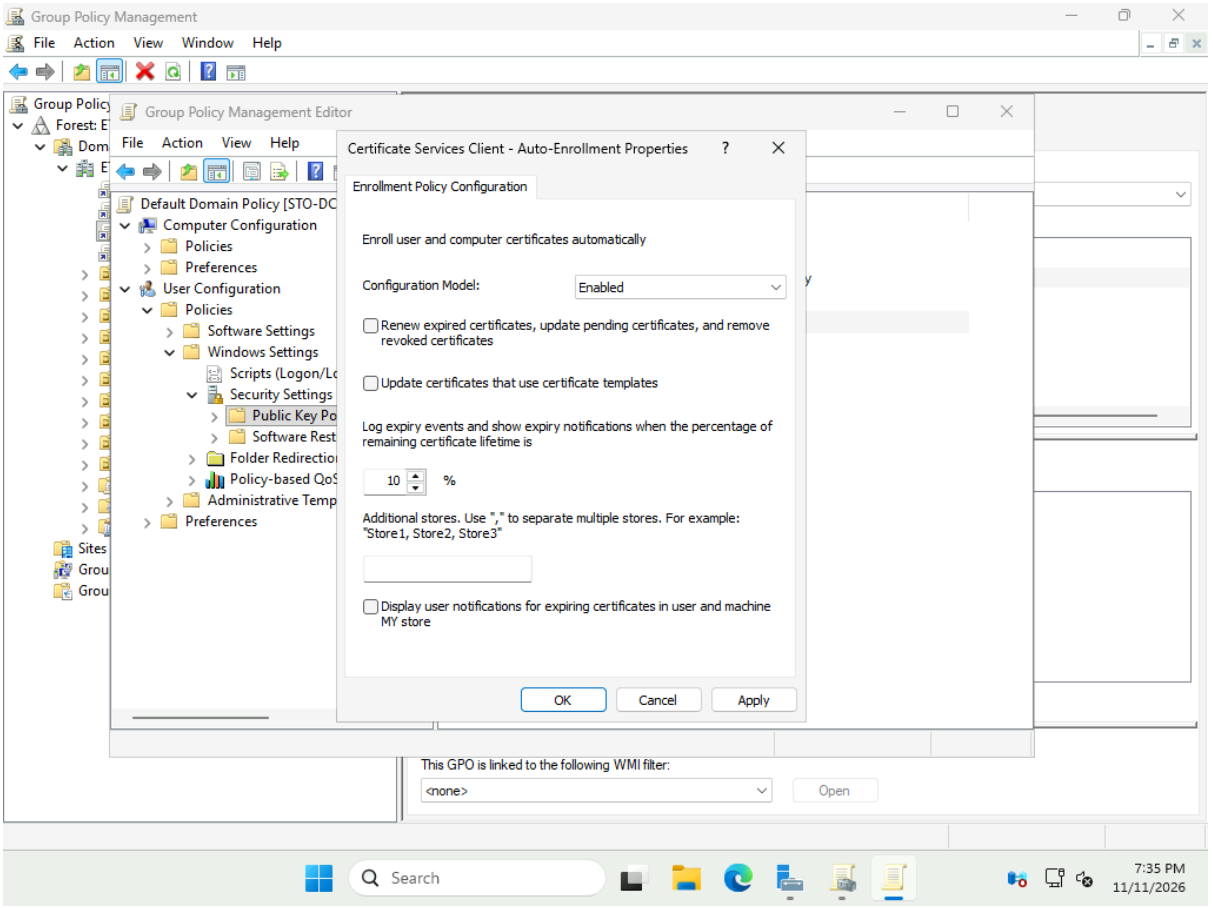

Server Manager
Create Certificate
Distinguished Name Properties
Specify the required information for the certificate. State/province and City/locality must be specified as official names and they cannot contain abbreviations.
Common name: STO-SVR1
Organization: ETHA Corporation
Organizational unit: IT
City/locality: Malmö
State/province: M
Country/region: US
Previous
Next
Finish
Cancel
Ready
Hide
Events
Services
Performance
BPA results
Search
6:18 PM
11/11/2026

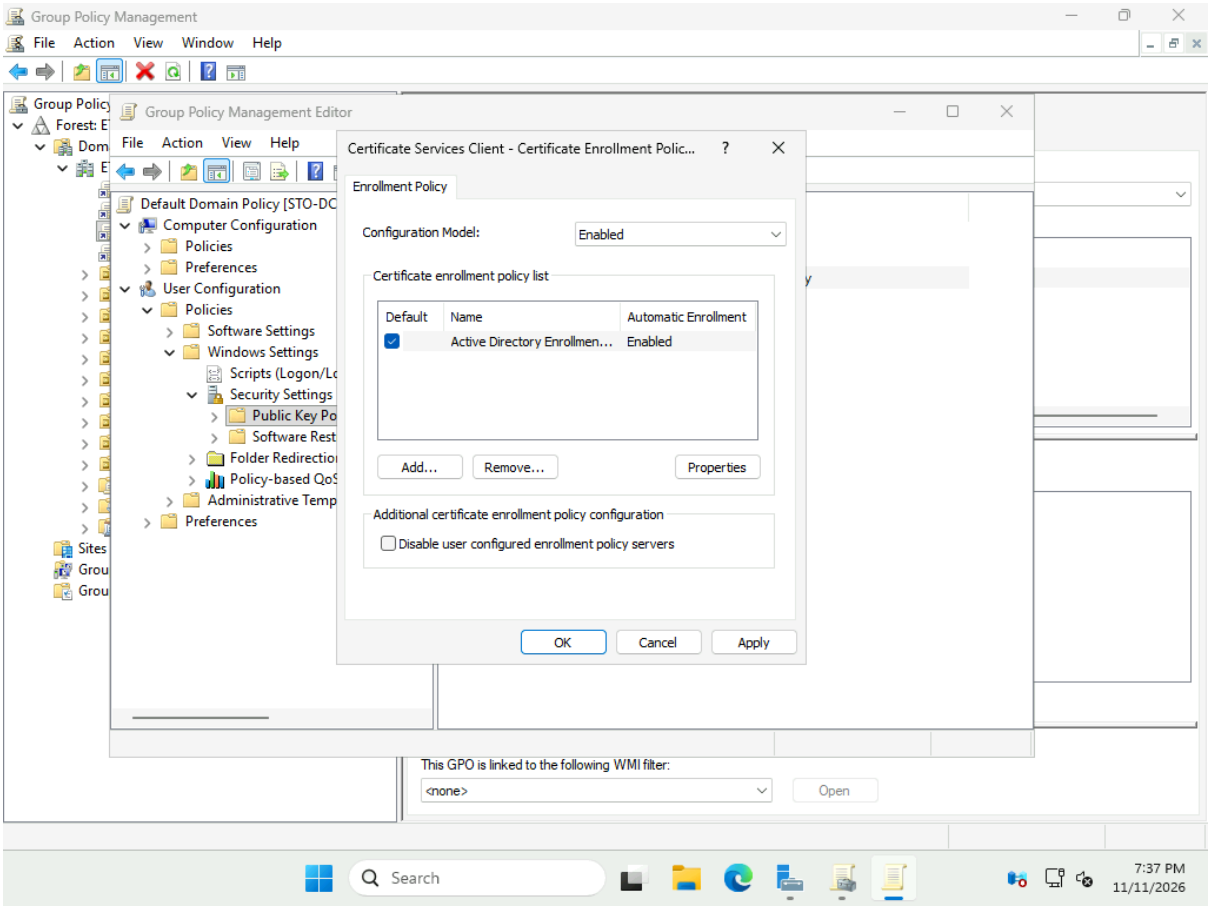

Server Manager
Server Manager • Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD DS
AD RMS
DNS
File and Storage Ser
IIS
WELCOME TO SERVER MANAGER
Internet Information Services (IIS) Manager
STO-SVR1
File
View
Help
Connections
Start Page
STO-SVR1 (ETHA\Administrat
Application Pools
Sites
Server Certificates
Use this feature to request and manage certificates that the Web server can use with websites configured for SSL.
Filter:
Go
Show All
Name
Issued To
STO-SVR1
STO-SVR1.ETHA.com
Actions
Import...
Create Certificate Request...
Complete Certificate Request...
Create Domain Certificate...
Create Self-Signed Certificate...
View...
Export...
Renew...
Remove
Enable Automatic Rebind of Renewed Certificate
Help
Features View
Content View
Ready
Hide
BPA results
Search
7:05 PM
11/11/2026



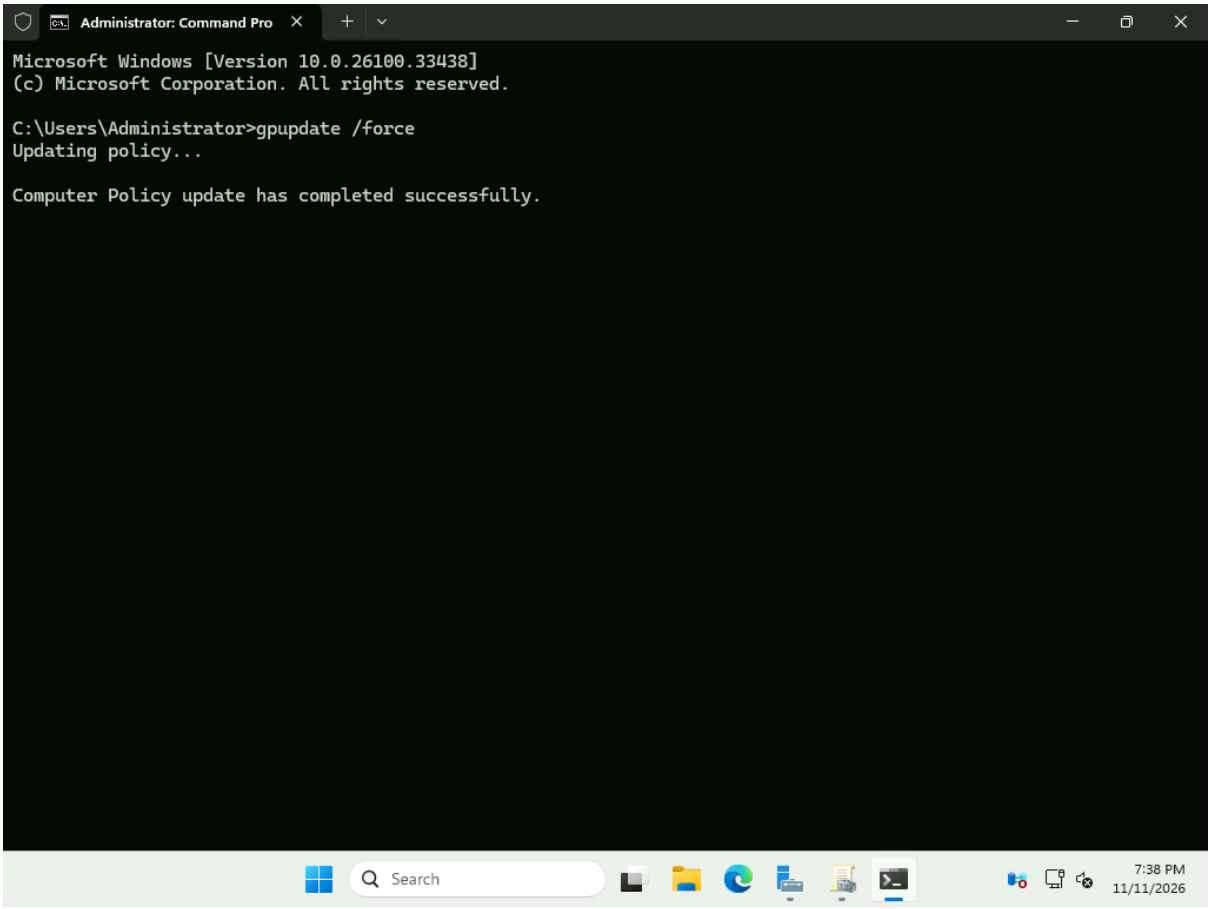
Practice 4: Configuring Certificate Enrollment

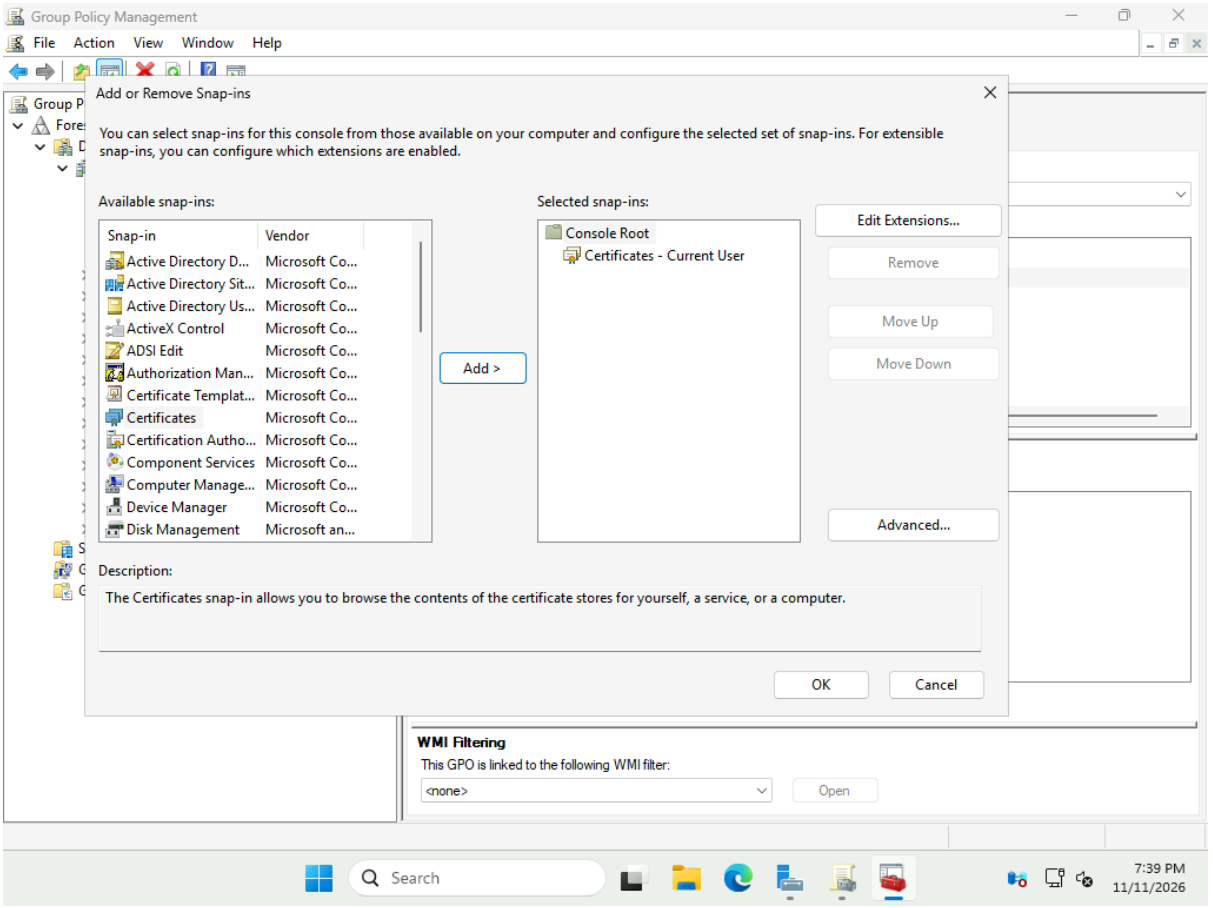
Task 9: Configure autoenrollment for users

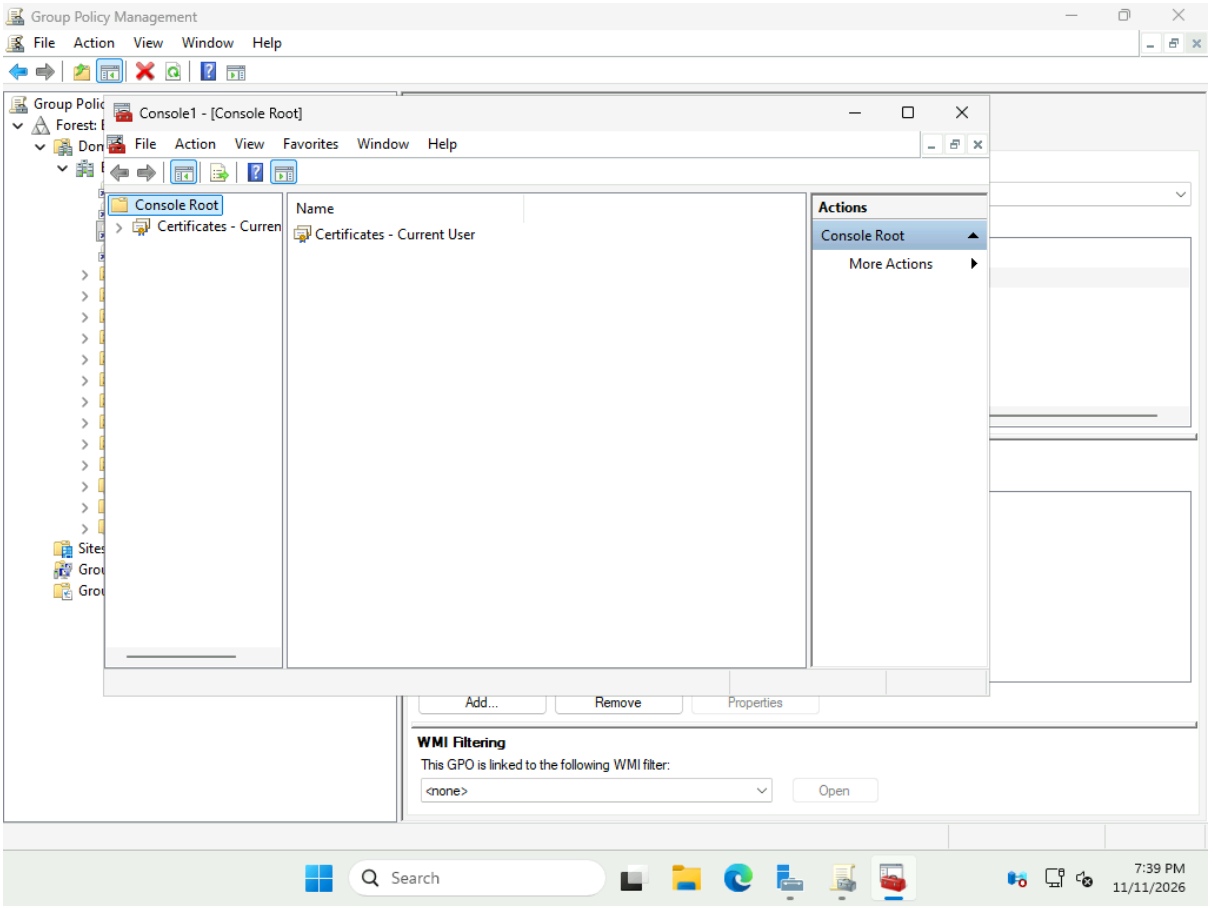

Group Policy Management
File Action View Window Help
Group Policy Management Editor
File Action View Help
Certificate Services Client - Auto-Enrollment Properties
Enrollment Policy Configuration
Enroll user and computer certificates automatically
Configuration Model:
Enabled
Renew expired certificates, update pending certificates, and remove revoked certificates
Update certificates that use certificate templates
Log expiry events and show expiry notifications when the percentage of remaining certificate lifetime is
10
%
Additional stores. Use "," to separate multiple stores. For example: "Store1, Store2, Store3"
Display user notifications for expiring certificates in user and machine MY store
OK
Cancel
Apply
Default Domain Policy [STO-DC
Computer Configuration
Policies
Preferences
User Configuration
Policies
Software Settings
Windows Settings
Scripts (Logon/Lo
Security Settings
Public Key Po
Software Rest
Folder Redirectio
Policy-based QoS
Administrative Temp
Preferences
Sites
Grou
Grou
This GPO is linked to the following WMI filter:
<none>
Open
Search
7:35 PM
11/11/2026

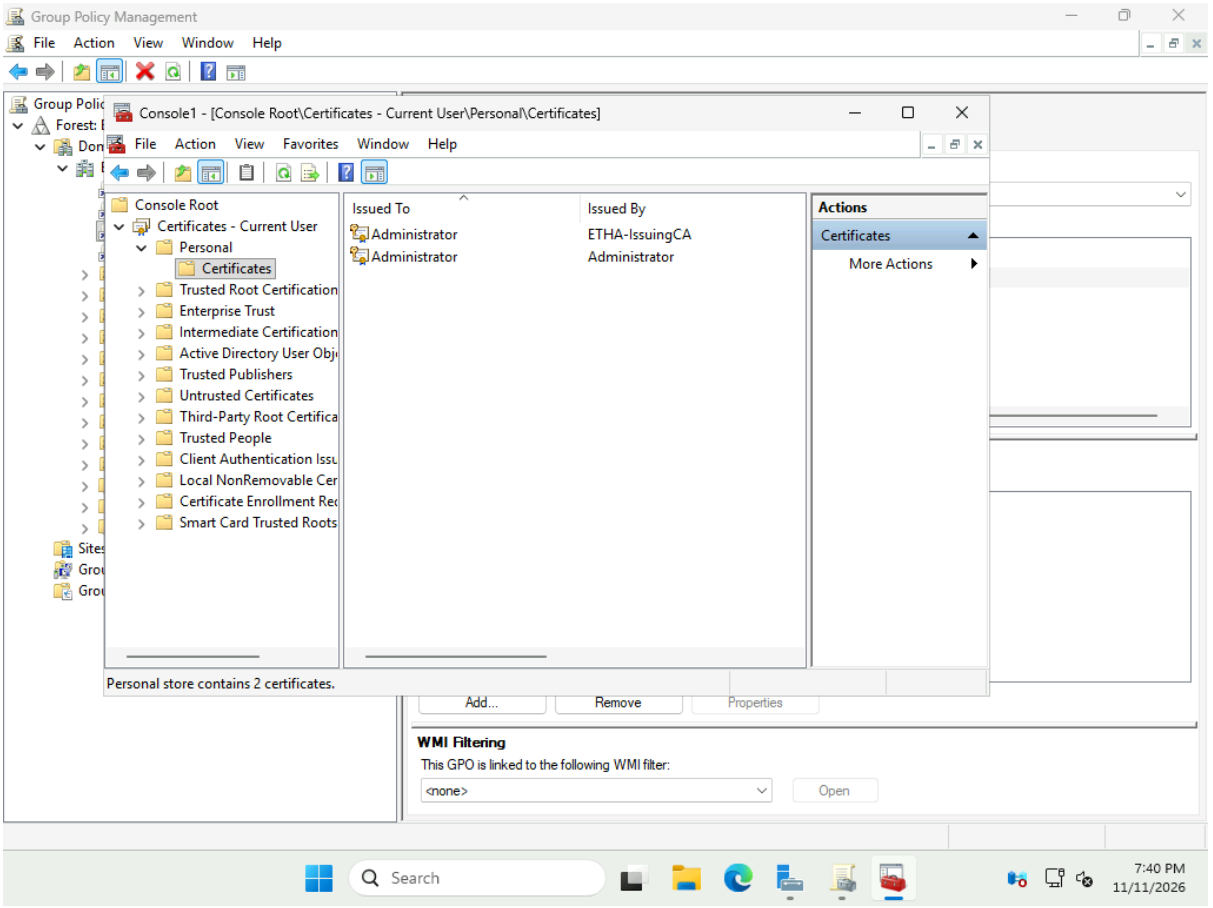


Task 10: Apply the GPOs changes and verify autoenrollment

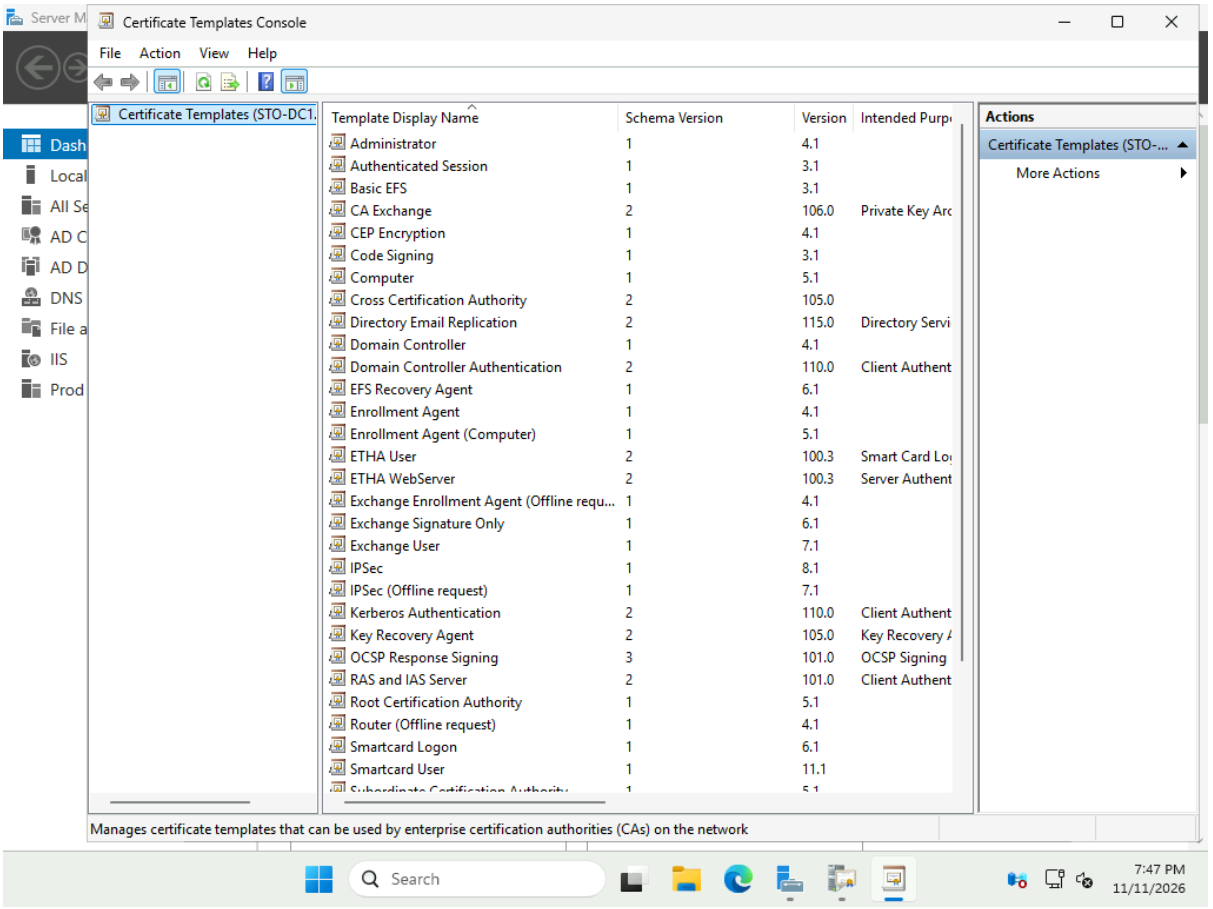

Administrator: Command Pro
Microsoft Windows [Version 10.0.26100.33438]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>gpupdate /force
Updating policy...
Computer Policy update has completed successfully.
Search
7:38 PM
11/11/2026

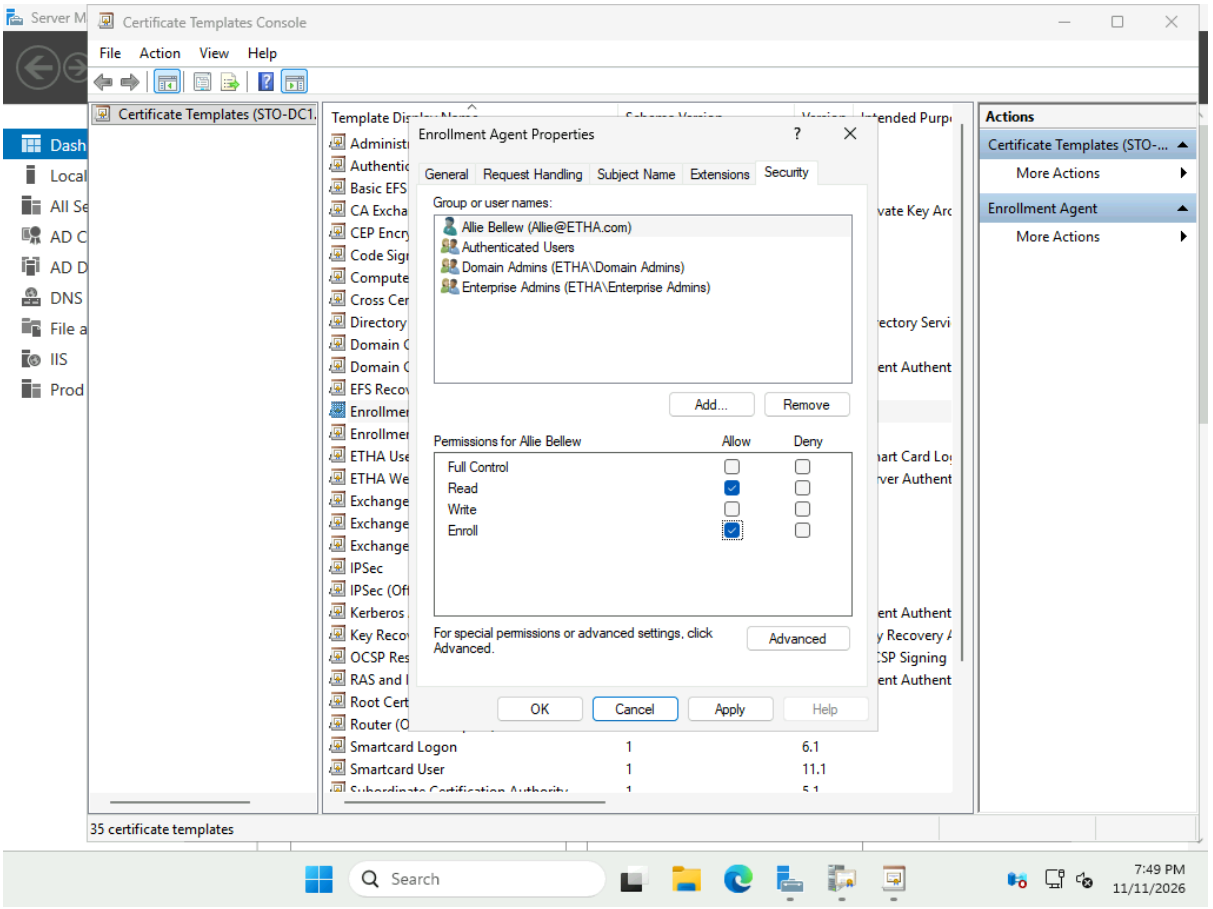

Group Policy Management
File Action View Window Help
Add or Remove Snap-ins
You can select snap-ins for this console from those available on your computer and configure the selected set of snap-ins. For extensible snap-ins, you can configure which extensions are enabled.
Available snap-ins:
Snap-in
Vendor
Active Directory D... Microsoft Co...
Active Directory Sit... Microsoft Co...
Active Directory Us... Microsoft Co...
ActiveX Control Microsoft Co...
ADSI Edit Microsoft Co...
Authorization Man... Microsoft Co...
Certificate Templat... Microsoft Co...
Certificates Microsoft Co...
Certification Autho... Microsoft Co...
Component Services Microsoft Co...
Computer Manage... Microsoft Co...
Device Manager Microsoft Co...
Disk Management Microsoft an...
Add >
Selected snap-ins:
Console Root
Certificates - Current User
Edit Extensions...
Remove
Move Up
Move Down
Advanced...
Description:
The Certificates snap-in allows you to browse the contents of the certificate stores for yourself, a service, or a computer.
OK
Cancel
WMI Filtering
This GPO is linked to the following WMI filter:
<none>
Open
Search
7:39 PM
11/11/2026

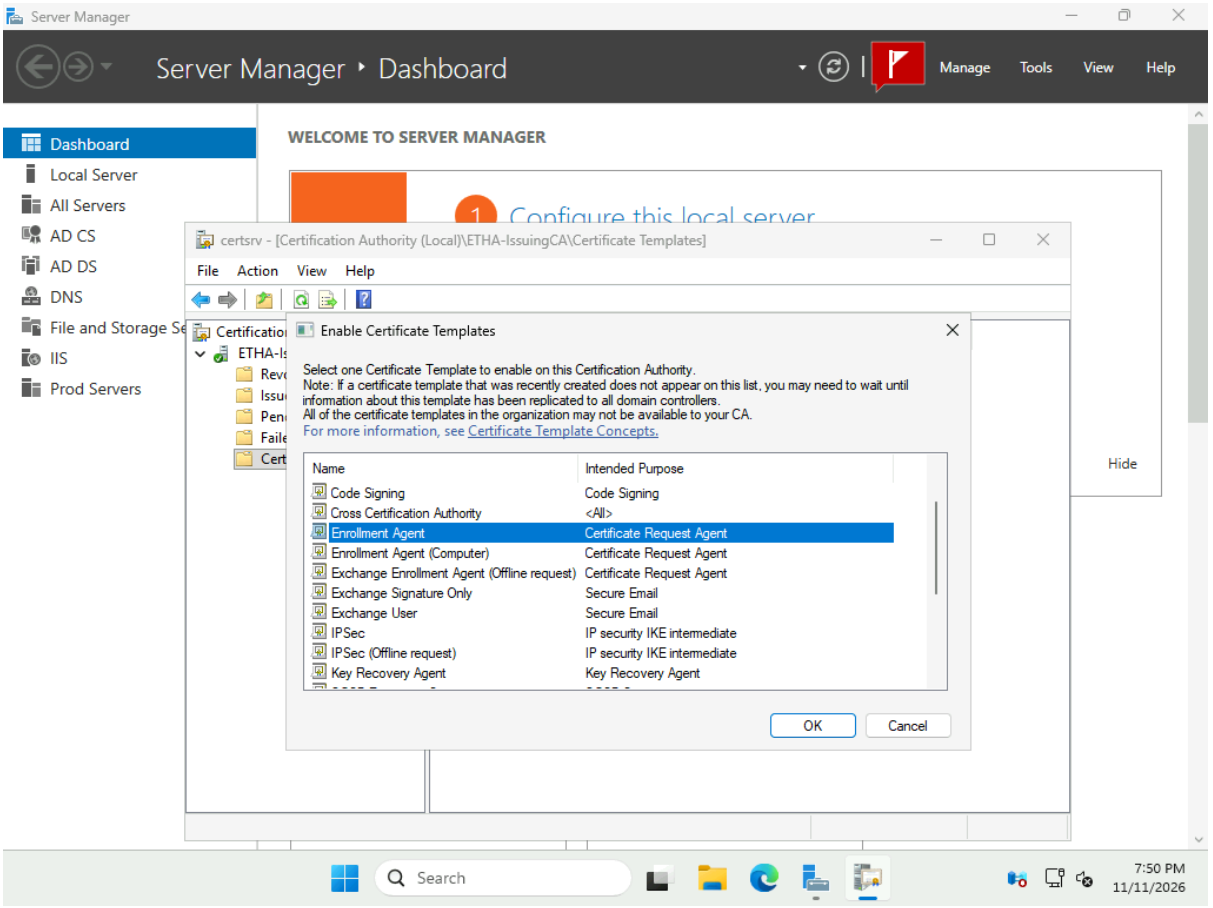

Group Policy Management
File Action View Window Help
Console1 - [Console Root]
File Action View Favorites Window Help
Console Root
Certificates - Curren
Name
Certificates - Current User
Actions
Console Root
More Actions
Add...
Remove
Properties
WMI Filtering
This GPO is linked to the following WMI filter:
<none>
Open
Search
7:39 PM
11/11/2026

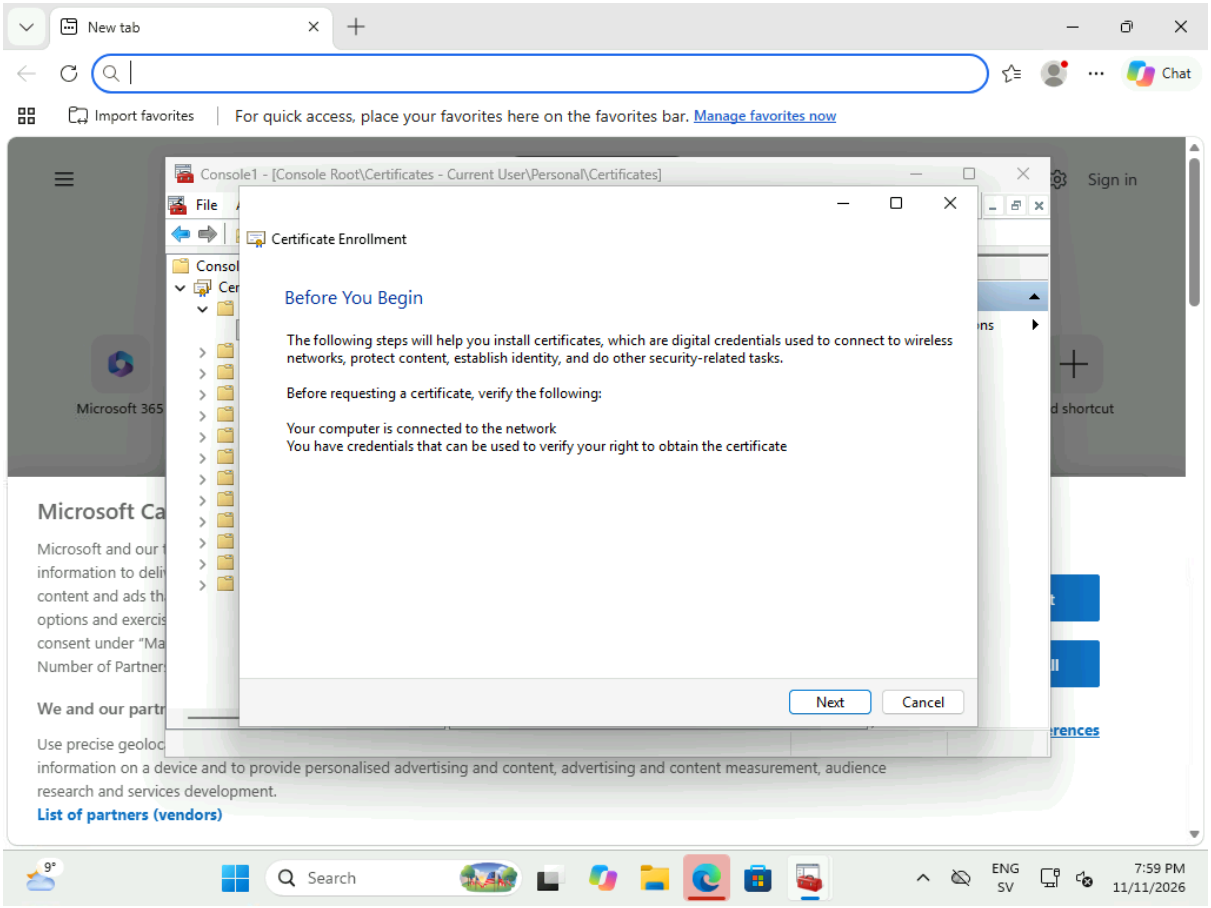


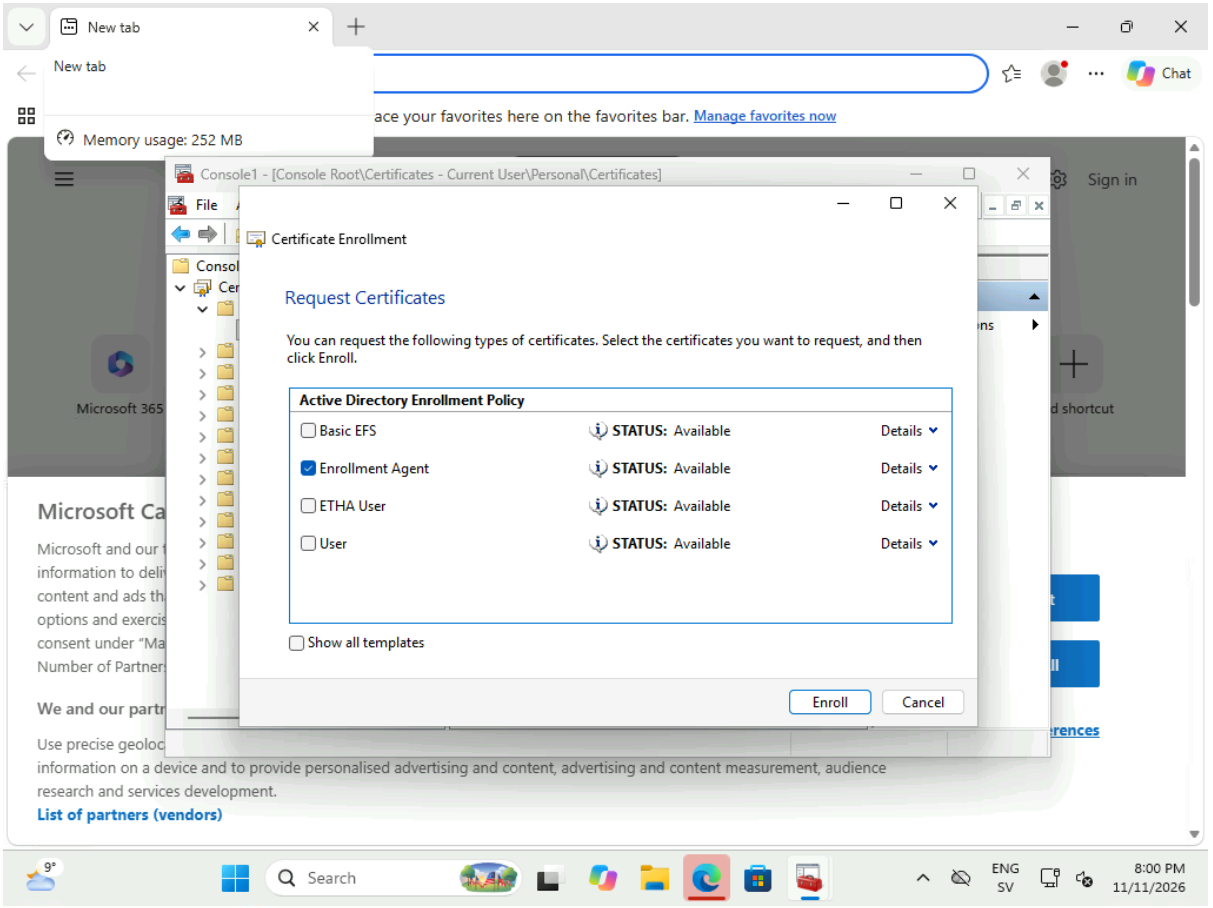
Task 11: Configure the CA Enrollment Agent for smart card certificates

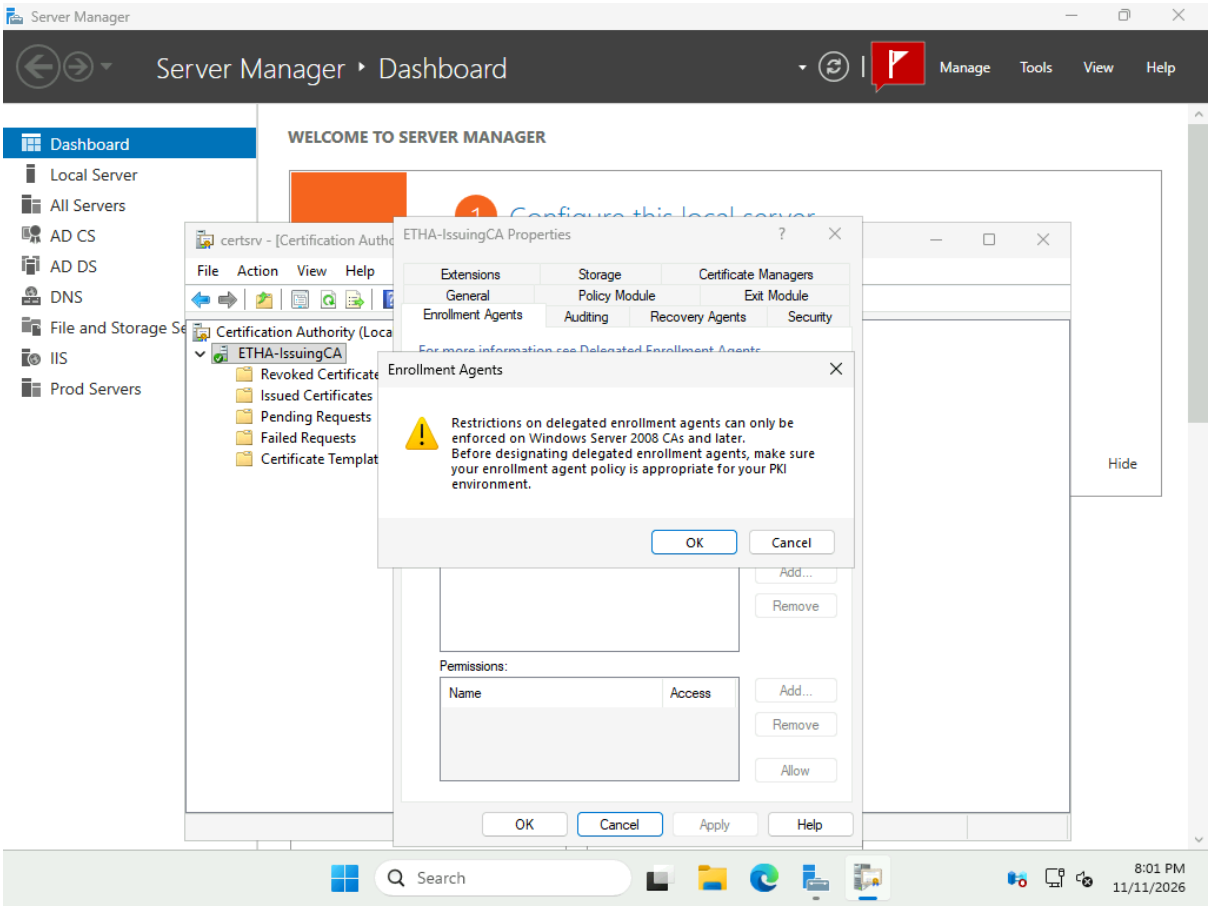

Certificate Templates Console
File Action View Help
Certificate Templates (STO-DC1.
Template Display Name
Schema Version
Version
Intended Purp
Administrator 1 4.1
Authenticated Session 1 3.1
Basic EFS 1 3.1
CA Exchange 2 106.0 Private Key Arc
CEP Encryption 1 4.1
Code Signing 1 3.1
Computer 1 5.1
Cross Certification Authority 2 105.0
Directory Email Replication 2 115.0 Directory Servi
Domain Controller 1 4.1
Domain Controller Authentication 2 110.0 Client Authent
EFS Recovery Agent 1 6.1
Enrollment Agent 1 4.1
Enrollment Agent (Computer) 1 5.1
ETHA User 2 100.3 Smart Card Lo
ETHA WebServer 2 100.3 Server Authent
Exchange Enrollment Agent (Offline requ... 1 4.1
Exchange Signature Only 1 6.1
Exchange User 1 7.1
IPSec 1 8.1
IPSec (Offline request) 1 7.1
Kerberos Authentication 2 110.0 Client Authent
Key Recovery Agent 2 105.0 Key Recovery A
OCSP Response Signing 3 101.0 OCSP Signing
RAS and IAS Server 2 101.0 Client Authent
Root Certification Authority 1 5.1
Router (Offline request) 1 4.1
Smartcard Logon 1 6.1
Smartcard User 1 11.1
Actions
Certificate Templates (STO-...
More Actions
Manages certificate templates that can be used by enterprise certification authorities (CAs) on the network
Search
7:47 PM
11/11/2026

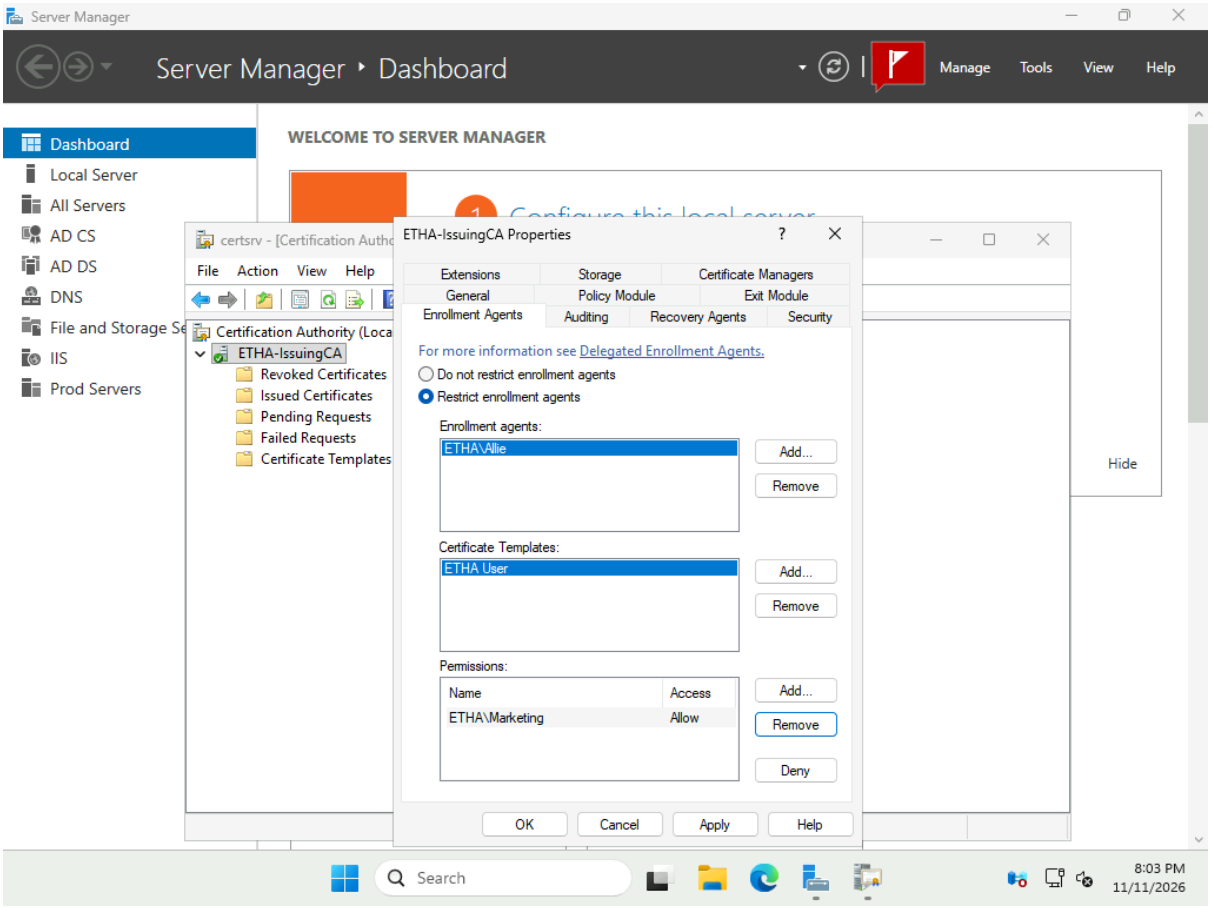

Certificate Templates Console
File Action View Help
Certificate Templates (STO-DC1.
Enrollment Agent Properties
General Request Handling Subject Name Extensions Security
Group or user names:
Allie Bellew (Allie@ETHA.com)
Authenticated Users
Domain Admins (ETHA\Domain Admins)
Enterprise Admins (ETHA\Enterprise Admins)
Add... Remove
Permissions for Allie Bellew Allow Deny
Full Control
Read
Write
Enroll
For special permissions or advanced settings, click Advanced.
Advanced
OK Cancel Apply Help
Smartcard Logon 1 6.1
Smartcard User 1 11.1
Actions
Certificate Templates (STO-...
More Actions
Enrollment Agent
More Actions
35 certificate templates
Search
7:49 PM
11/11/2026


Server Manager
Server Manager • Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storage Se
IIS
Prod Servers
WELCOME TO SERVER MANAGER
Configure this local server
certsrv - [Certification Authority (Local)\ETHA-IssuingCA\Certificate Templates]
File
Action
View
Help
Enable Certificate Templates
Select one Certificate Template to enable on this Certification Authority.
Note: If a certificate template that was recently created does not appear on this list, you may need to wait until information about this template has been replicated to all domain controllers.
All of the certificate templates in the organization may not be available to your CA.
For more information, see Certificate Template Concepts.
Name
Intended Purpose
Code Signing
Code Signing
Cross Certification Authority
<All>
Enrollment Agent
Certificate Request Agent
Enrollment Agent (Computer)
Certificate Request Agent
Exchange Enrollment Agent (Offline request)
Certificate Request Agent
Exchange Signature Only
Secure Email
Exchange User
Secure Email
IPSec
IP security IKE intermediate
IPSec (Offline request)
IP security IKE intermediate
Key Recovery Agent
Key Recovery Agent
OK
Cancel
Hide
Search
7:50 PM
11/11/2026


New tab
Chat
Import favorites
For quick access, place your favorites here on the favorites bar. Manage favorites now
Console1 - [Console Root\Certificates - Current User\Personal\Certificates]
File
Certificate Enrollment
Before You Begin
The following steps will help you install certificates, which are digital credentials used to connect to wireless networks, protect content, establish identity, and do other security-related tasks.
Before requesting a certificate, verify the following:
Your computer is connected to the network
You have credentials that can be used to verify your right to obtain the certificate
Next
Cancel
Sign in
Microsoft 365
Microsoft Ca
We and our partn
List of partners (vendors)
Search
ENG
SV
7:59 PM
11/11/2026

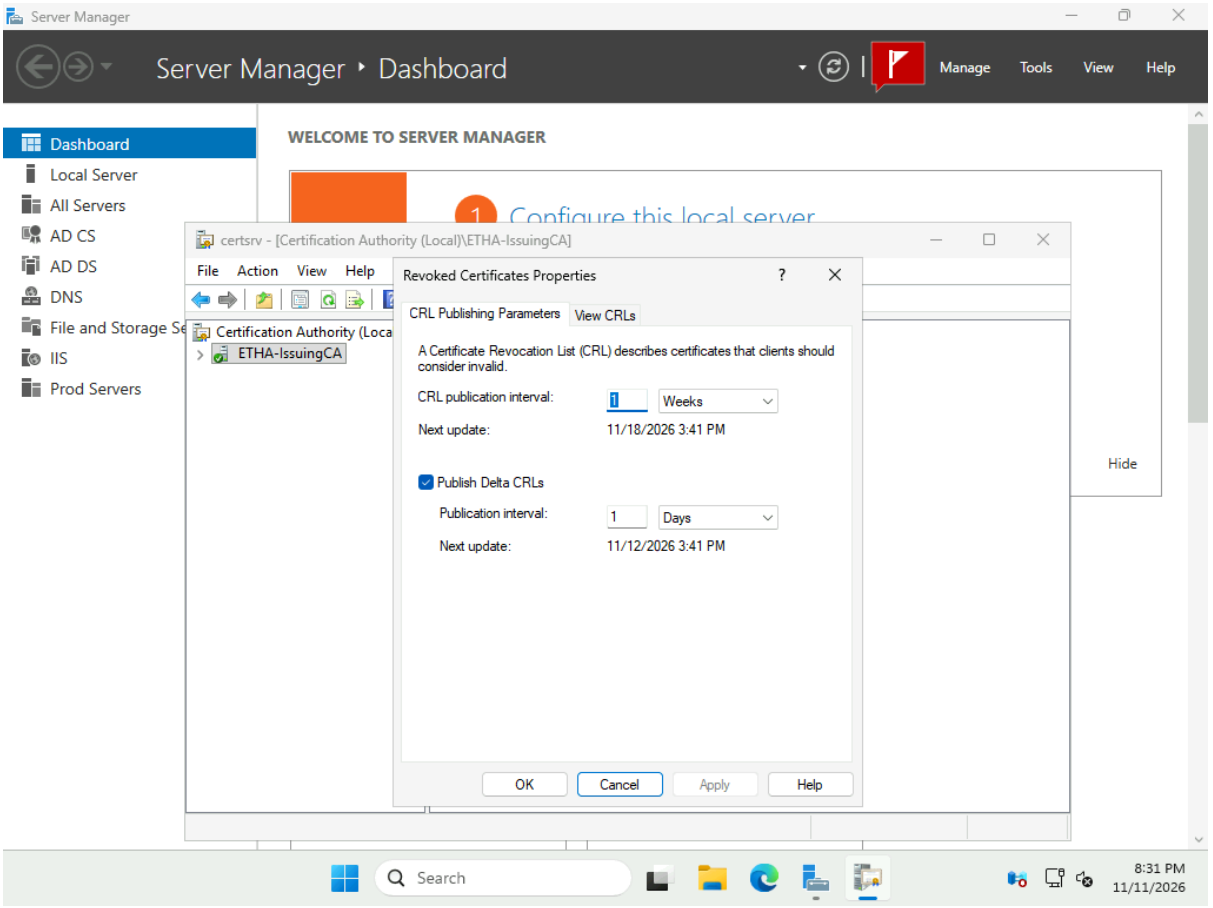

New tab
New tab
Memory usage: 252 MB
Chat
ace your favorites here on the favorites bar. Manage favorites now
Sign in
Console1 - [Console Root\Certificates - Current User\Personal\Certificates]
File
Certificate Enrollment
Request Certificates
You can request the following types of certificates. Select the certificates you want to request, and then click Enroll.
Active Directory Enrollment Policy
Basic EFS
STATUS: Available
Details
Enrollment Agent
STATUS: Available
Details
ETHA User
STATUS: Available
Details
User
STATUS: Available
Details
Show all templates
Enroll
Cancel
Microsoft 365
Microsoft Ca
We and our partn
List of partners (vendors)
Search
ENG
SV
8:00 PM
11/11/2026

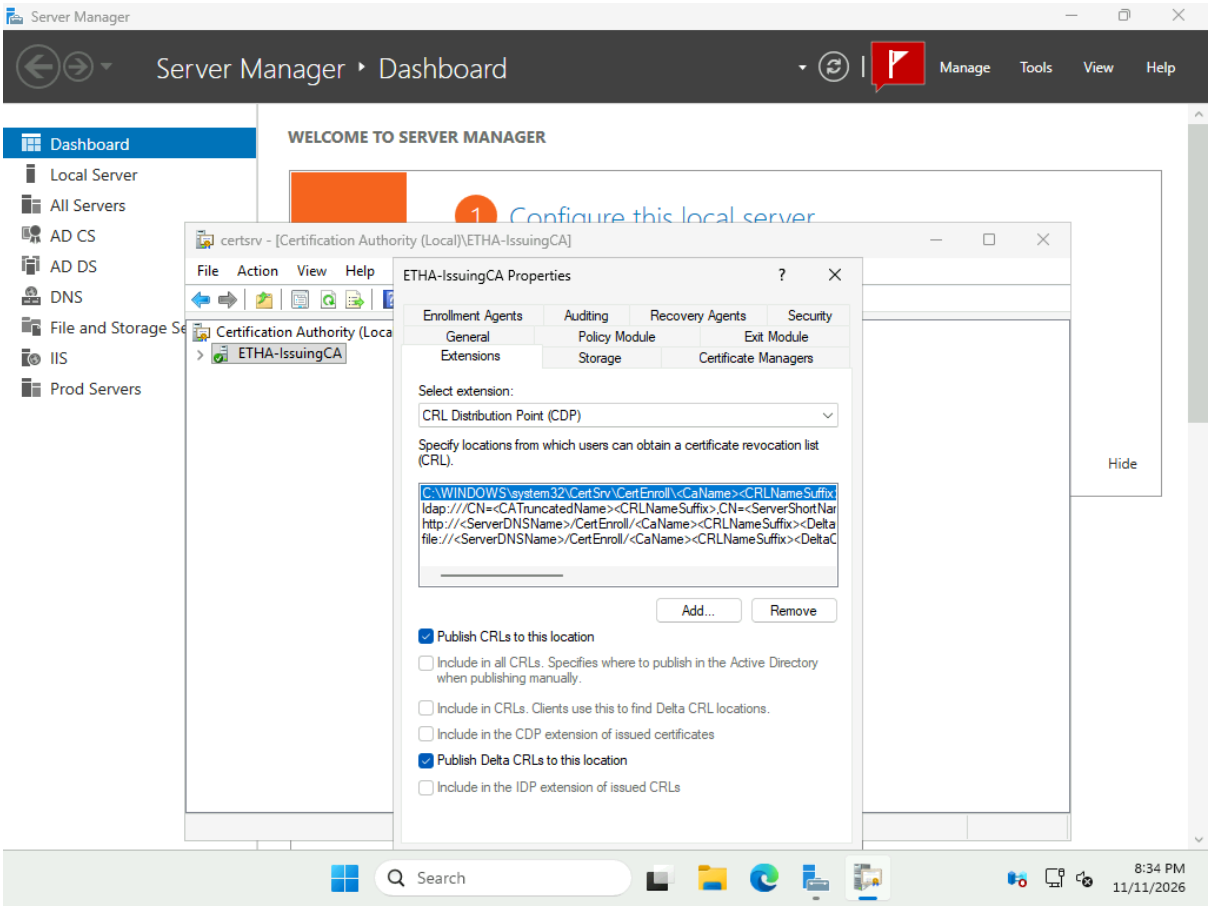

Server Manager
Server Manager • Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storage Se
IIS
Prod Servers
WELCOME TO SERVER MANAGER
Hide
certsrv - [Certification Autho
File
Action
View
Help
Certification Authority (Loca
ETHA-IssuingCA
Revoked Certificate
Issued Certificates
Pending Requests
Failed Requests
Certificate Templat
ETHA-IssuingCA Properties
Extensions
Storage
Certificate Managers
General
Policy Module
Exit Module
Enrollment Agents
Auditing
Recovery Agents
Security
Enrollment Agents
Restrictions on delegated enrollment agents can only be enforced on Windows Server 2008 CAs and later.
Before designating delegated enrollment agents, make sure your enrollment agent policy is appropriate for your PKI environment.
OK
Cancel
Add...
Remove
Permissions:
Name
Access
Add...
Remove
Allow
OK
Cancel
Apply
Help
Search
8:01 PM
11/11/2026



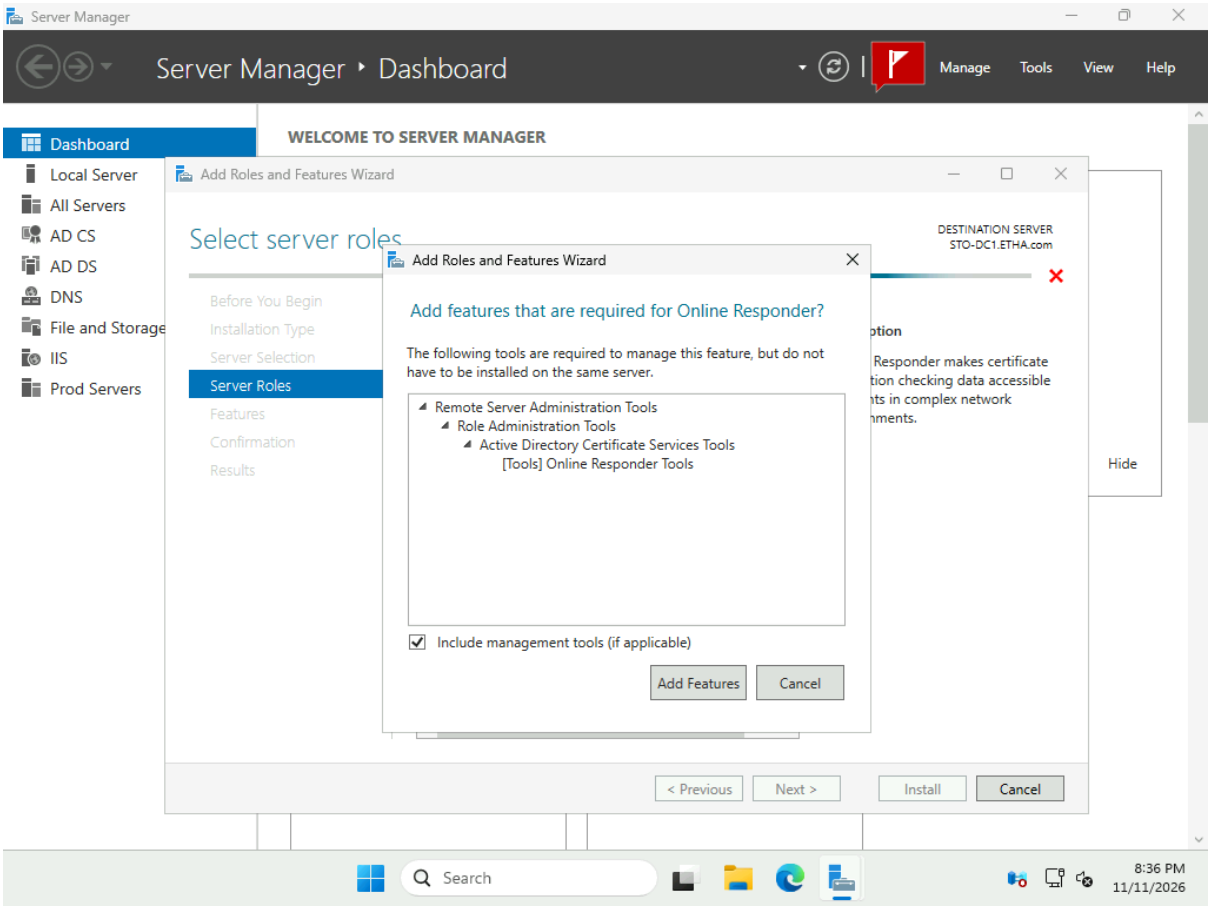
Practice 5: Configuring Certificate Revocation

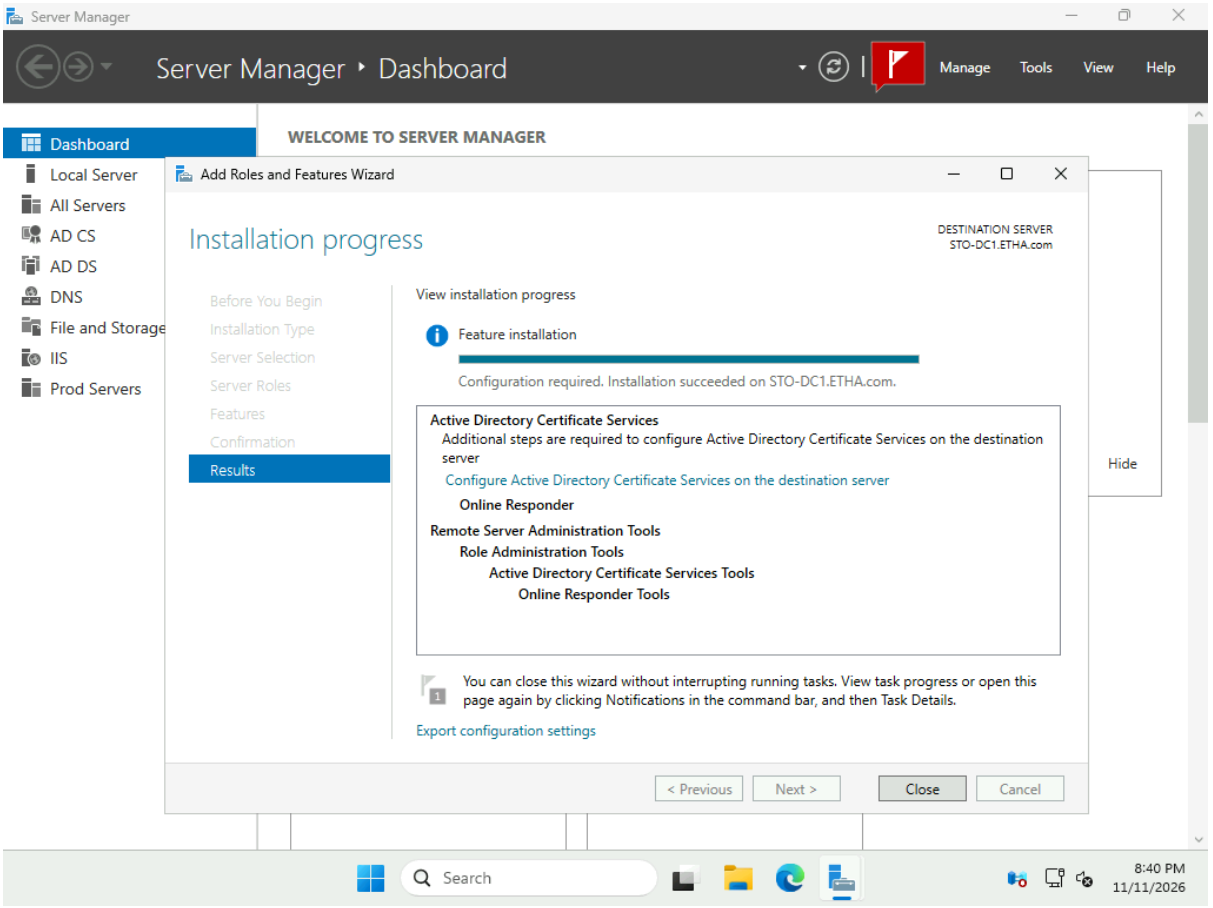
Task 12: Configure certified revocation list (CRL) distribution

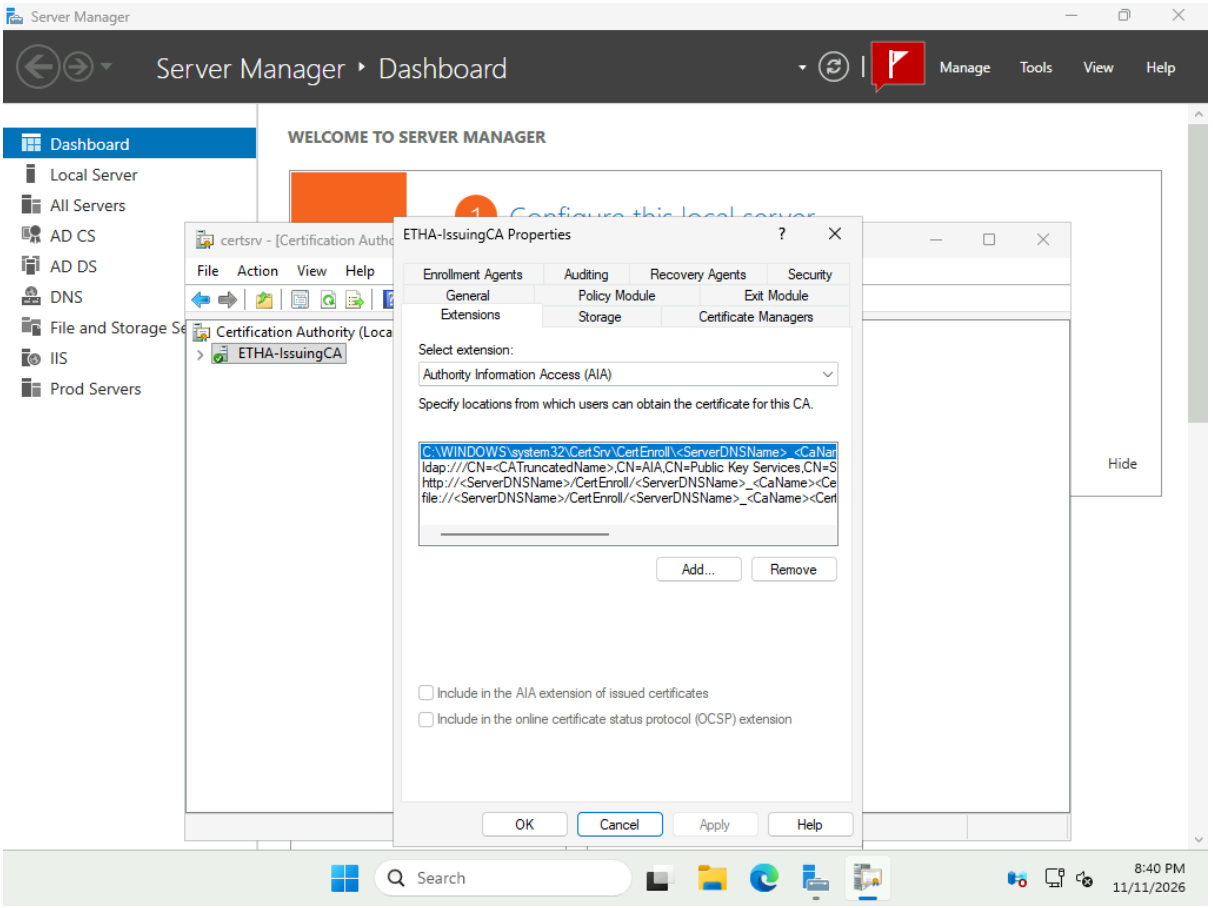

Server Manager
Server Manager • Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storage Se
IIS
Prod Servers
WELCOME TO SERVER MANAGER
Configure this local server
certsrv - [Certification Authority (Local)\ETHA-IssuingCA]
File
Action
View
Help
Certification Authority (Loca
ETHA-IssuingCA
Revoked Certificates Properties
CRL Publishing Parameters
View CRLs
A Certificate Revocation List (CRL) describes certificates that clients should consider invalid.
CRL publication interval:
1
Weeks
Next update:
11/18/2026 3:41 PM
Publish Delta CRLs
Publication interval:
1
Days
Next update:
11/12/2026 3:41 PM
OK
Cancel
Apply
Help
Hide
Search
8:31 PM
11/11/2026

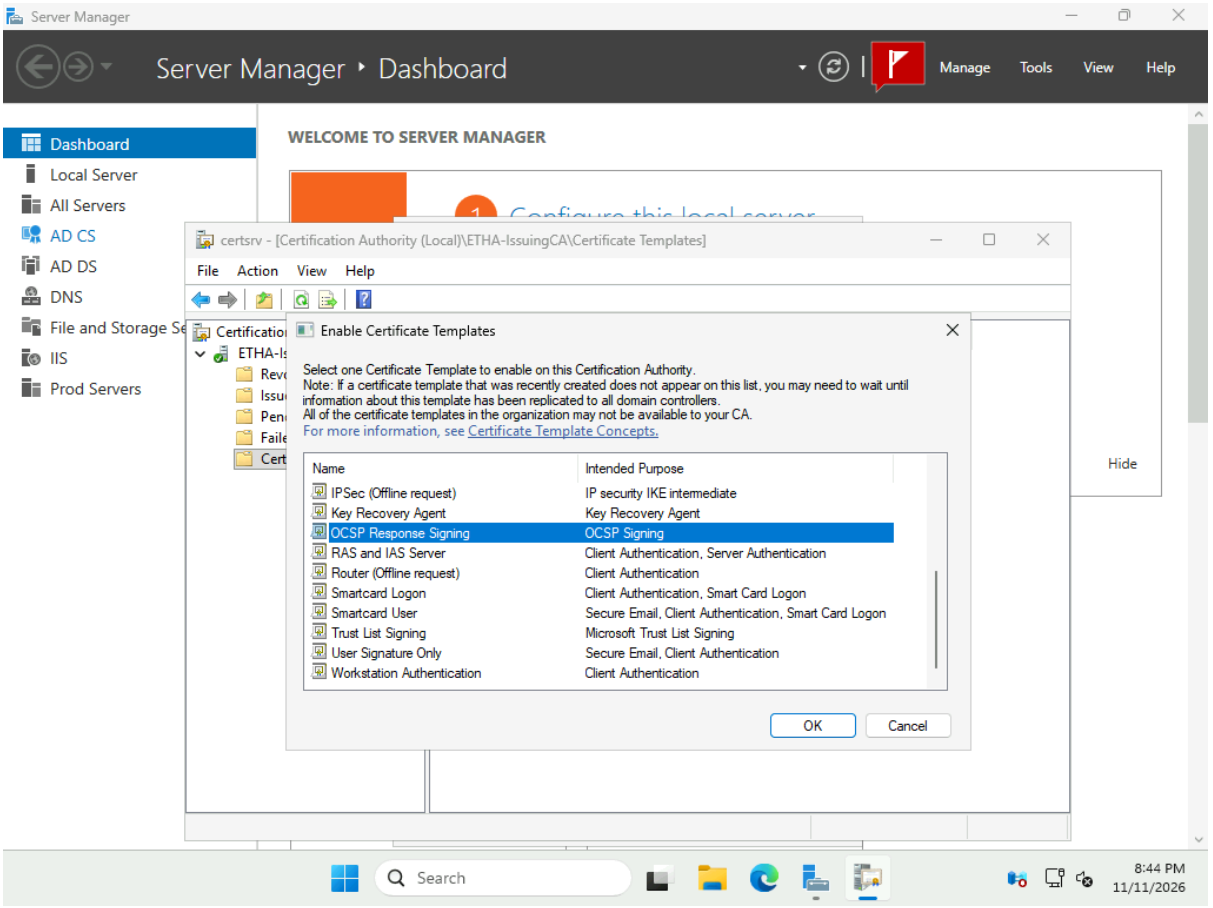


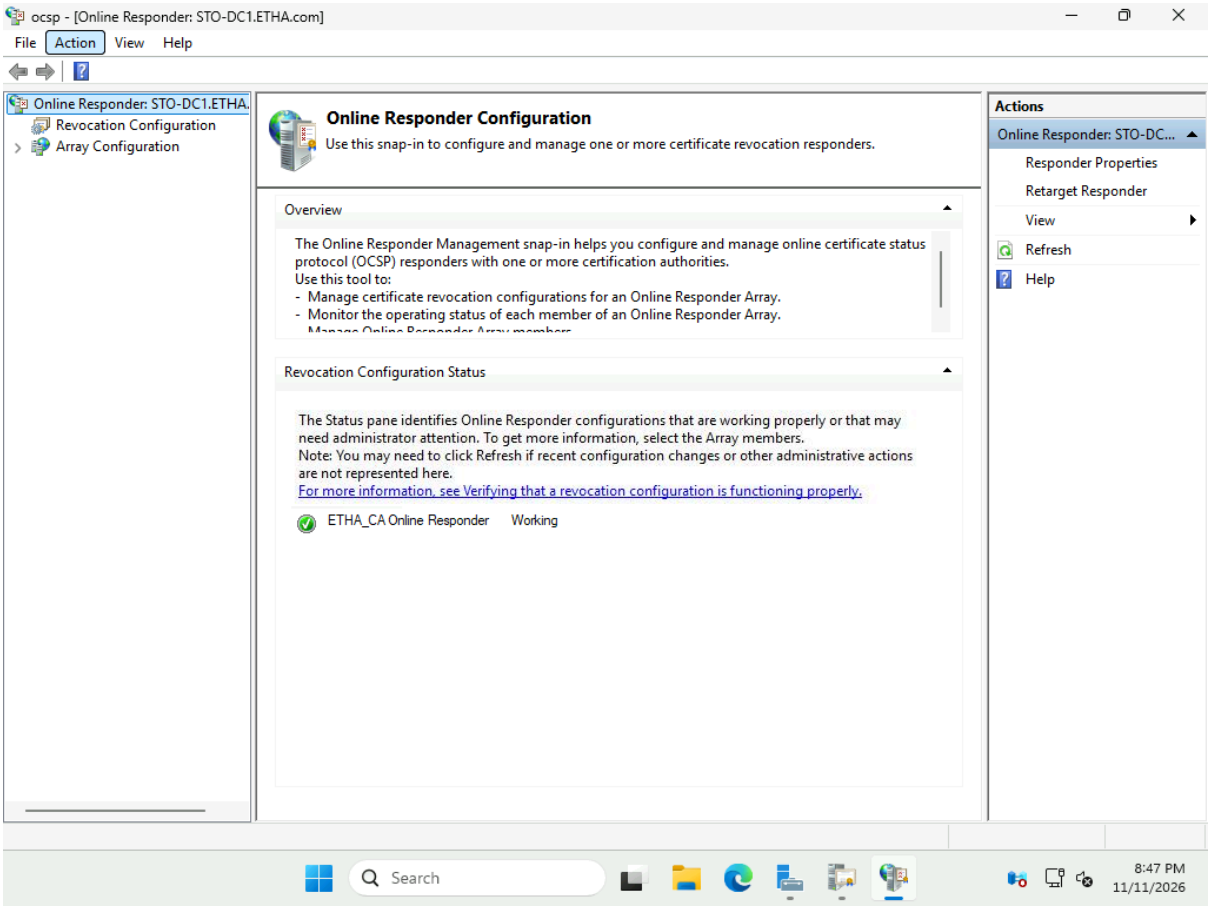
Task 13: Install and configure an Online Responder


Server Manager
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storage
IIS
Prod Servers
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Select server roles
DESTINATION SERVER
STO-DC1.ETHA.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Add Roles and Features Wizard
Add features that are required for Online Responder?
The following tools are required to manage this feature, but do not have to be installed on the same server.
Remote Server Administration Tools
Role Administration Tools
Active Directory Certificate Services Tools
[Tools] Online Responder Tools
Include management tools (if applicable)
Add Features
Cancel
Responder makes certificate
tion checking data accessible
ts in complex network
nments.
Hide
< Previous
Next >
Install
Cancel
Search
8:36 PM
11/11/2026


Server Manager
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storage
IIS
Prod Servers
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Installation progress
DESTINATION SERVER
STO-DC1.ETHA.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
View installation progress
Feature installation
Configuration required. Installation succeeded on STO-DC1.ETHA.com.
Active Directory Certificate Services
Additional steps are required to configure Active Directory Certificate Services on the destination server
Configure Active Directory Certificate Services on the destination server
Online Responder
Remote Server Administration Tools
Role Administration Tools
Active Directory Certificate Services Tools
Online Responder Tools
You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.
Export configuration settings
< Previous
Next >
Close
Cancel
Hide
Search
8:40 PM
11/11/2026

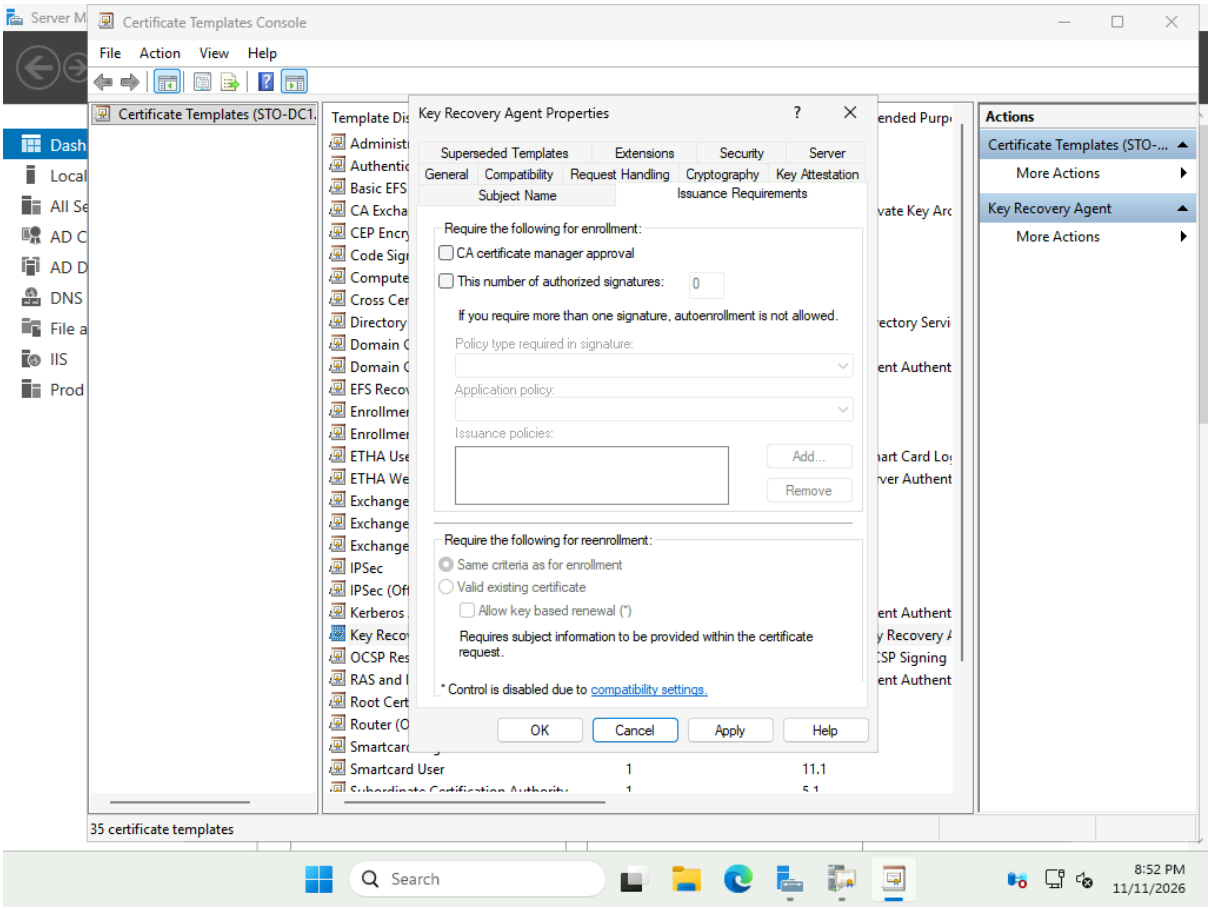

Server Manager
Server Manager • Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storage Se
IIS
Prod Servers
WELCOME TO SERVER MANAGER
Hide
certsrv - [Certification Autho
File
Action
View
Help
Certification Authority (Loca
ETHA-IssuingCA
ETHA-IssuingCA Properties
Enrollment Agents
Auditing
Recovery Agents
Security
General
Policy Module
Exit Module
Extensions
Storage
Certificate Managers
Select extension:
Authority Information Access (AIA)
Specify locations from which users can obtain the certificate for this CA.
C:\WINDOWS\system32\CertSrv\CertEnroll\<ServerDNSName>_<CaNar
ldap:///CN=<CATruncatedName>,CN=AIA,CN=Public Key Services,CN=S
http://<ServerDNSName>/CertEnroll/<ServerDNSName>_<CaName><Ce
file://<ServerDNSName>/CertEnroll/<ServerDNSName>_<CaName><Cerl
Add...
Remove
Include in the AIA extension of issued certificates
Include in the online certificate status protocol (OCSP) extension
OK
Cancel
Apply
Help
Search
8:40 PM
11/11/2026

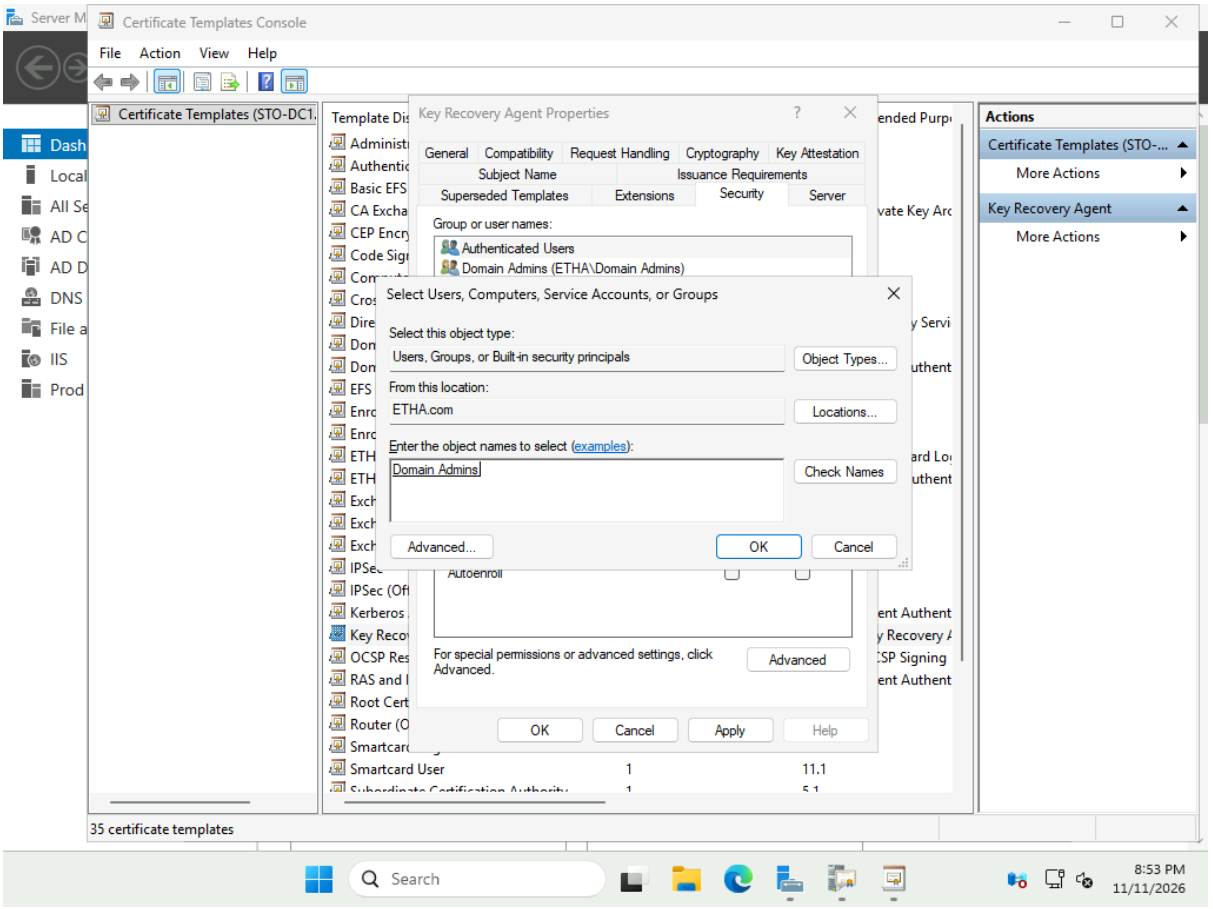

Server Manager
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD CS
AD DS
DNS
File and Storage Se
IIS
Prod Servers
WELCOME TO SERVER MANAGER
certsrv - [Certification Authority (Local)\ETHA-IssuingCA\Certificate Templates]
File
Action
View
Help
Enable Certificate Templates
Select one Certificate Template to enable on this Certification Authority.
Note: If a certificate template that was recently created does not appear on this list, you may need to wait until information about this template has been replicated to all domain controllers.
All of the certificate templates in the organization may not be available to your CA.
For more information, see Certificate Template Concepts.
Name
Intended Purpose
IPSec (Offline request)
IP security IKE intermediate
Key Recovery Agent
Key Recovery Agent
OCSP Response Signing
OCSP Signing
RAS and IAS Server
Client Authentication, Server Authentication
Router (Offline request)
Client Authentication
Smartcard Logon
Client Authentication, Smart Card Logon
Smartcard User
Secure Email, Client Authentication, Smart Card Logon
Trust List Signing
Microsoft Trust List Signing
User Signature Only
Secure Email, Client Authentication
Workstation Authentication
Client Authentication
OK
Cancel
Hide
Search
8:44 PM
11/11/2026

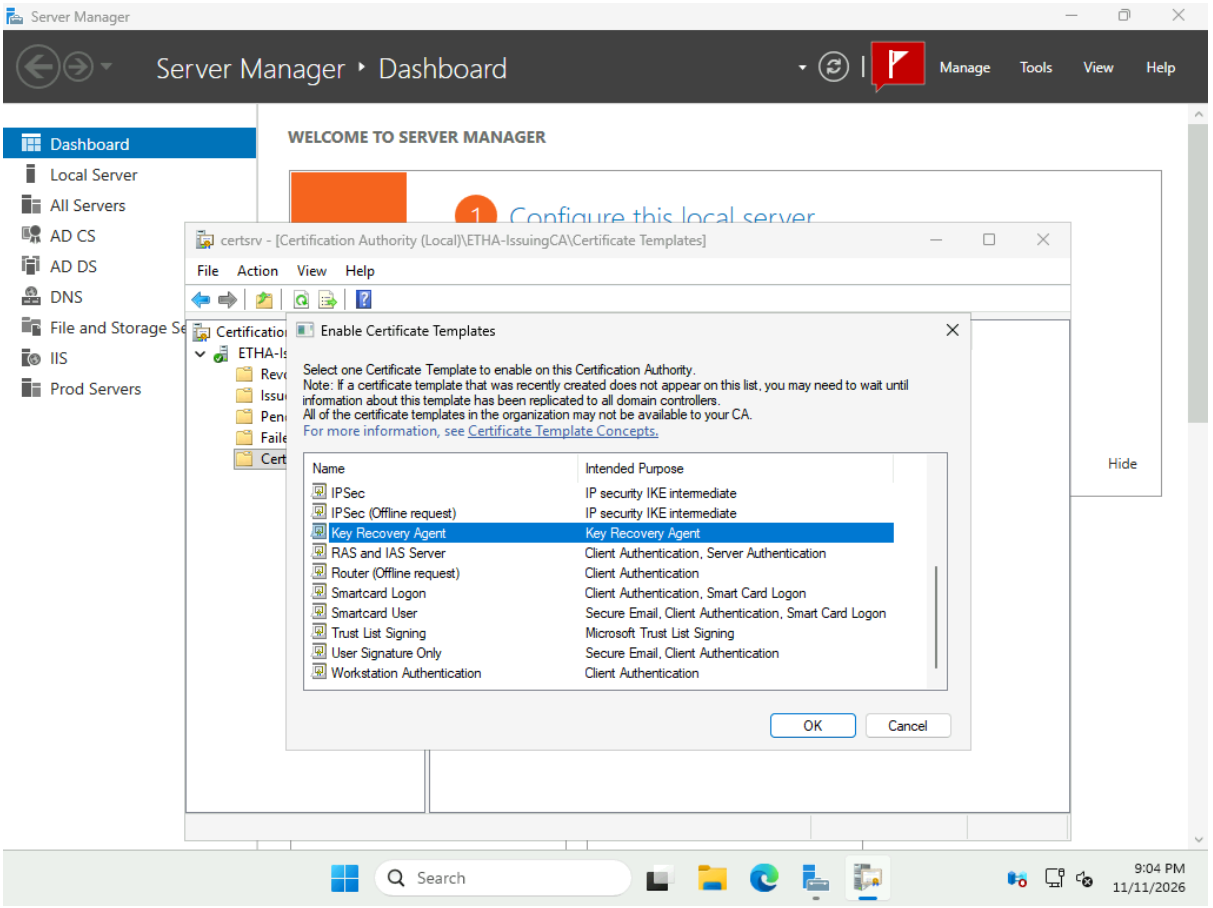


Practice 6: Configuring Key Recovery

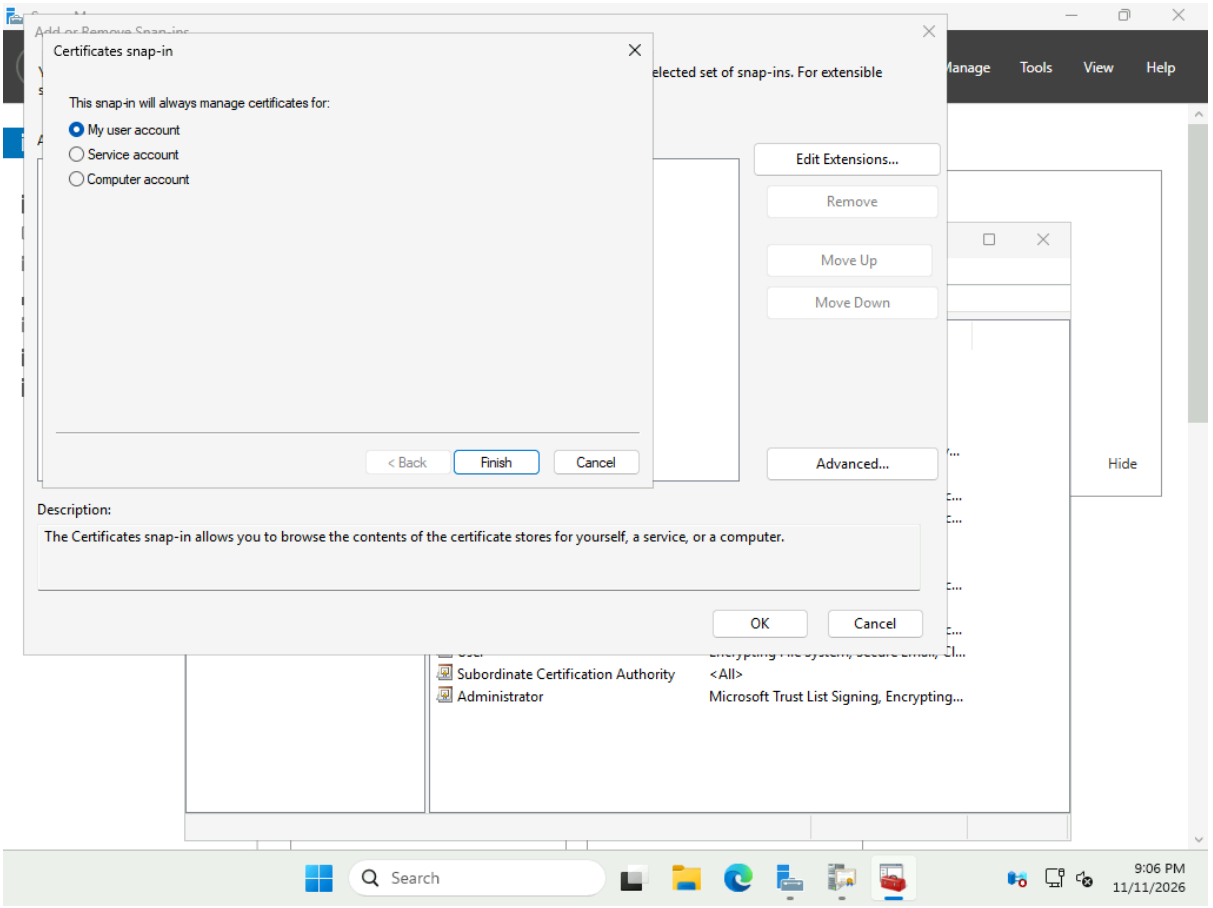
Task 14: Configure the CA to issue KRA certificates

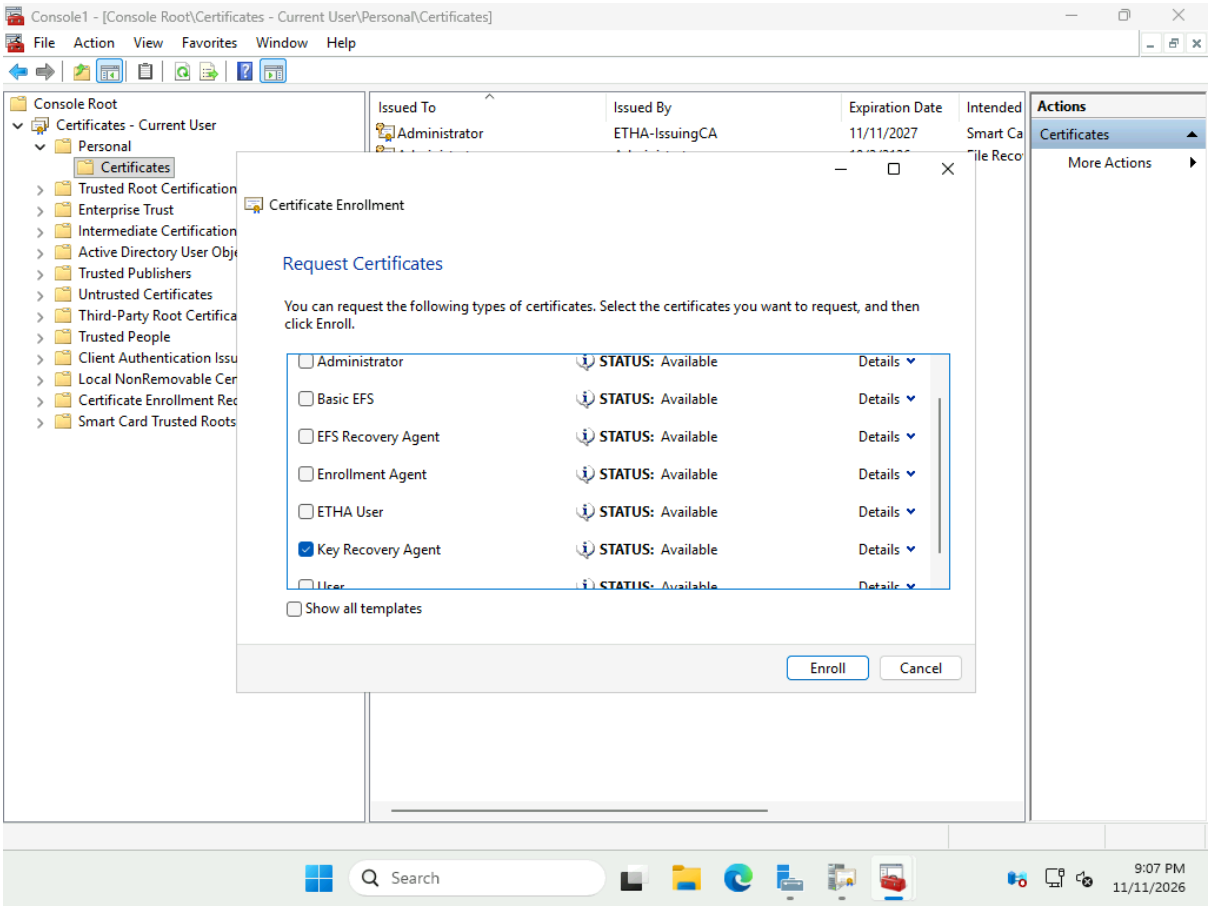

Certificate Templates Console
File Action View Help
Certificate Templates (STO-DC1...
Key Recovery Agent Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Require the following for enrollment:
CA certificate manager approval
This number of authorized signatures:
If you require more than one signature, autoenrollment is not allowed.
Policy type required in signature:
Application policy:
Issuance policies:
Add...
Remove
Require the following for reenrollment:
Same criteria as for enrollment
Valid existing certificate
Allow key based renewal (*)
Requires subject information to be provided within the certificate request.
* Control is disabled due to compatibility settings.
OK
Cancel
Apply
Help
Smartcard User
1
11.1
Actions
Certificate Templates (STO-...
More Actions
Key Recovery Agent
More Actions
35 certificate templates
Search
8:52 PM
11/11/2026


Certificate Templates Console
File Action View Help
Certificate Templates (STO-DC1...
35 certificate templates
Key Recovery Agent Properties
General Compatibility Request Handling Cryptography Key Attestation
Subject Name Issuance Requirements
Superseded Templates Extensions Security Server
Group or user names:
Authenticated Users
Domain Admins (ETHA\Domain Admins)
Autoenroll
For special permissions or advanced settings, click Advanced.
Advanced
OK Cancel Apply Help
Select Users, Computers, Service Accounts, or Groups
Select this object type:
Users, Groups, or Built-in security principals
Object Types...
From this location:
ETHA.com
Locations...
Enter the object names to select (examples):
Domain Admins
Check Names
Advanced... OK Cancel
Smartcard User 1 11.1
Actions
Certificate Templates (STO-...
More Actions
Key Recovery Agent
More Actions
Search
8:53 PM
11/11/2026

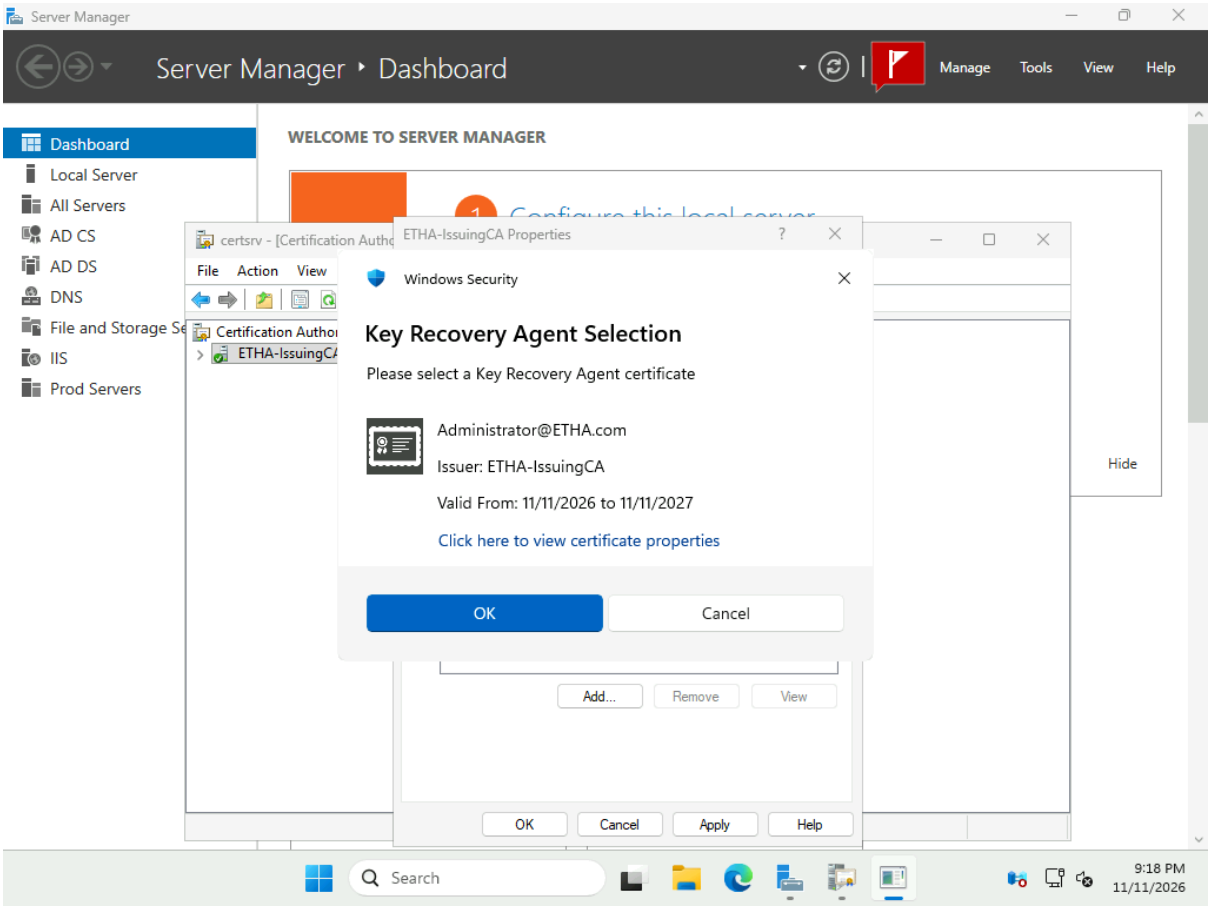


Task 15: Acquire the KRA certificate

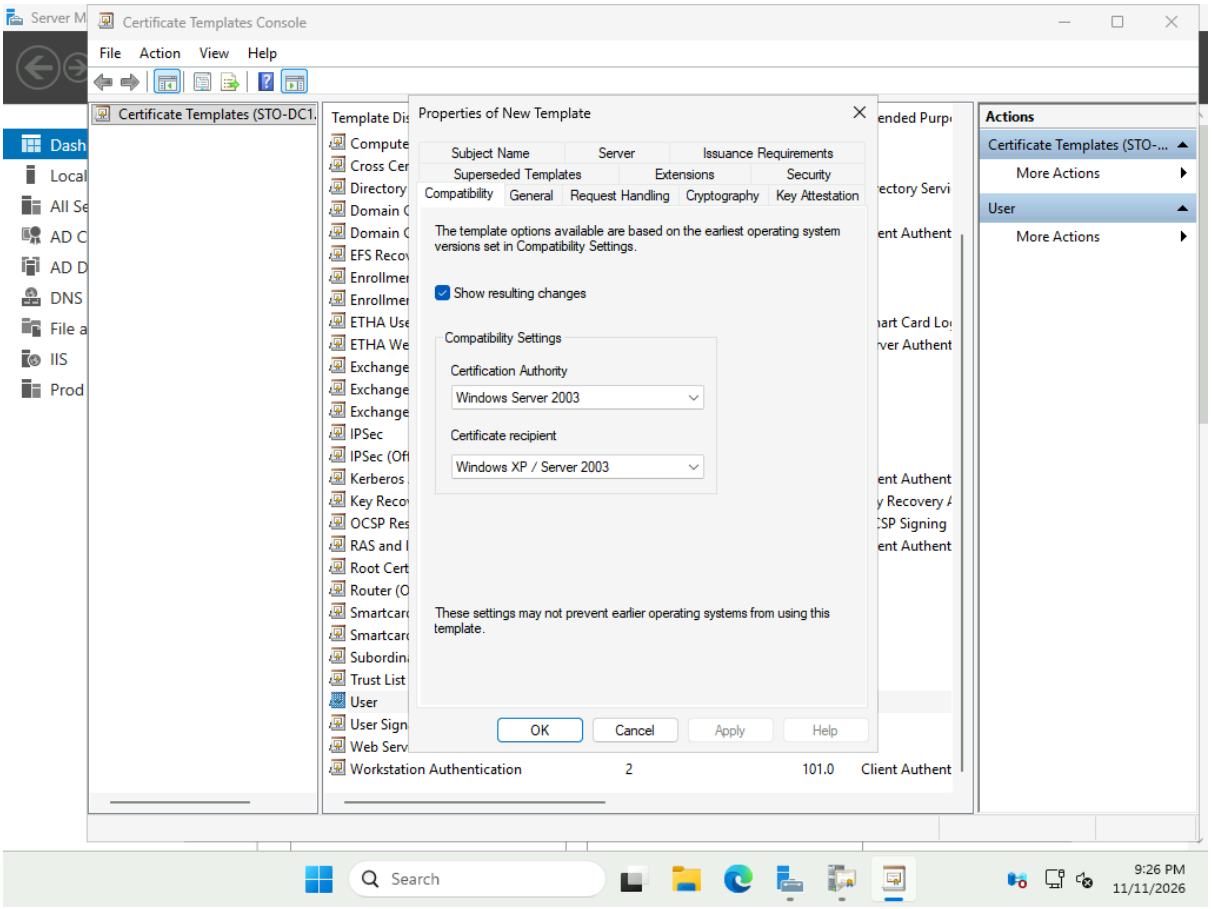

Certificates snap-in
This snap-in will always manage certificates for:
My user account
Service account
Computer account
< Back
Finish
Cancel
Edit Extensions...
Remove
Move Up
Move Down
Advanced...
Description:
The Certificates snap-in allows you to browse the contents of the certificate stores for yourself, a service, or a computer.
OK
Cancel
Subordinate Certification Authority
<All>
Administrator
Microsoft Trust List Signing, Encrypting...
Hide
Search
9:06 PM
11/11/2026

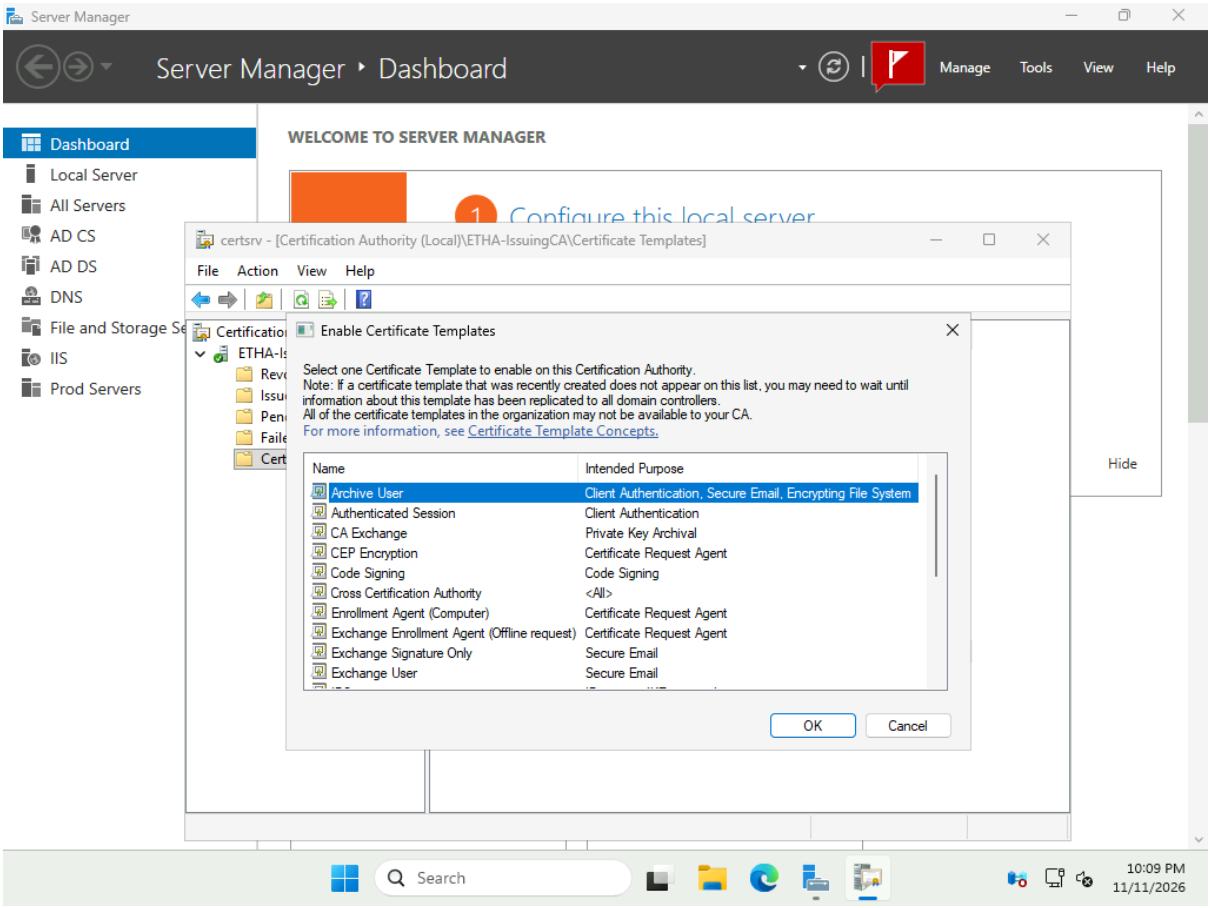


Task 16: Configure the CA to allow key recovery

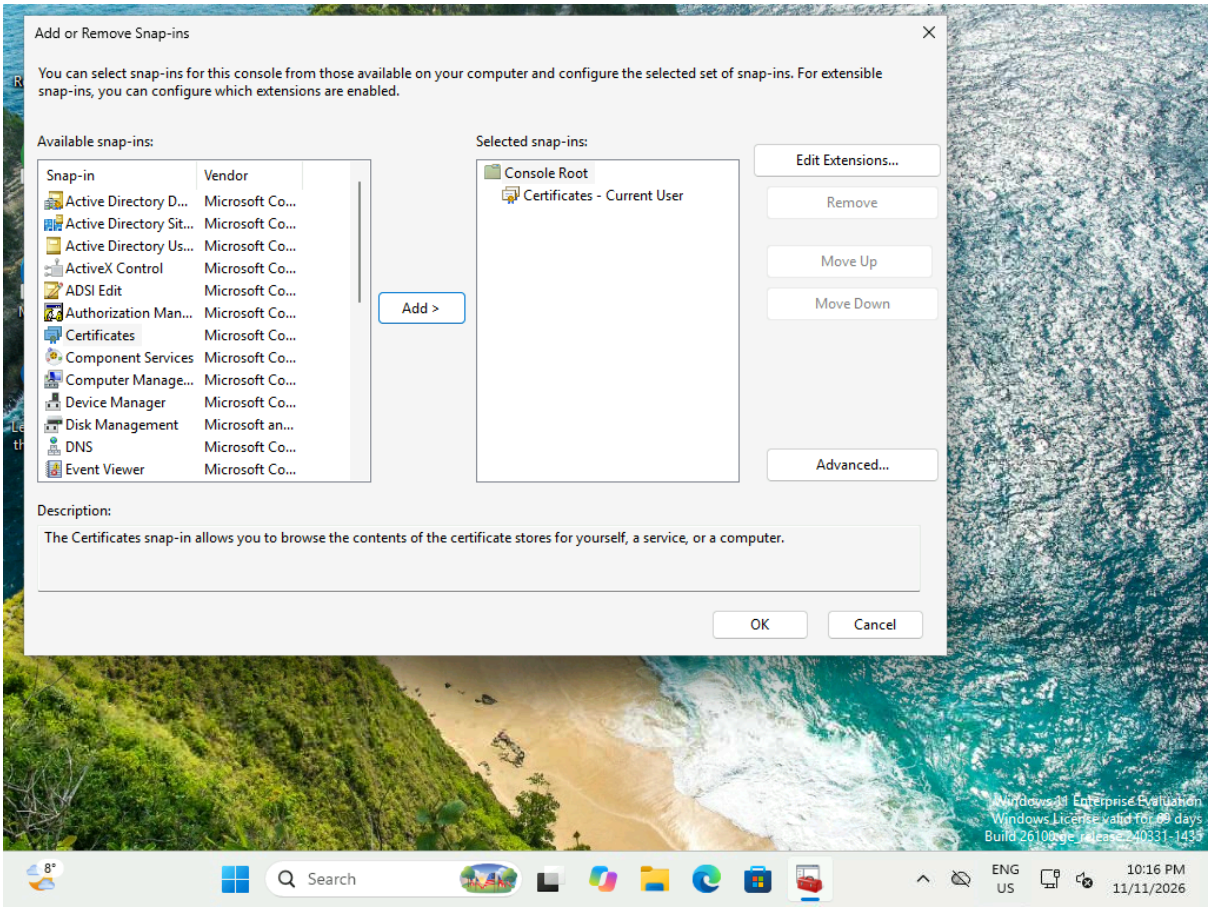


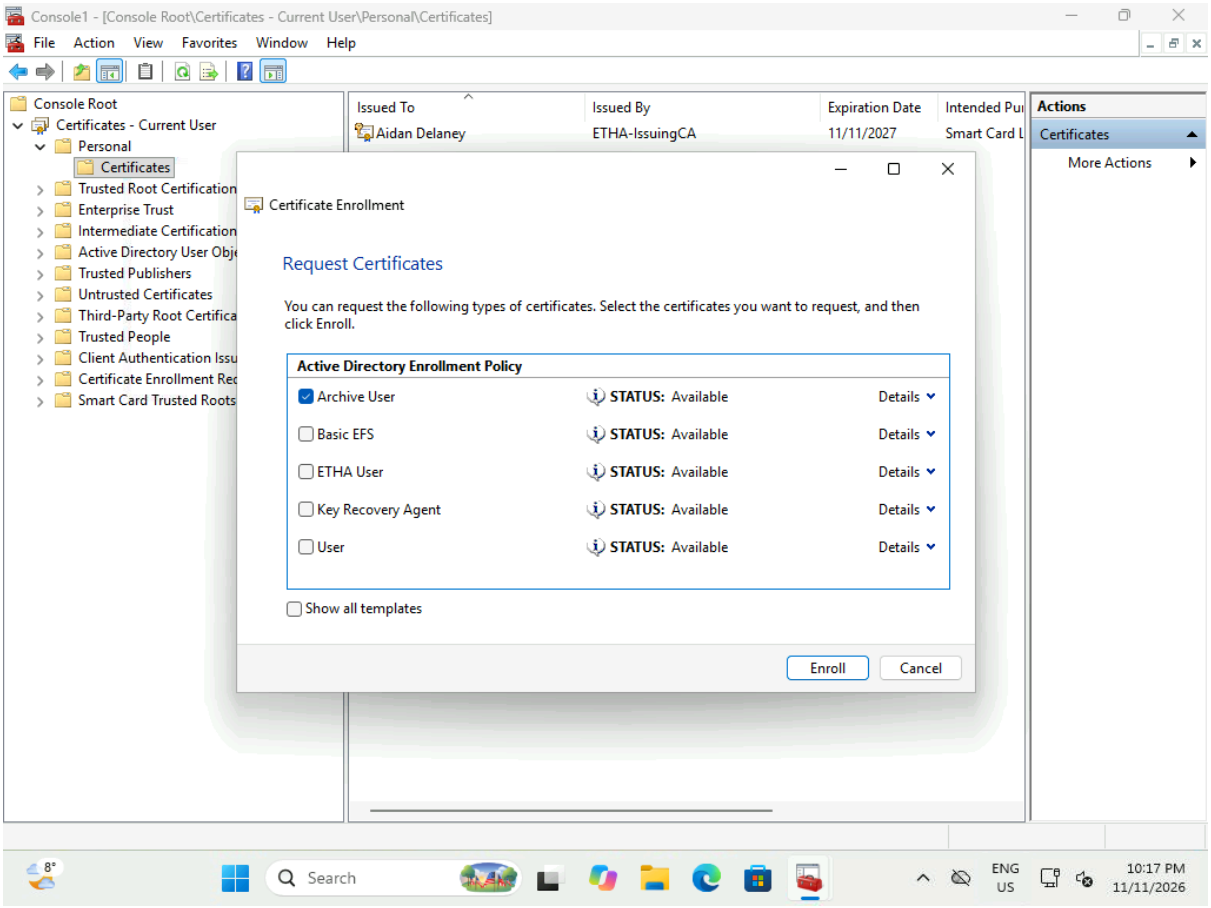
Task 17: Configure a custom template for key archival

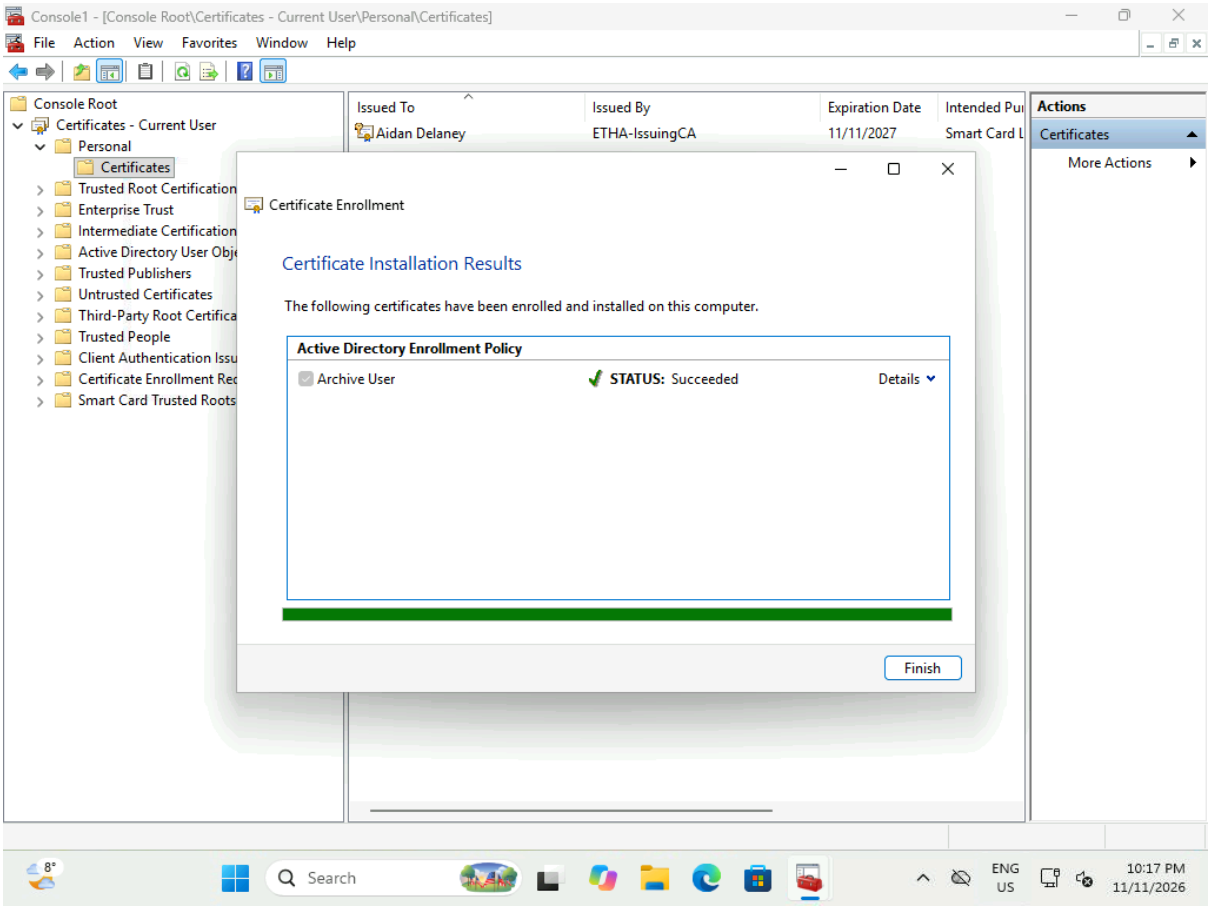

Certificate Templates Console
File Action View Help
Certificate Templates (STO-DC1...
Properties of New Template
Subject Name
Server
Issuance Requirements
Superseded Templates
Extensions
Security
Compatibility
General
Request Handling
Cryptography
Key Attestation
The template options available are based on the earliest operating system versions set in Compatibility Settings.
Show resulting changes
Compatibility Settings
Certification Authority
Windows Server 2003
Certificate recipient
Windows XP / Server 2003
These settings may not prevent earlier operating systems from using this template.
OK
Cancel
Apply
Help
Workstation Authentication
2
101.0
Client Authent
Actions
Certificate Templates (STO-...
More Actions
User
More Actions
Search
9:26 PM
11/11/2026

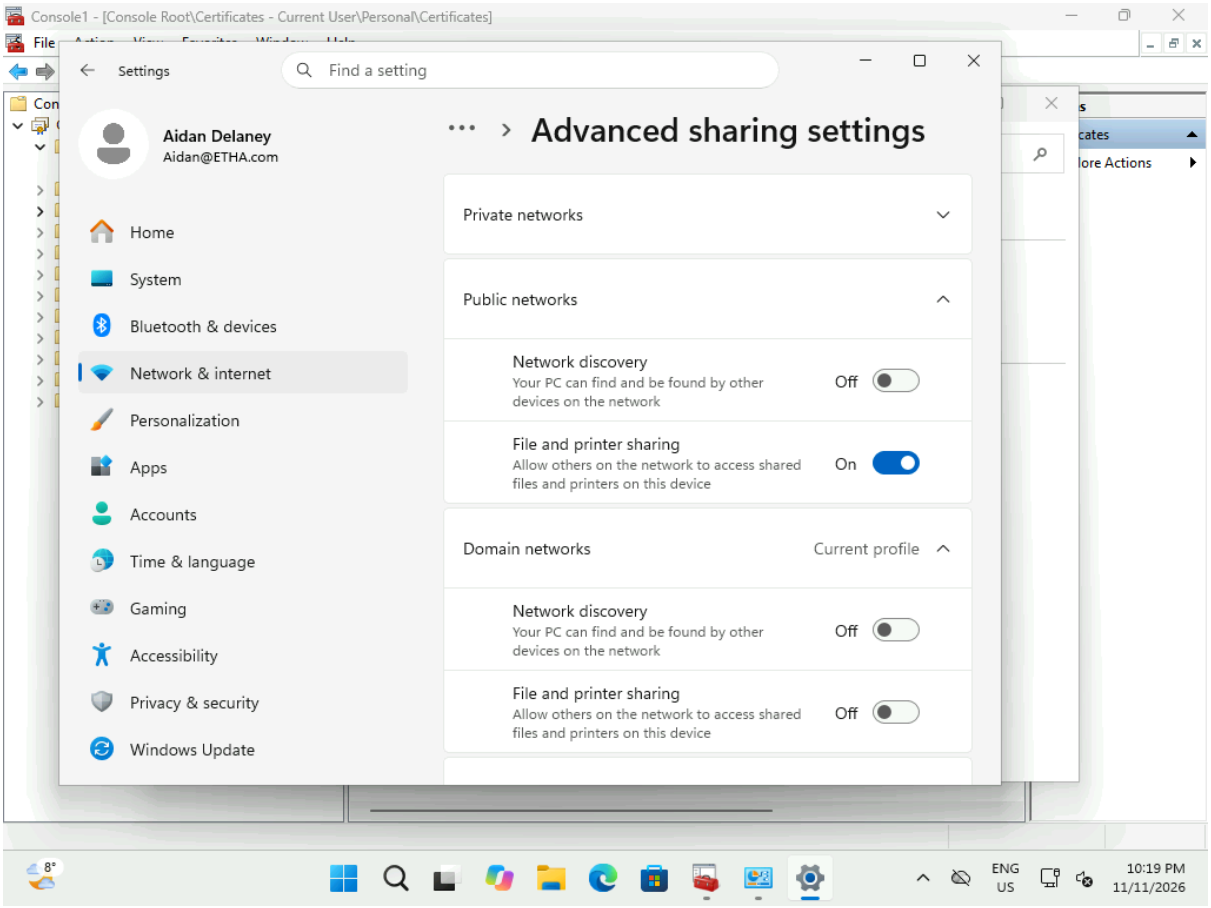


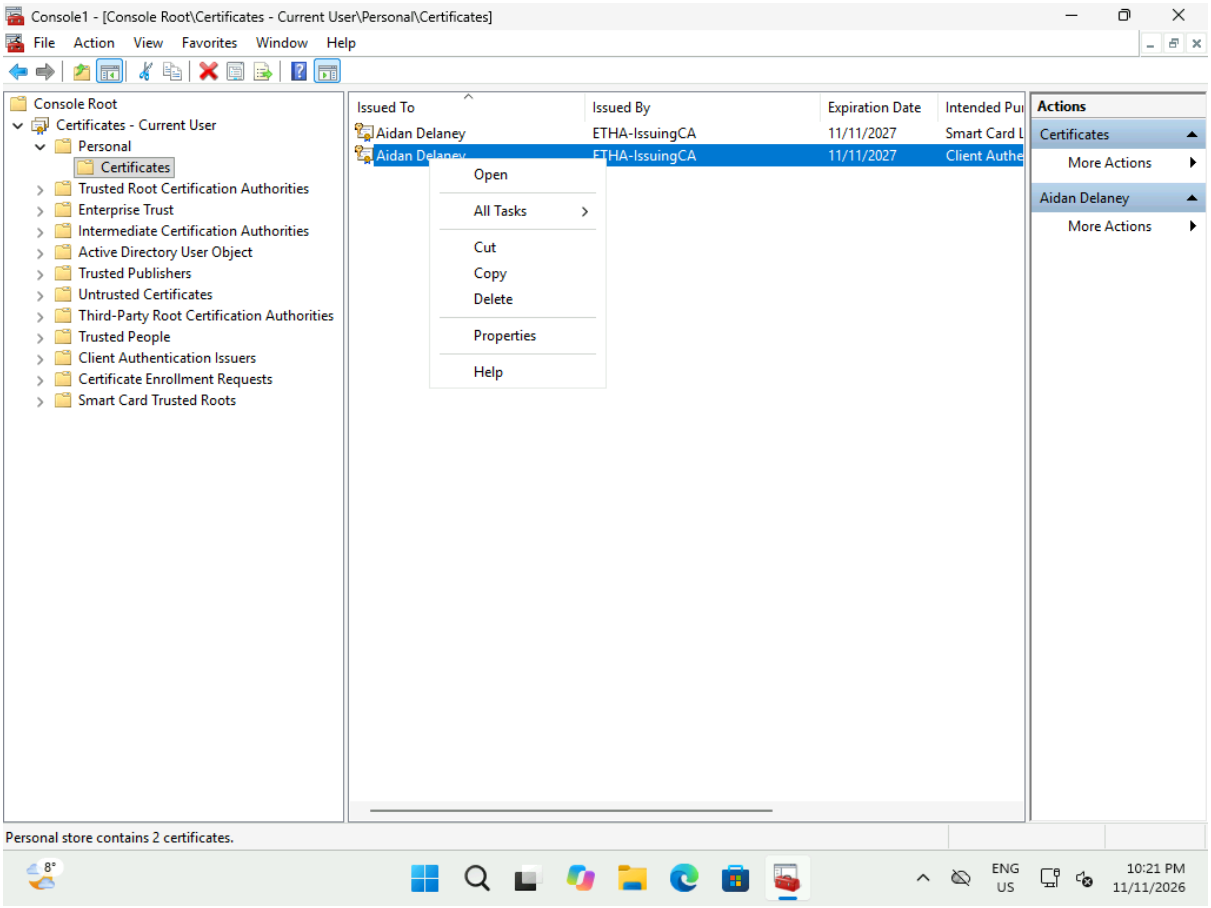
Task 18: Verify key archival functionality

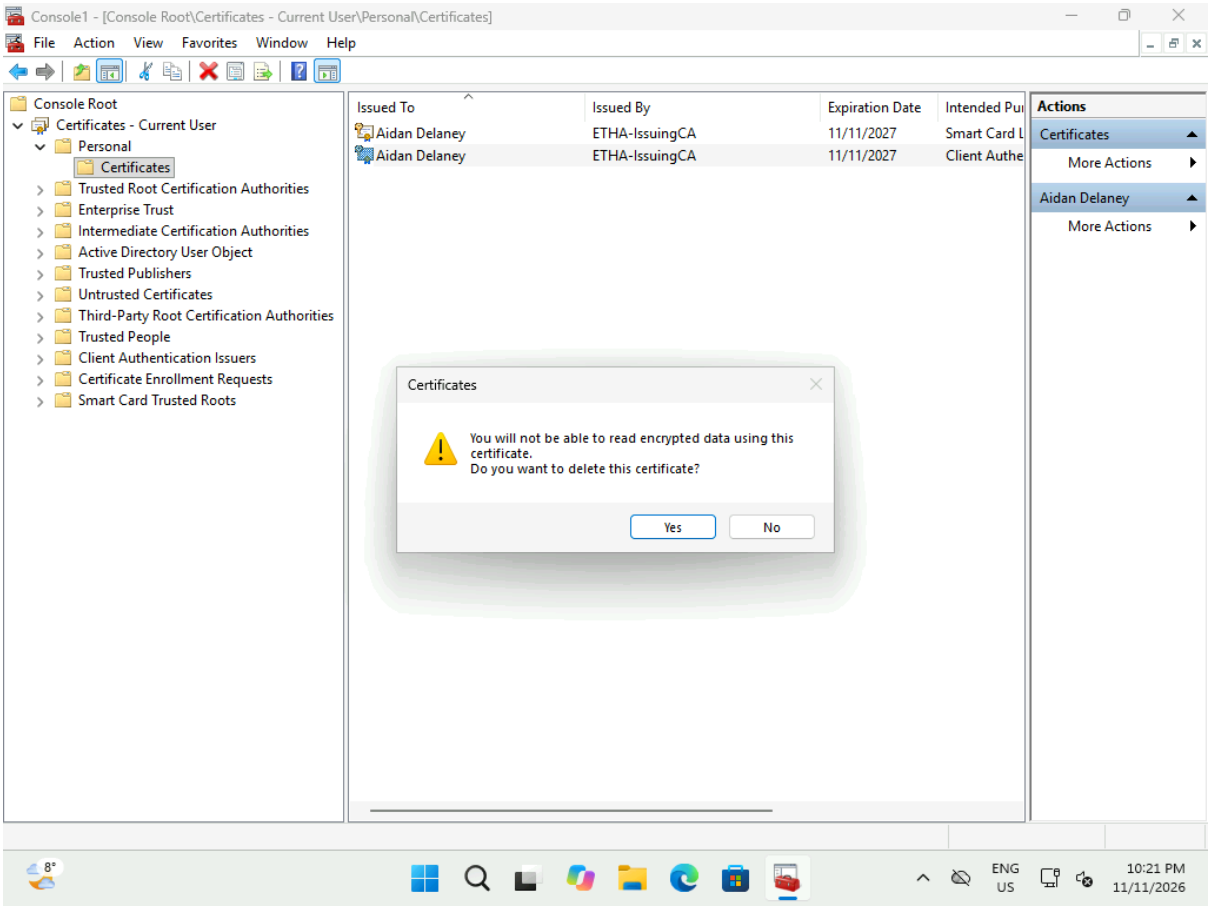

Add or Remove Snap-ins
You can select snap-ins for this console from those available on your computer and configure the selected set of snap-ins. For extensible snap-ins, you can configure which extensions are enabled.
Available snap-ins:
Snap-in
Vendor
Active Directory D... Microsoft Co...
Active Directory Sit... Microsoft Co...
Active Directory Us... Microsoft Co...
ActiveX Control Microsoft Co...
ADSI Edit Microsoft Co...
Authorization Man... Microsoft Co...
Certificates Microsoft Co...
Component Services Microsoft Co...
Computer Manage... Microsoft Co...
Device Manager Microsoft Co...
Disk Management Microsoft an...
DNS Microsoft Co...
Event Viewer Microsoft Co...
Add >
Selected snap-ins:
Console Root
Certificates - Current User
Edit Extensions...
Remove
Move Up
Move Down
Advanced...
Description:
The Certificates snap-in allows you to browse the contents of the certificate stores for yourself, a service, or a computer.
OK
Cancel
Windows 11 Enterprise Evaluation
Windows License valid for 89 days
Build 26100.ge_release.240331-1435
8°
Search
ENG
US
10:16 PM
11/11/2026

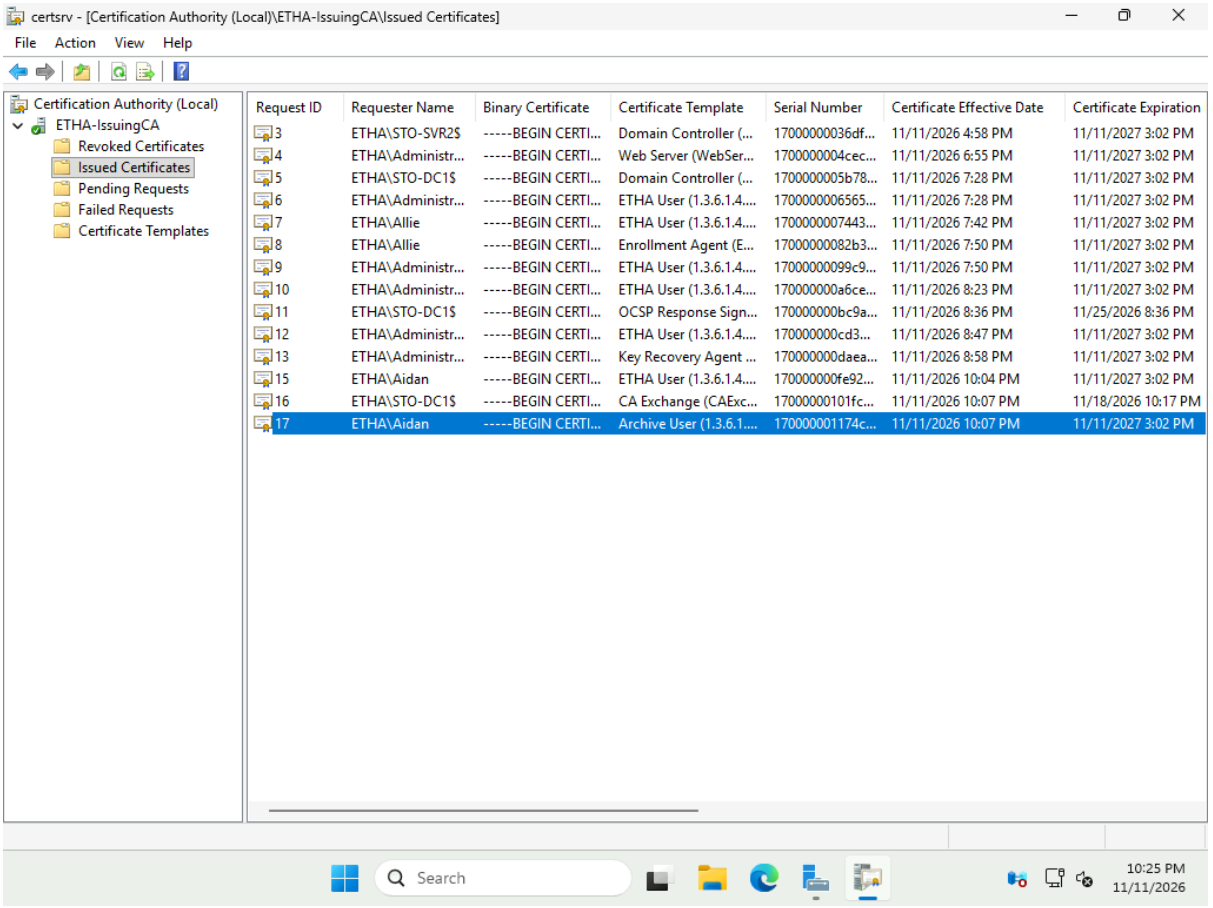

Console1 - [Console Root\Certificates - Current User\Personal\Certificates]
File Action View Favorites Window Help
Console Root
Certificates - Current User
Personal
Certificates
Trusted Root Certification
Enterprise Trust
Intermediate Certification
Active Directory User Obj
Trusted Publishers
Untrusted Certificates
Third-Party Root Certifica
Trusted People
Client Authentication Issu
Certificate Enrollment Re
Smart Card Trusted Roots
Issued To
Issued By
Expiration Date
Intended Pu
Aidan Delaney
ETHA-IssuingCA
11/11/2027
Smart Card L
Actions
Certificates
More Actions
Certificate Enrollment
Request Certificates
You can request the following types of certificates. Select the certificates you want to request, and then click Enroll.
Active Directory Enrollment Policy
Archive User
STATUS: Available
Details
Basic EFS
STATUS: Available
Details
ETHA User
STATUS: Available
Details
Key Recovery Agent
STATUS: Available
Details
User
STATUS: Available
Details
Show all templates
Enroll
Cancel
8°
Search
ENG
US
10:17 PM
11/11/2026

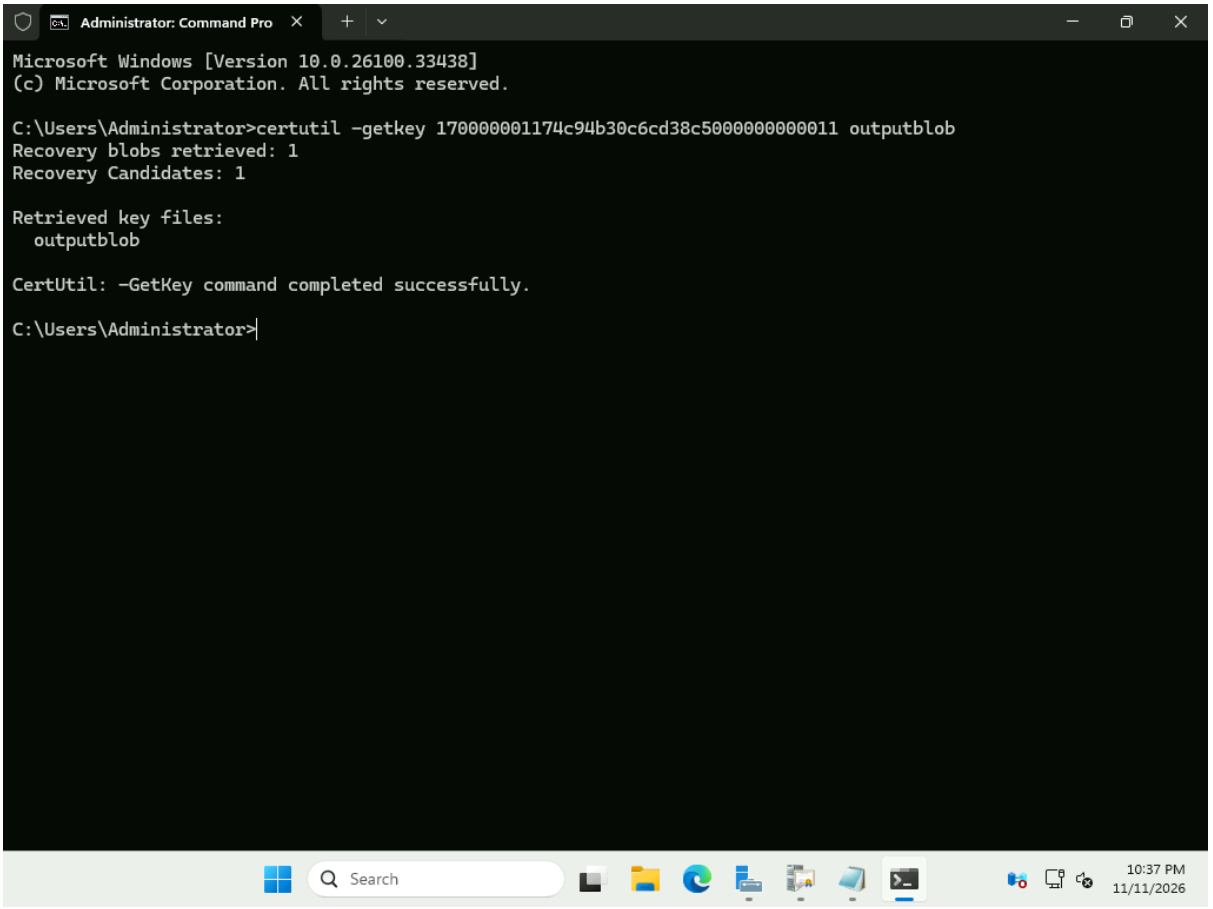

Console1 - [Console Root\Certificates - Current User\Personal\Certificates]
File Action View Favorites Window Help
Console Root
Certificates - Current User
Personal
Certificates
Trusted Root Certification
Enterprise Trust
Intermediate Certification
Active Directory User Obj
Trusted Publishers
Untrusted Certificates
Third-Party Root Certifica
Trusted People
Client Authentication Issu
Certificate Enrollment Re
Smart Card Trusted Roots
Issued To
Issued By
Expiration Date
Intended Pu
Aidan Delaney
ETHA-IssuingCA
11/11/2027
Smart Card L
Actions
Certificates
More Actions
Certificate Enrollment
Certificate Installation Results
The following certificates have been enrolled and installed on this computer.
Active Directory Enrollment Policy
Archive User
STATUS: Succeeded
Details
Finish
Search
ENG
US
10:17 PM
11/11/2026

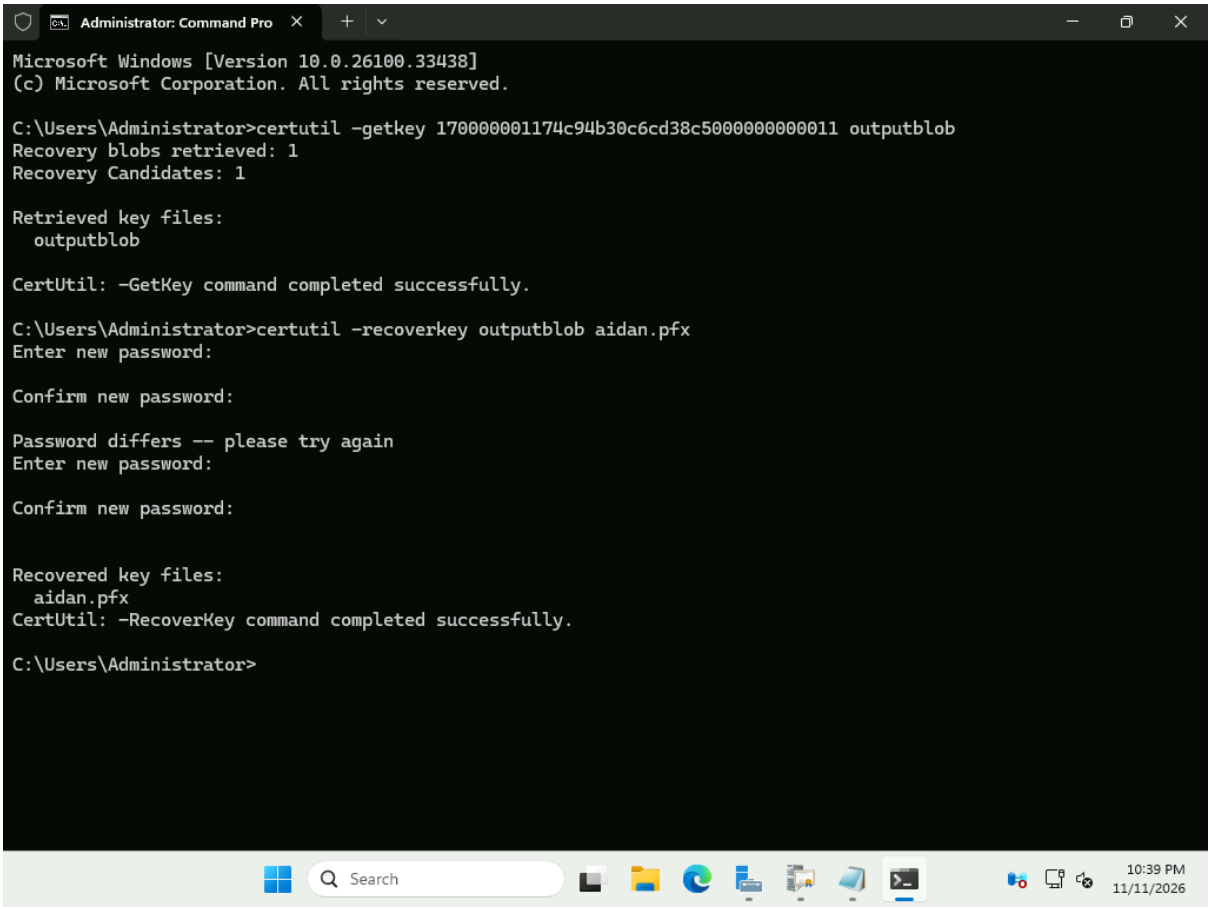

Console1 - [Console Root\Certificates - Current User\Personal\Certificates]
File
Settings
Find a setting
Aidan Delaney
Aidan@ETHA.com
Home
System
Bluetooth & devices
Network & internet
Personalization
Apps
Accounts
Time & language
Gaming
Accessibility
Privacy & security
Windows Update
Advanced sharing settings
Private networks
Public networks
Network discovery
Your PC can find and be found by other devices on the network
Off
File and printer sharing
Allow others on the network to access shared files and printers on this device
On
Domain networks
Current profile
Network discovery
Your PC can find and be found by other devices on the network
Off
File and printer sharing
Allow others on the network to access shared files and printers on this device
Off
ENG
US
10:19 PM
11/11/2026

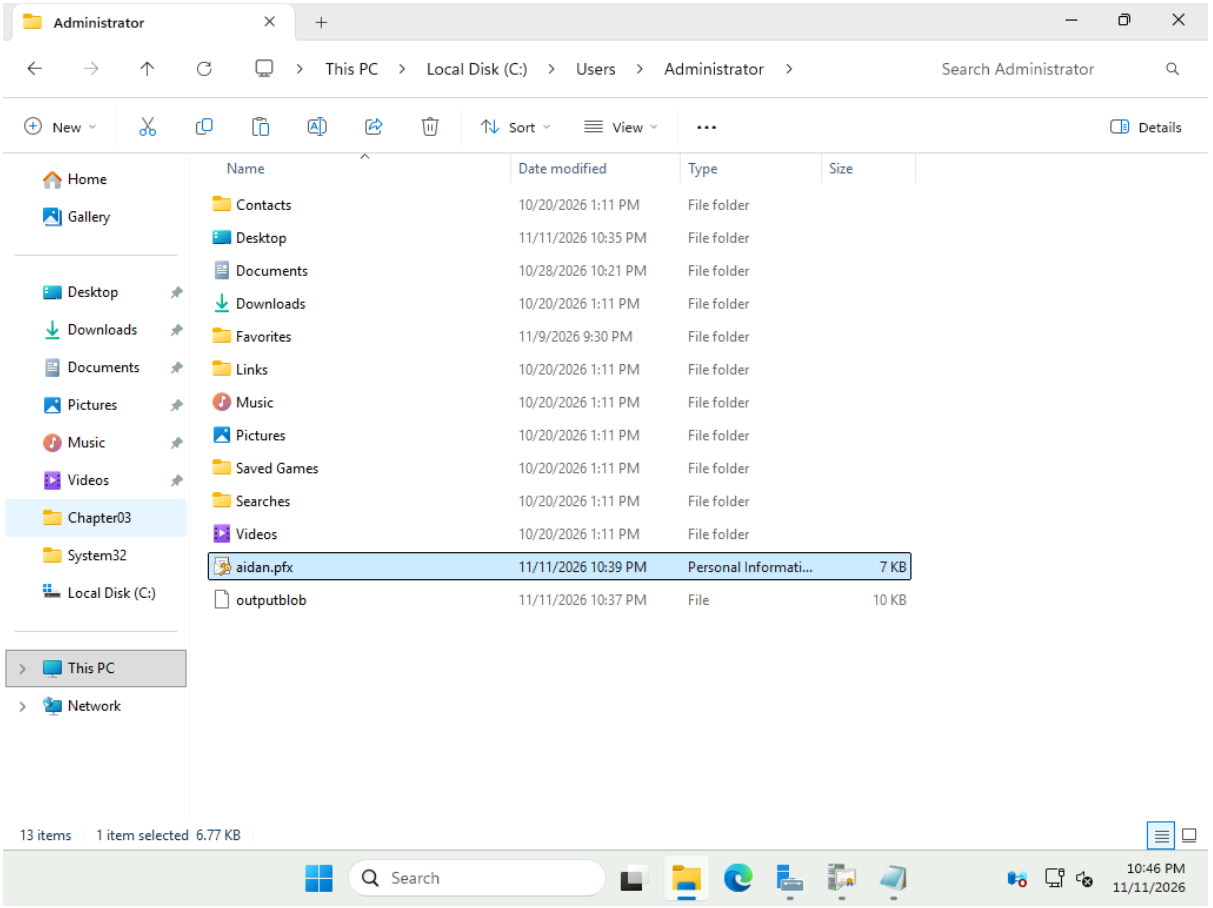

Console1 - [Console Root\Certificates - Current User\Personal\Certificates]
File Action View Favorites Window Help
Console Root
Certificates - Current User
Personal
Certificates
Trusted Root Certification Authorities
Enterprise Trust
Intermediate Certification Authorities
Active Directory User Object
Trusted Publishers
Untrusted Certificates
Third-Party Root Certification Authorities
Trusted People
Client Authentication Issuers
Certificate Enrollment Requests
Smart Card Trusted Roots
Issued To
Issued By
Expiration Date
Intended Pu
Aidan Delaney
ETHA-IssuingCA
11/11/2027
Smart Card L
Aidan Delaney
ETHA-IssuingCA
11/11/2027
Client Authe
Open
All Tasks
Cut
Copy
Delete
Properties
Help
Actions
Certificates
More Actions
Aidan Delaney
More Actions
Personal store contains 2 certificates.
8°
ENG
US
10:21 PM
11/11/2026

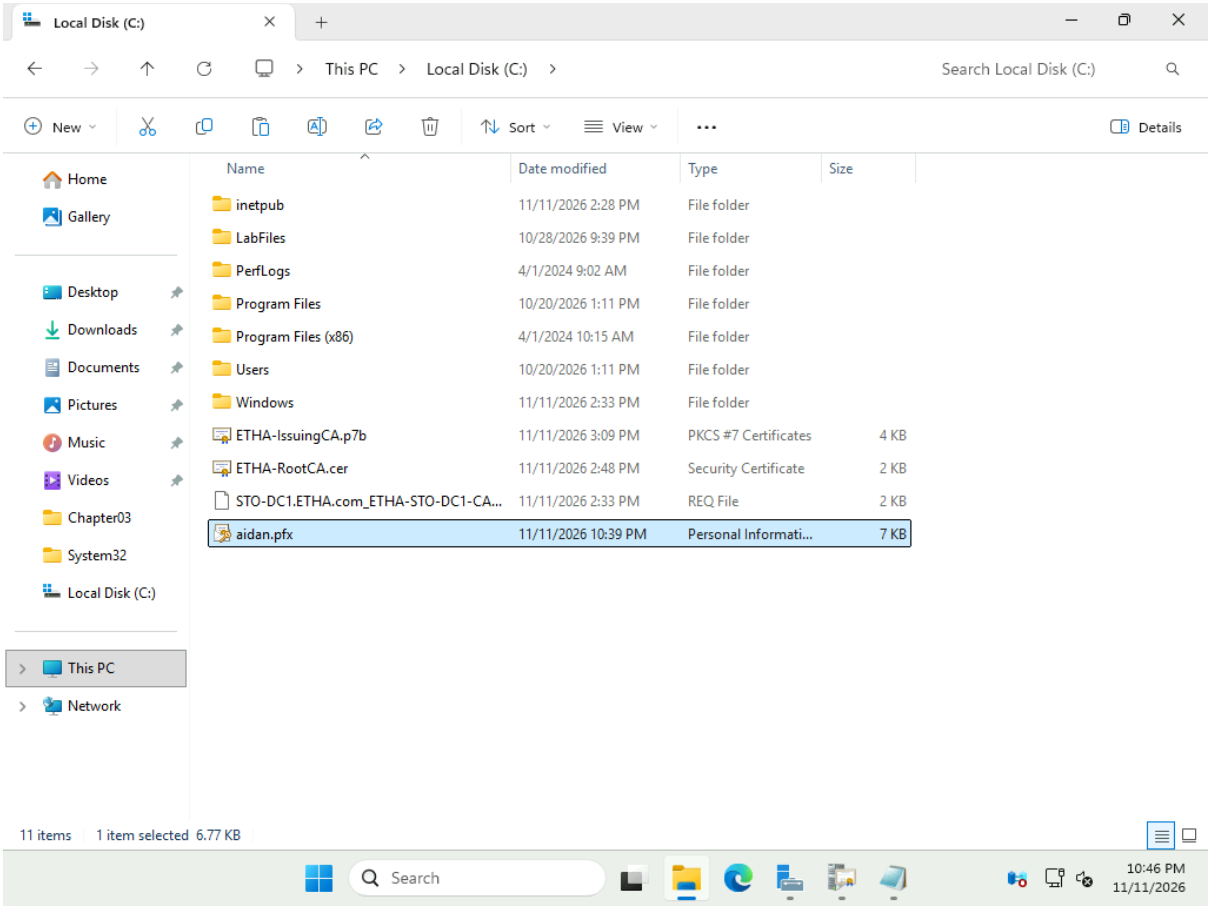

Console1 - [Console Root\Certificates - Current User\Personal\Certificates]
File Action View Favorites Window Help
Console Root
Certificates - Current User
Personal
Certificates
Trusted Root Certification Authorities
Enterprise Trust
Intermediate Certification Authorities
Active Directory User Object
Trusted Publishers
Untrusted Certificates
Third-Party Root Certification Authorities
Trusted People
Client Authentication Issuers
Certificate Enrollment Requests
Smart Card Trusted Roots
Issued To
Issued By
Expiration Date
Intended Pu
Aidan Delaney
ETHA-IssuingCA
11/11/2027
Smart Card L
Aidan Delaney
ETHA-IssuingCA
11/11/2027
Client Authe
Actions
Certificates
More Actions
Aidan Delaney
More Actions
Certificates
You will not be able to read encrypted data using this certificate.
Do you want to delete this certificate?
Yes
No
8°
ENG
US
10:21 PM
11/11/2026

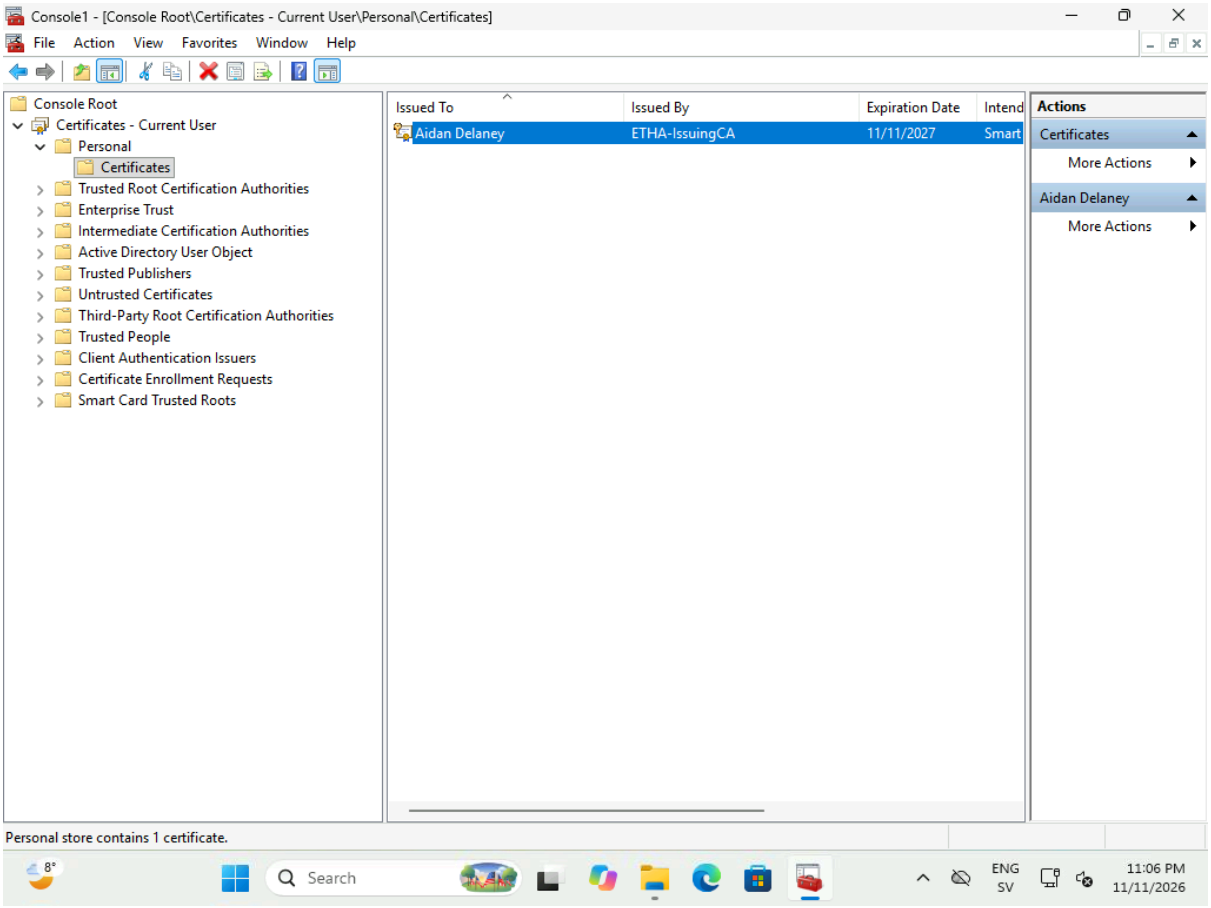

certsrv - [Certification Authority (Local)\ETHA-IssuingCA\Issued Certificates]
File Action View Help
Certification Authority (Local)
ETHA-IssuingCA
Revoked Certificates
Issued Certificates
Pending Requests
Failed Requests
Certificate Templates
Request ID | Requester Name | Binary Certificate | Certificate Template | Serial Number | Certificate Effective Date | Certificate Expiration
3 | ETHA\STO-SVR2$ | -----BEGIN CERTI... | Domain Controller (... | 17000000036df... | 11/11/2026 4:58 PM | 11/11/2027 3:02 PM
4 | ETHA\Administr... | -----BEGIN CERTI... | Web Server (WebSer... | 1700000004cec... | 11/11/2026 6:55 PM | 11/11/2027 3:02 PM
5 | ETHA\STO-DC1$ | -----BEGIN CERTI... | Domain Controller (... | 1700000005b78... | 11/11/2026 7:28 PM | 11/11/2027 3:02 PM
6 | ETHA\Administr... | -----BEGIN CERTI... | ETHA User (1.3.6.1.4.... | 1700000006565... | 11/11/2026 7:28 PM | 11/11/2027 3:02 PM
7 | ETHA\Allie | -----BEGIN CERTI... | ETHA User (1.3.6.1.4.... | 1700000007443... | 11/11/2026 7:42 PM | 11/11/2027 3:02 PM
8 | ETHA\Allie | -----BEGIN CERTI... | Enrollment Agent (E... | 17000000082b3... | 11/11/2026 7:50 PM | 11/11/2027 3:02 PM
9 | ETHA\Administr... | -----BEGIN CERTI... | ETHA User (1.3.6.1.4.... | 17000000099c9... | 11/11/2026 7:50 PM | 11/11/2027 3:02 PM
10 | ETHA\Administr... | -----BEGIN CERTI... | ETHA User (1.3.6.1.4.... | 170000000a6ce... | 11/11/2026 8:23 PM | 11/11/2027 3:02 PM
11 | ETHA\STO-DC1$ | -----BEGIN CERTI... | OCSP Response Sign... | 170000000bc9a... | 11/11/2026 8:36 PM | 11/25/2026 8:36 PM
12 | ETHA\Administr... | -----BEGIN CERTI... | ETHA User (1.3.6.1.4.... | 170000000cd3... | 11/11/2026 8:47 PM | 11/11/2027 3:02 PM
13 | ETHA\Administr... | -----BEGIN CERTI... | Key Recovery Agent ... | 170000000daea... | 11/11/2026 8:58 PM | 11/11/2027 3:02 PM
15 | ETHA\Aidan | -----BEGIN CERTI... | ETHA User (1.3.6.1.4.... | 170000000fe92... | 11/11/2026 10:04 PM | 11/11/2027 3:02 PM
16 | ETHA\STO-DC1$ | -----BEGIN CERTI... | CA Exchange (CAExc... | 17000000101fc... | 11/11/2026 10:07 PM | 11/18/2026 10:17 PM
17 | ETHA\Aidan | -----BEGIN CERTI... | Archive User (1.3.6.1.... | 170000001174c... | 11/11/2026 10:07 PM | 11/11/2027 3:02 PM
Search
10:25 PM
11/11/2026


Administrator: Command Pro
Microsoft Windows [Version 10.0.26100.33438]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>certutil -getkey 170000001174c94b30c6cd38c5000000000011 outputblob
Recovery blobs retrieved: 1
Recovery Candidates: 1
Retrieved key files:
outputblob
CertUtil: -GetKey command completed successfully.
C:\Users\Administrator>
Search
10:37 PM
11/11/2026


Administrator: Command Pro
Microsoft Windows [Version 10.0.26100.33438]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>certutil -getkey 170000001174c94b30c6cd38c5000000000011 outputblob
Recovery blobs retrieved: 1
Recovery Candidates: 1
Retrieved key files:
outputblob
CertUtil: -GetKey command completed successfully.
C:\Users\Administrator>certutil -recoverkey outputblob aidan.pfx
Enter new password:
Confirm new password:
Password differs -- please try again
Enter new password:
Confirm new password:
Recovered key files:
aidan.pfx
CertUtil: -RecoverKey command completed successfully.
C:\Users\Administrator>
Search
10:39 PM
11/11/2026


Administrator
This PC
Local Disk (C:)
Users
Administrator
Search Administrator
New
Sort
View
Details
Home
Gallery
Desktop
Downloads
Documents
Pictures
Music
Videos
Chapter03
System32
Local Disk (C:)
This PC
Network
Name
Date modified
Type
Size
Contacts 10/20/2026 1:11 PM File folder
Desktop 11/11/2026 10:35 PM File folder
Documents 10/28/2026 10:21 PM File folder
Downloads 10/20/2026 1:11 PM File folder
Favorites 11/9/2026 9:30 PM File folder
Links 10/20/2026 1:11 PM File folder
Music 10/20/2026 1:11 PM File folder
Pictures 10/20/2026 1:11 PM File folder
Saved Games 10/20/2026 1:11 PM File folder
Searches 10/20/2026 1:11 PM File folder
Videos 10/20/2026 1:11 PM File folder
aidan.pfx 11/11/2026 10:39 PM Personal Informati... 7 KB
outputblob 11/11/2026 10:37 PM File 10 KB
13 items
1 item selected 6.77 KB
Search
10:46 PM
11/11/2026


Local Disk (C:)
This PC
Local Disk (C:)
Search Local Disk (C:)
New
Sort
View
Details
Home
Gallery
Desktop
Downloads
Documents
Pictures
Music
Videos
Chapter03
System32
Local Disk (C:)
This PC
Network
Name
Date modified
Type
Size
inetpub 11/11/2026 2:28 PM File folder
LabFiles 10/28/2026 9:39 PM File folder
PerfLogs 4/1/2024 9:02 AM File folder
Program Files 10/20/2026 1:11 PM File folder
Program Files (x86) 4/1/2024 10:15 AM File folder
Users 10/20/2026 1:11 PM File folder
Windows 11/11/2026 2:33 PM File folder
ETHA-IssuingCA.p7b 11/11/2026 3:09 PM PKCS #7 Certificates 4 KB
ETHA-RootCA.cer 11/11/2026 2:48 PM Security Certificate 2 KB
STO-DC1.ETHA.com_ETHA-STO-DC1-CA... 11/11/2026 2:33 PM REQ File 2 KB
aidan.pfx 11/11/2026 10:39 PM Personal Informati... 7 KB
11 items
1 item selected 6.77 KB
Search
10:46 PM
11/11/2026


Console1 - [Console Root\Certificates - Current User\Personal\Certificates]
File Action View Favorites Window Help
Console Root
Certificates - Current User
Personal
Certificates
Trusted Root Certification Authorities
Enterprise Trust
Intermediate Certification Authorities
Active Directory User Object
Trusted Publishers
Untrusted Certificates
Third-Party Root Certification Authorities
Trusted People
Client Authentication Issuers
Certificate Enrollment Requests
Smart Card Trusted Roots
Issued To
Issued By
Expiration Date
Intend
Aidan Delaney
ETHA-IssuingCA
11/11/2027
Smart
Actions
Certificates
More Actions
Aidan Delaney
More Actions
Personal store contains 1 certificate.
8°
Search
ENG
SV
11:06 PM
11/11/2026